ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
GRID CONTROLLER OF INDIA LIMITED
(A Government of India Enterprise)
[formerly Power System Operation Corporation Limited (POSOCO)]
राष्ट्रीय भार प्रेषण केन्द्र / National Load Despatch Centre
कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
Office : 1st and 2nd Floor, B-9, Qutab Institutional Area, Katwaria Sarai, New Delhi -110016
CIN : U40105DL2009GOI188682, Website : www.grid-india.in, E-mail : gridindiacc@grid-india.in, Tel.: 011- 42785855

संदर्भ संख्या:- GRID-INDIA/NLDC/MR/

दिनांक: **23.08.2024**

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – **जुलाई 2024**

महोदय,

आई०ई०जी०सी०-2023 की धारा-38 के अनुपालन में, **जुलाई 2024** माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय

एस.सी. दंभारे
वरिष्ठ महाप्रबन्धक
प्रणाली प्रचालन, रा.भा.प्रे.कें.

संलग्नक: मासिक प्रणाली रिपोर्ट

वितरण सूची
Distribution List

1 सचिव, के.वि.नि.आ. तीसरा एवं चौथा तल, चंद्रलोक भवन, 36, जनपथ, नई दिल्ली–110001
Secretary, CERC, 3rd & 4th Floor, Chanderlok Building,36,Janpath,New Delhi-110001

2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली–110066
Chief Engineer (GM), CEA,Sewa Bhavan, R.K.Puram , New Delhi-110066

3 सदस्य सचिव, उ. क्षे. वि. स., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
Member Secretary, NRPC,18/A,SJSS Marg, Katwaria Sarai, New Delhi-110016

4 सदस्य सचिव, प. क्षे. वि. स., एफ–3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई – 400093
Member Secretary, WRPC, F-3,MIDC Area, Andheri (East),Mumbai-400093

5 सदस्य सचिव, द. क्षे. वि. स., 29, रेस कोर्स क्रॉस रोड, बंगलूरु – 560009
Member Secretary, SRPC, 29,Race Course Cross Road,Bangalore-560009

6 सदस्य सचिव, पू. क्षे. वि. स., 14, गोल्फ क्लब रोड, कोलकाता – 700033
Member Secretary, ERPC, 14,Golf Club Road , Kolkata-700033

7 सदस्य सचिव,उ. पू. क्षे. वि. स.,मेघालय राज्य आवासीय वित्त सहकारी समिति लिमिटेड भवन, नोंग्रिम हिल्स, शिलोंग – 793003
Member Secretary, NERPC, Meghalaya State Housing Finance Cooperative Society Ltd. Building, Nongrim Hills , Shillong -793003

8 मुख्य अभियंता, राष्ट्रीय विद्युत समिति, एनआरपीसी भवन, तृतीय तल, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
Chief Engineer, National Power Committee, NRPC Building, 3rd floor, 18/A,SJSS Marg, Katwaria Sarai, New Delhi-110016

9 कार्यपालक निदेशक, पू. क्षे. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता – 700033
Executive Director, ERLDC,14, Golf Club Road , Kolkata-700033

10 कार्यपालक निदेशक, प. क्षे. भा. प्रे. के., एफ–3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई – 400093
Executive Director, WRLDC,F-3,MIDC Area, Andheri (East),Mumbai-400093

11 कार्यपालक निदेशक , उ.क्षे.भा.प्रे.के, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली – 110016
Executive Director, NRLDC,18/A,SJSS Marg, Katwaria Sarai, New Delhi-110016

12 कार्यपालक निदेशक, द. क्षे. भा. प्रे. के, 29, रेस कोर्स क्रॉस रोड, बंगलूरु – 560009
Executive Director, SRLDC, 29,Race Course Cross Road,Bangalore-560009

13 कार्यपालक निदेशक, पू. क्षे. भा. प्रे. के., डोंगतिएह, लोअर नोंग्रह, लापलंग, शिलोंग- 793006
Executive Director, NERLDC, Dongtieh, Lower Nongrah, Laplang, Shillong-793006

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly Operation Report



जुलाई 2024

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED
Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI

माह जुलाई 2024 के लिए प्रचालन निष्पादन रिपोर्ट

OPERATIONAL PERFORMANCE REPORT FOR THE MONTH OF JULY -2024

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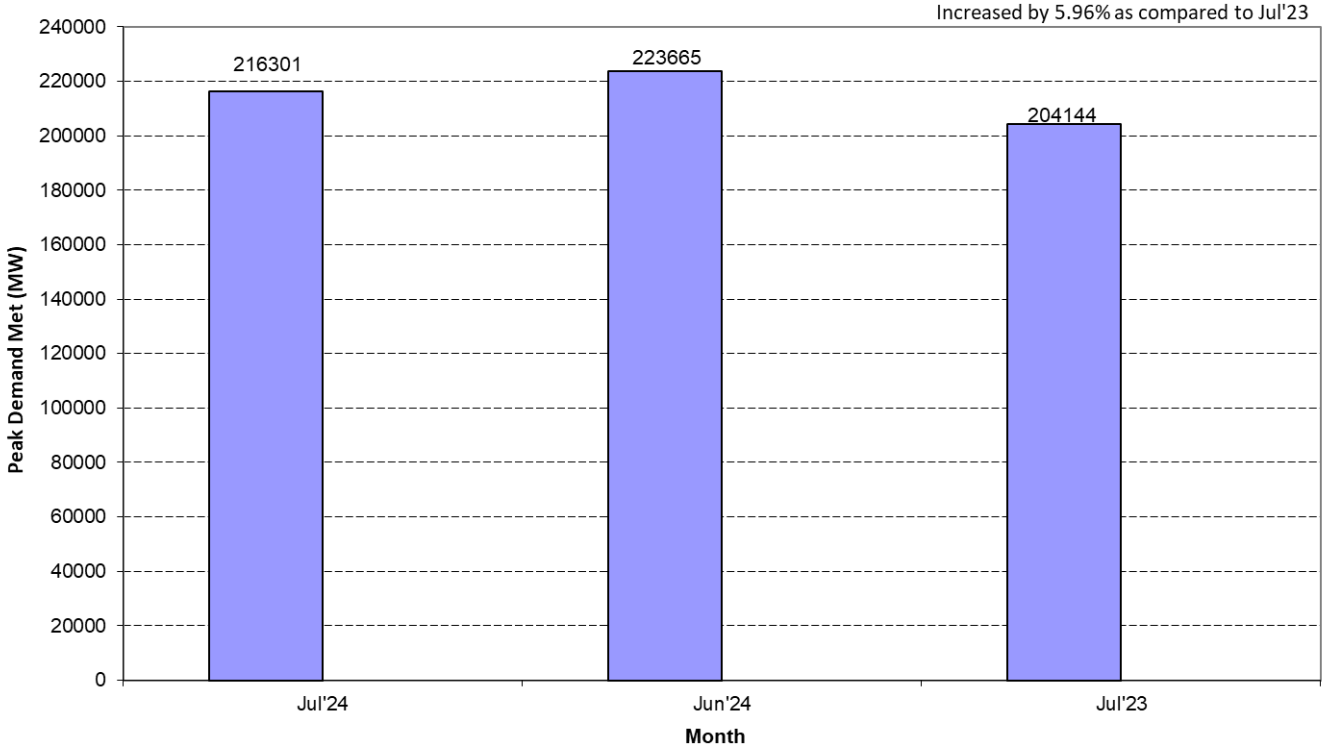
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1. SUMMARY OF REPORT FOR THE MONTH OF JULY-2024

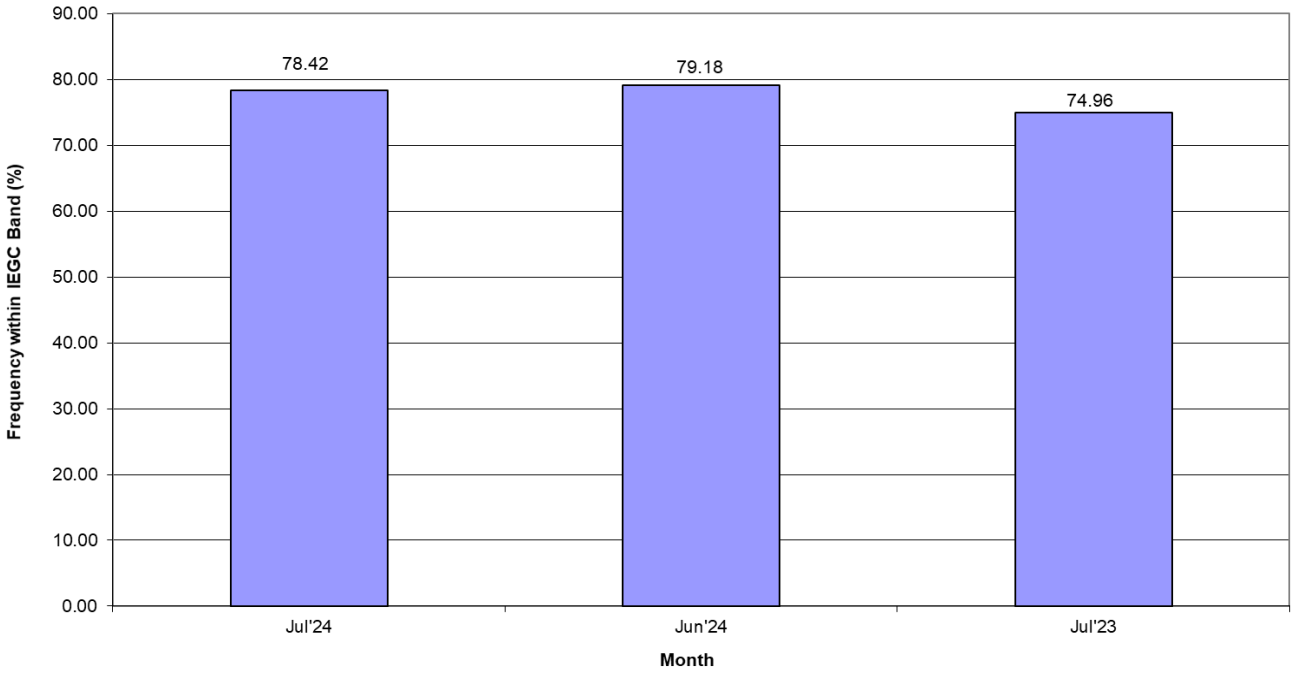
EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)

Decreased by 3.29% as compared to Jun'24

Increased by 5.96% as compared to Jul'23

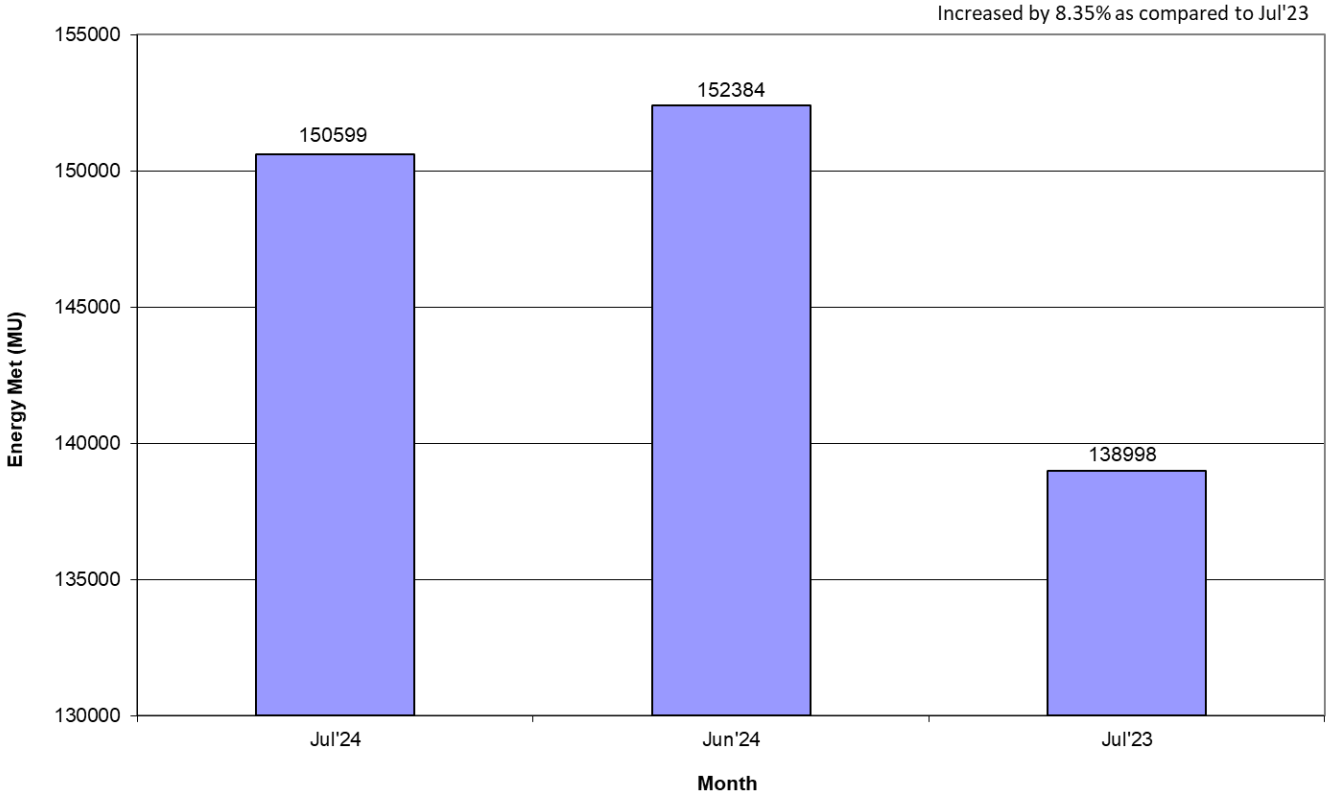


Frequency within IEGC Band



ENERGY MET AT NATIONAL LEVEL (MU)

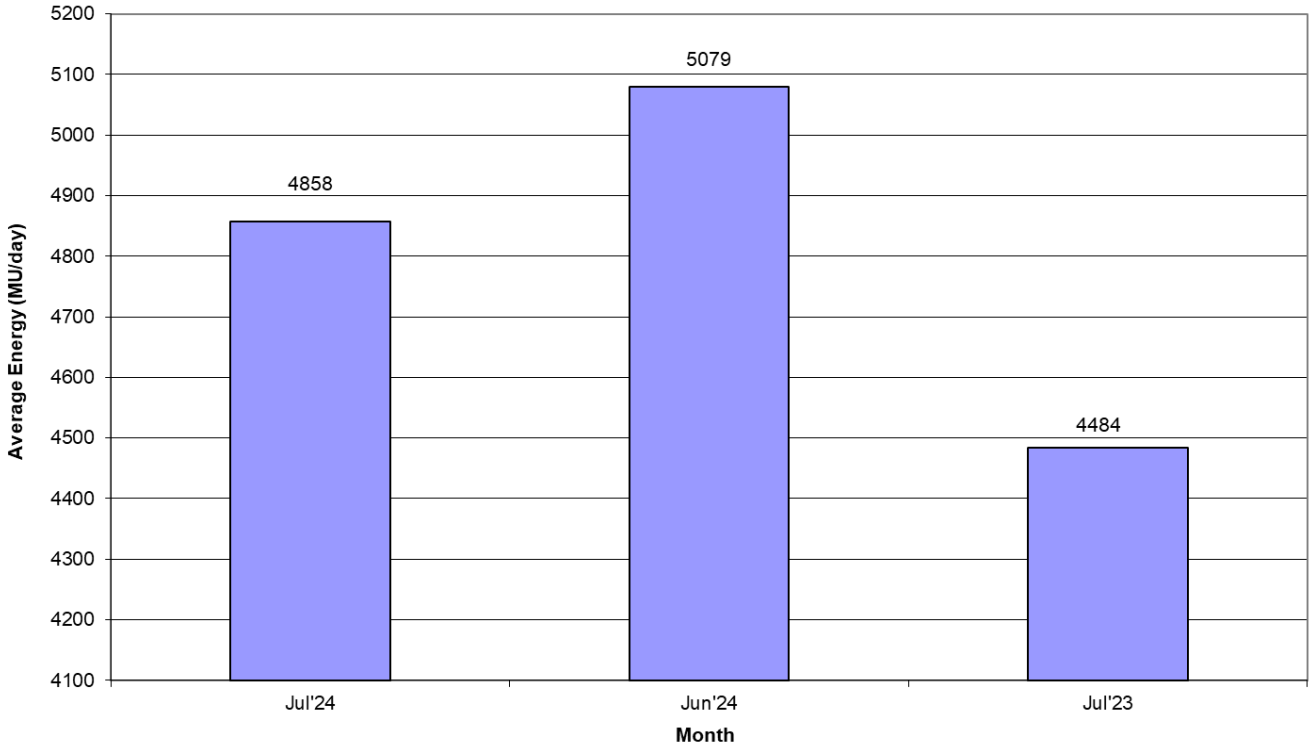
Increased by 8.35% as compared to Jul'23



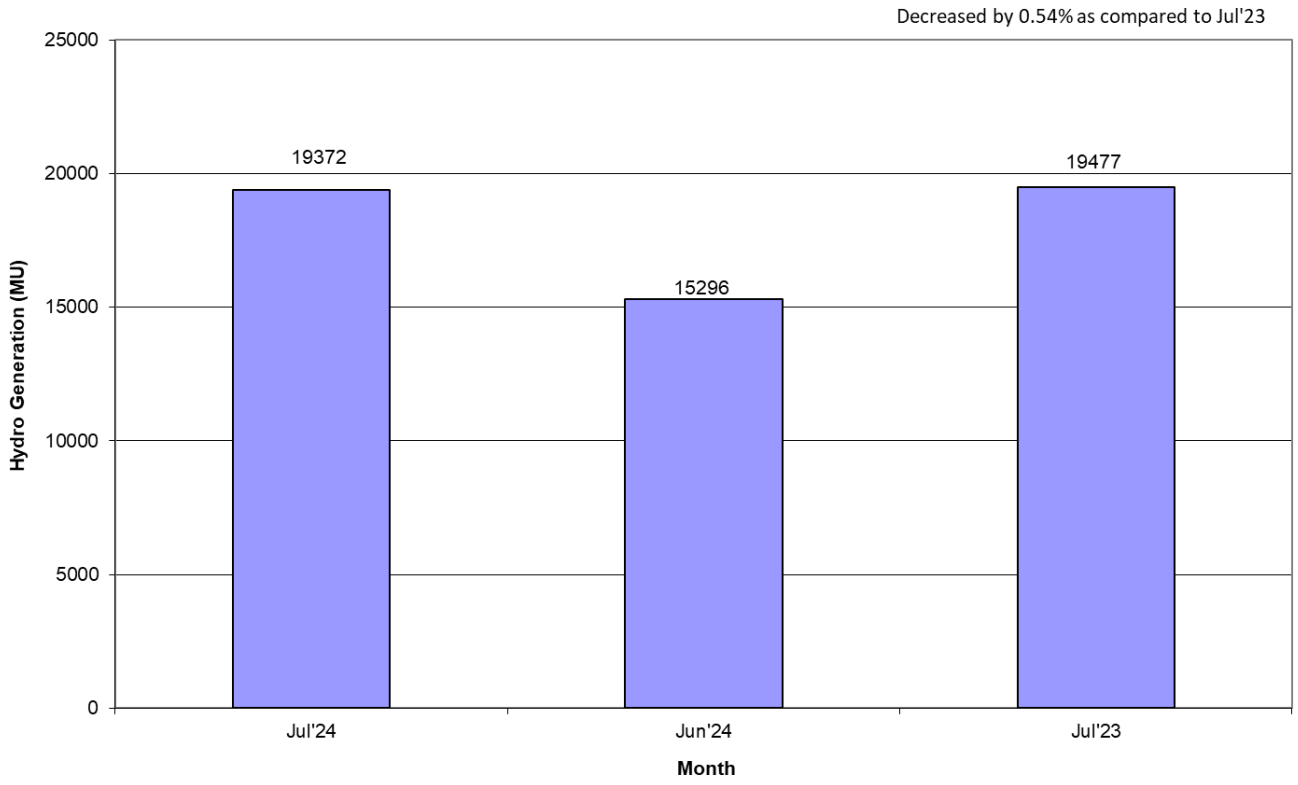
AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)

Decreased by 4.36% as compared to Jun'24

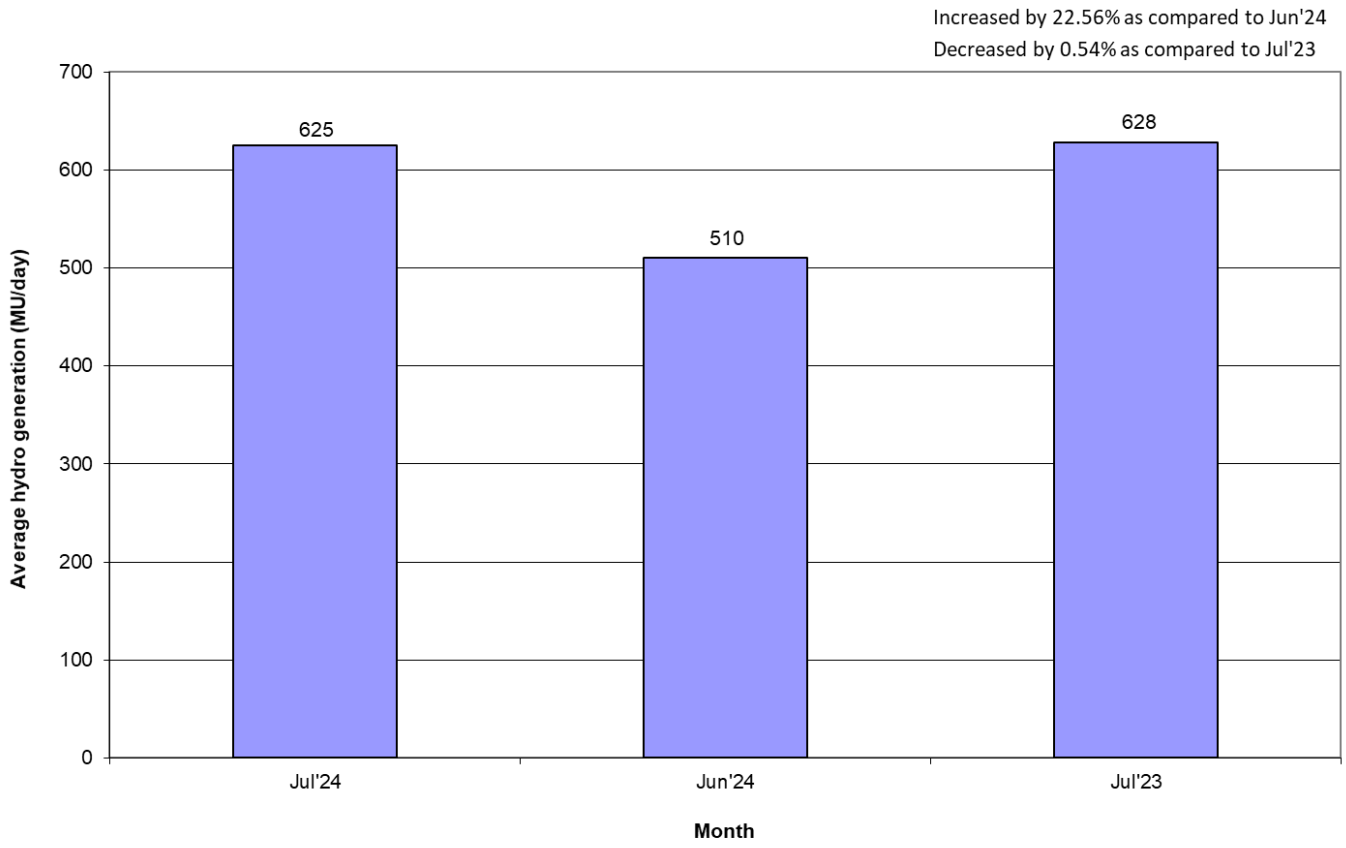
Increased by 8.35% as compared to Jul'23



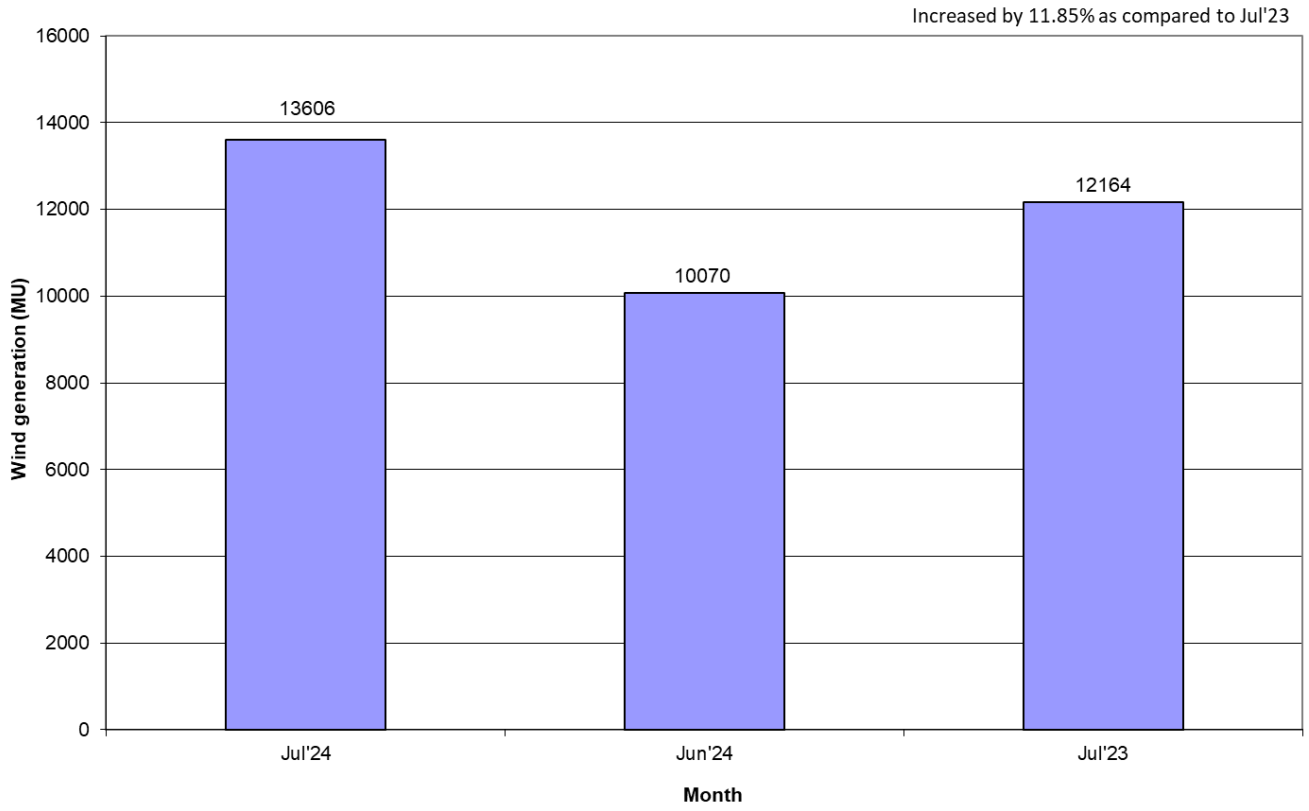
HYDRO GENERATION AT NATIONAL LEVEL (MU)



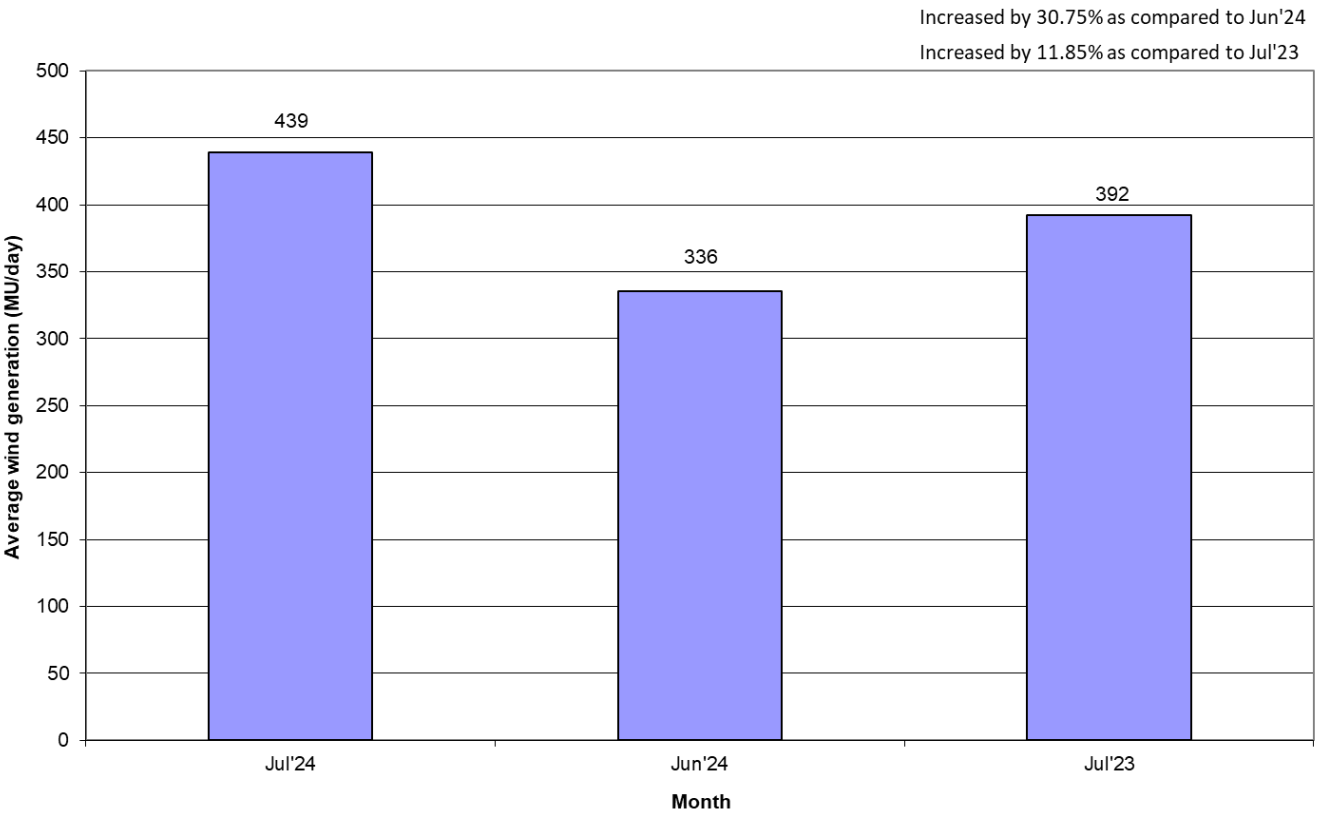
AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)



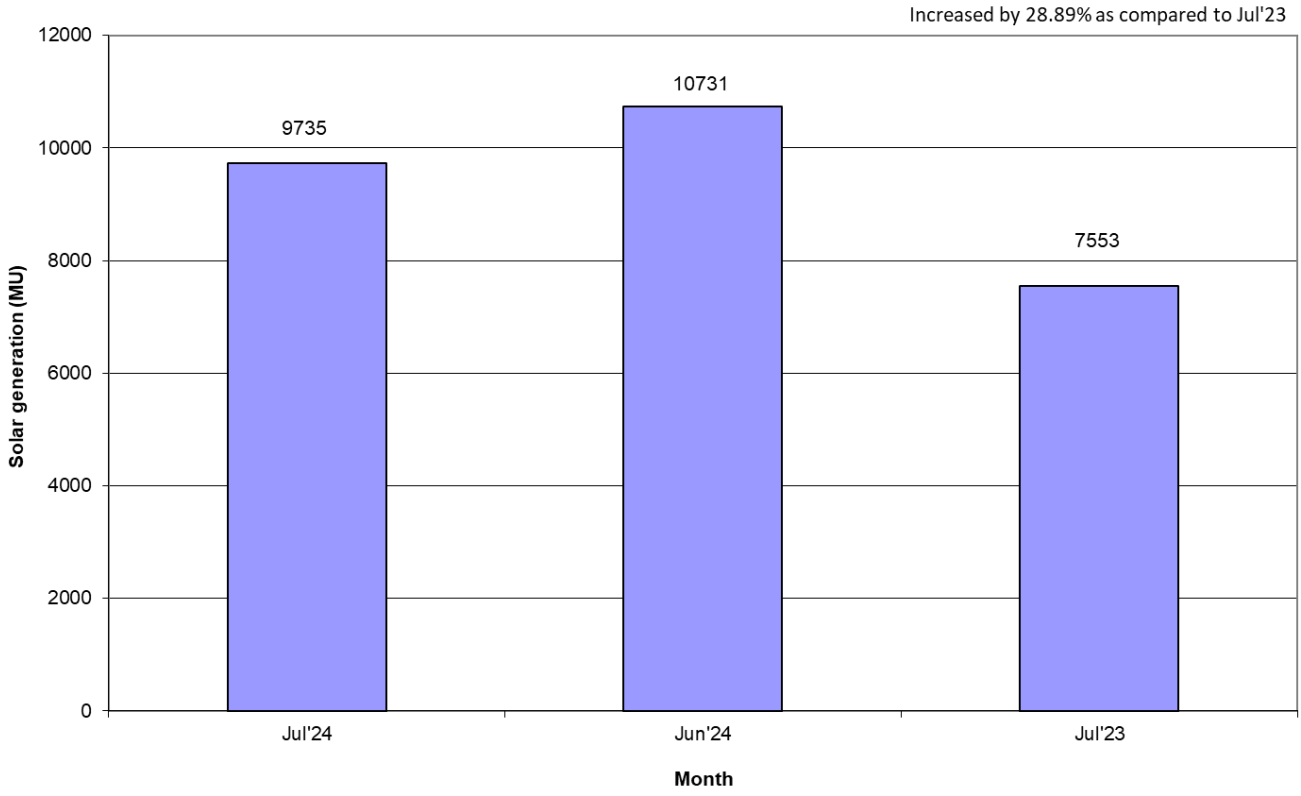
WIND GENERATION AT NATIONAL LEVEL (MU)



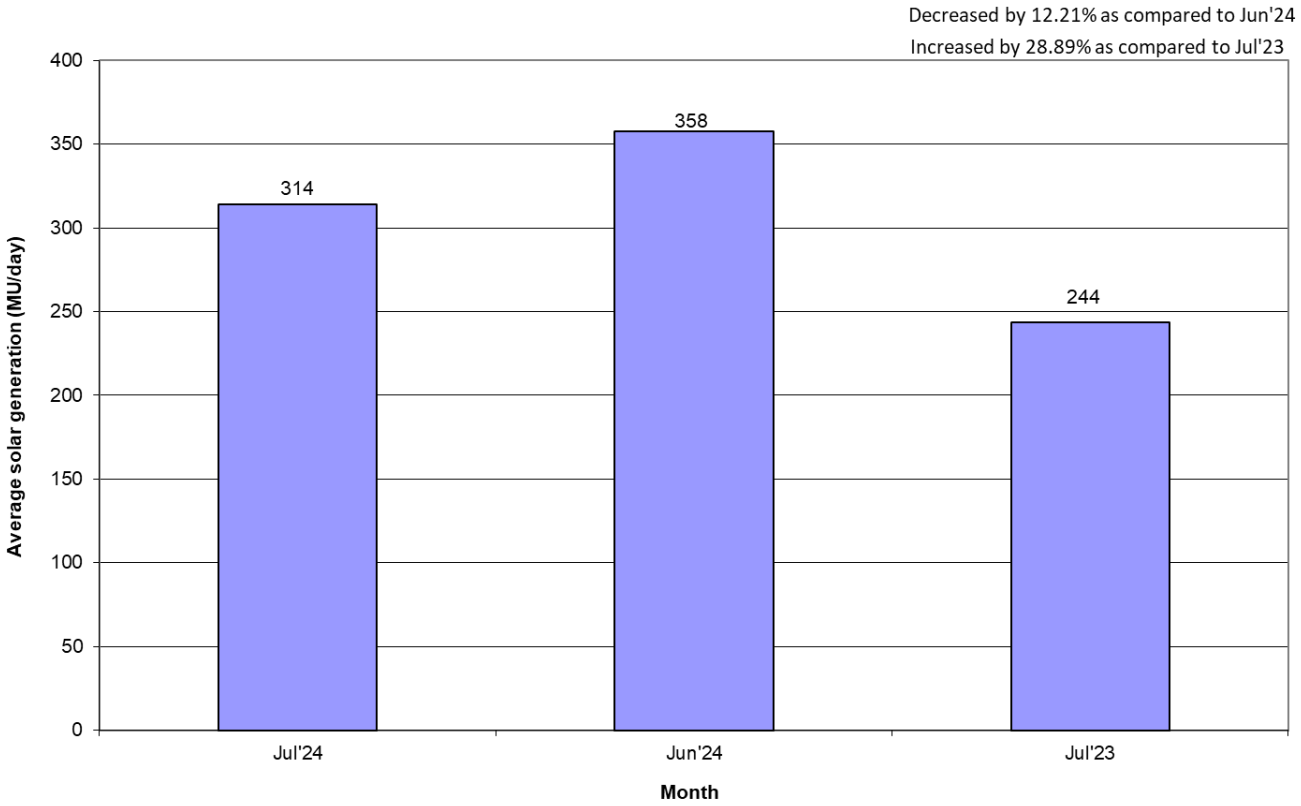
AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)



SOLAR GENERATION AT NATIONAL LEVEL (MU)



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 31.07.2024
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	56658	1580	5995	0	64233	1620	20830	40671	127353
2	WR	74663	1400	10806	0	86869	3240	7563	51187	148858
3	SR	49717	3640	6272	434	60063	3320	11827	55734	130944
4	ER	28690	0	80	0	28770	0	4764	2036	35570
5	NER	1242	0	1665	36	2943	0	1944	609	5496
6	ISLANDS	0	0	0	120	120	0	0	40	160
	ALL INDIA	210970	6620	24818	589	242997	8180	46928	150276	448381

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति

EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

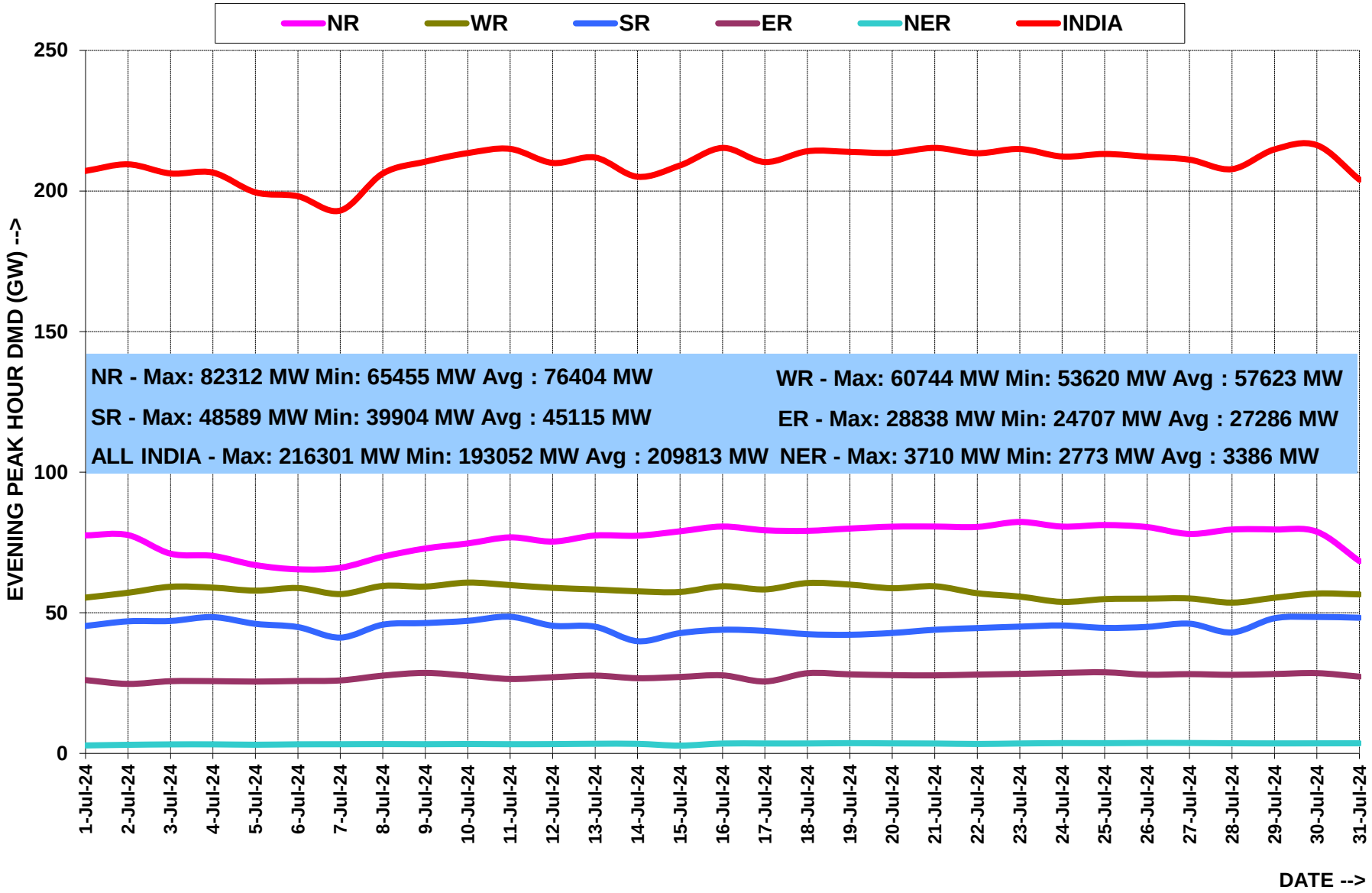
सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Jul-24	77536	55413	45317	26070	2826	207162
2-Jul-24	77643	57127	46970	24707	3045	209492
3-Jul-24	71032	59254	47091	25676	3197	206250
4-Jul-24	70231	58955	48459	25689	3215	206549
5-Jul-24	66963	57880	46072	25540	3075	199530
6-Jul-24	65455	58803	44914	25746	3228	198146
7-Jul-24	66013	56660	41178	25948	3253	193052
8-Jul-24	69957	59554	45766	27653	3309	206239
9-Jul-24	72872	59340	46348	28616	3261	210437
10-Jul-24	74670	60744	47128	27627	3327	213496
11-Jul-24	76825	59850	48589	26460	3261	214985
12-Jul-24	75336	58863	45404	27089	3300	209992
13-Jul-24	77483	58307	45057	27654	3417	211918
14-Jul-24	77418	57635	39904	26734	3386	205077
15-Jul-24	78986	57404	42765	27176	2773	209104
16-Jul-24	80673	59448	43962	27750	3498	215331
17-Jul-24	79351	58317	43553	25576	3513	210310
18-Jul-24	79145	60582	42394	28512	3522	214155
19-Jul-24	79959	60027	42191	28106	3639	213922
20-Jul-24	80633	58739	42824	27827	3552	213575
21-Jul-24	80673	59448	43962	27750	3498	215331
22-Jul-24	80534	56963	44567	28035	3343	213442
23-Jul-24	82312	55757	45079	28281	3523	214952
24-Jul-24	80656	53873	45496	28605	3652	212282
25-Jul-24	81215	54871	44627	28838	3631	213182
26-Jul-24	80487	55028	44983	27991	3710	212199
27-Jul-24	78042	55101	46121	28194	3691	211149
28-Jul-24	79605	53620	43015	27946	3597	207783
29-Jul-24	79624	55353	48057	28225	3549	214808
30-Jul-24	78821	56847	48516	28543	3574	216301
31-Jul-24	68369	56539	48249	27303	3586	204046
उच्चतम MAXIMUM	82312	60744	48589	28838	3710	216301
निम्नतम MINIMUM	65455	53620	39904	24707	2773	193052
औसत AVERAGE	76404	57623	45115	27286	3386	209813
अब तक का उच्चतम All Time Max.	**82312**	67482	54798	29695	**3710**	227354
दिनांक Date	**23.07.24**	30.05.24	29.04.24	29.05.24	**26.07.24**	29.05.24

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।

Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND(at 2000hrs)MET DURING THE MONTH OF JULY 2024



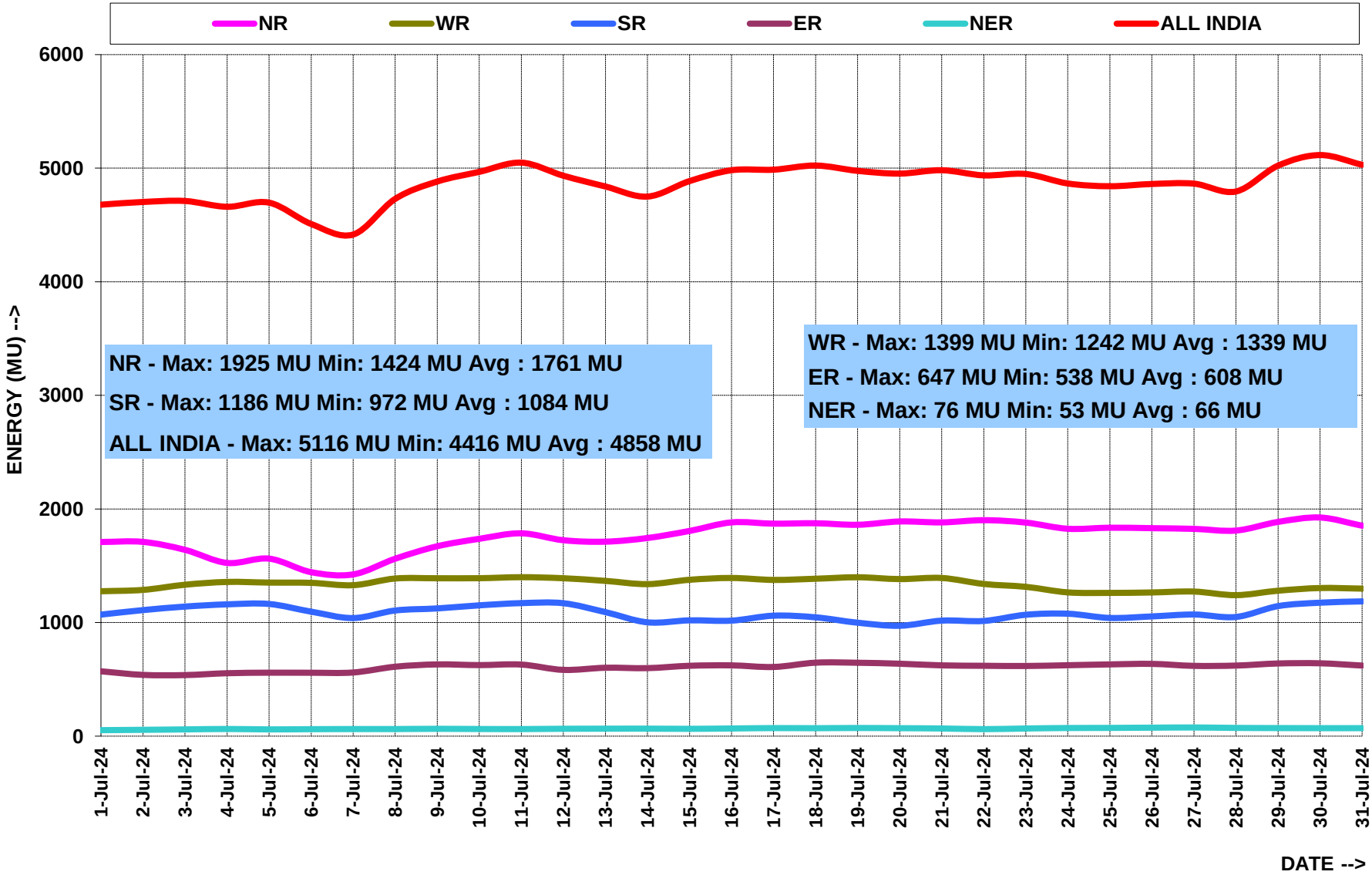
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-24	1709	1276	1070	571	53	4679
02-Jul-24	1710	1288	1110	539	56	4703
03-Jul-24	1640	1333	1141	538	59	4711
04-Jul-24	1525	1358	1160	554	63	4660
05-Jul-24	1563	1352	1164	559	60	4697
06-Jul-24	1444	1350	1096	558	61	4509
07-Jul-24	1424	1329	1040	561	62	4416
08-Jul-24	1562	1388	1106	611	62	4730
09-Jul-24	1672	1390	1125	632	65	4883
10-Jul-24	1737	1391	1152	626	63	4968
11-Jul-24	1786	1399	1171	630	62	5049
12-Jul-24	1725	1390	1170	583	65	4934
13-Jul-24	1712	1367	1092	602	66	4839
14-Jul-24	1744	1339	1002	599	66	4750
15-Jul-24	1806	1377	1019	620	65	4886
16-Jul-24	1882	1393	1017	623	68	4982
17-Jul-24	1871	1375	1061	609	71	4987
18-Jul-24	1874	1386	1046	647	70	5023
19-Jul-24	1861	1399	999	646	72	4977
20-Jul-24	1890	1383	972	638	70	4952
21-Jul-24	1882	1393	1017	623	68	4982
22-Jul-24	1901	1340	1014	620	62	4937
23-Jul-24	1880	1315	1069	618	68	4949
24-Jul-24	1826	1266	1078	625	72	4865
25-Jul-24	1835	1261	1041	632	73	4841
26-Jul-24	1831	1265	1054	636	75	4861
27-Jul-24	1825	1273	1071	619	76	4864
28-Jul-24	1810	1242	1049	621	73	4795
29-Jul-24	1886	1280	1145	640	71	5022
30-Jul-24	1925	1304	1175	641	70	5116
31-Jul-24	1853	1299	1186	622	70	5030
कुल TOTAL	54592	41497	33612	18842	2056	150599
उच्चतम MAXIMUM	1925	1399	1186	647	76	5116
निम्नतम MINIMUM	1424	1242	972	538	53	4416
औसत AVERAGE	1761	1339	1084	608	66	4858
संचयी 2024-25 Cumulative 2024-25	197126	181802	142436	74948	7260	603572
अब तक का उच्चतम All Time Max.	1990	1648	1416	692	**76**	5466
दिनांक Date	18.06.24	06.05.24	05.04.24	10.06.24	**27.07.24**	30.05.24

ENERGY MET DURING THE MONTH OF JULY' 2024



5. वर्ष 2024-25 के लिए आवृति रूपरेखा FREQUENCY PROFILE FOR YEAR 2024-25

राष्ट्रीय ग्रिड NATIONAL GRID									
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-24	All India Grid	5.25	78.56	16.19		50.43	49.55	50.00
	May-24	All India Grid	2.49	80.04	17.47		50.46	49.72	50.01
	Jun-24	All India Grid	4.50	79.18	16.32		50.67	49.63	50.00
	Jul-24	All India Grid	6.41	78.42	15.17		50.37	49.64	50.00
	2024-25 (upto Jul)	All India Grid	4.66	79.05	16.29		50.67	49.55	50.00

5.1 जुलाई 2024 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR JULY 2024

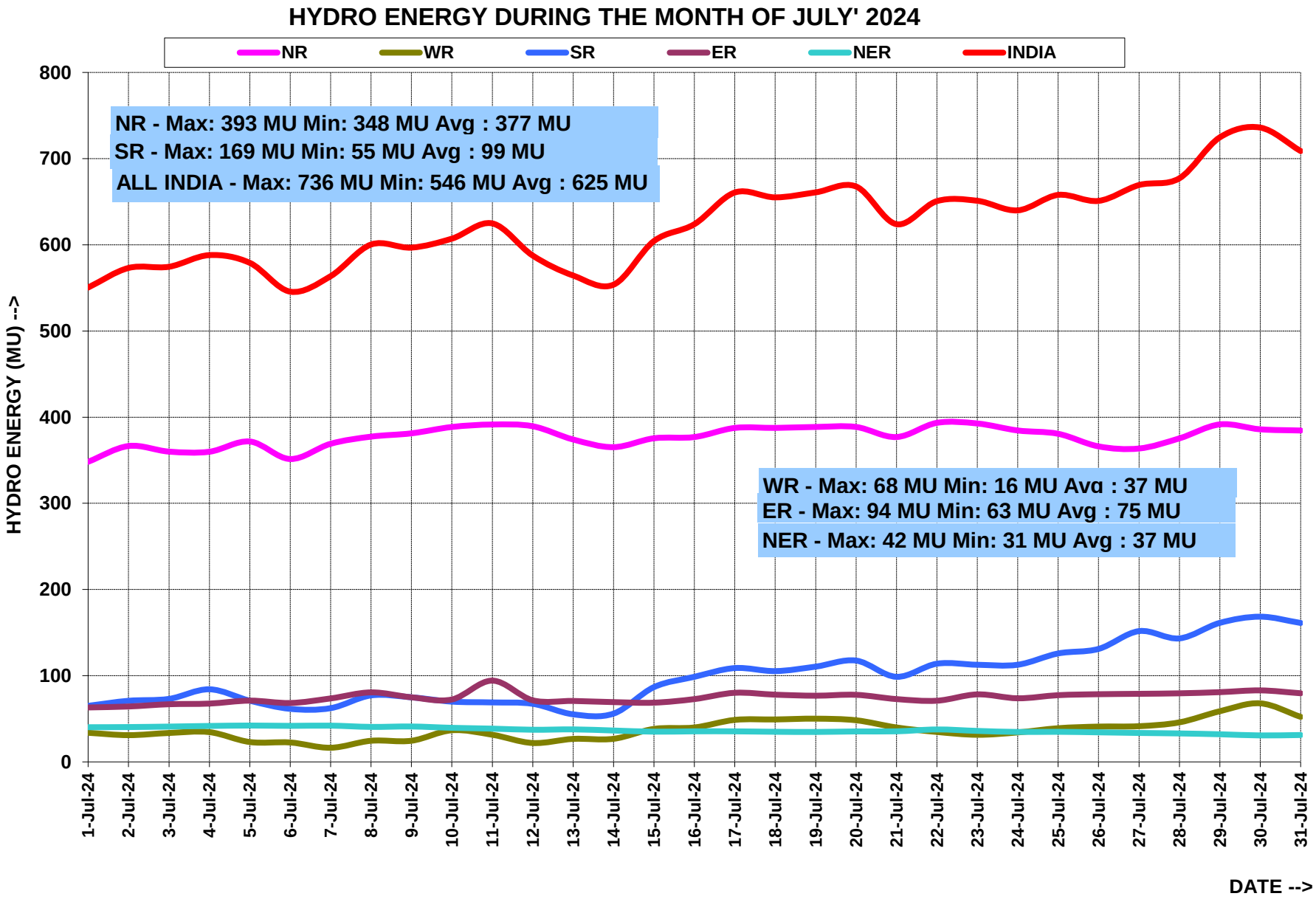
फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05		उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-Jul-24	1.97	77.29	20.74		50.22	49.84	50.01	0.03
	02-Jul-24	5.27	77.08	17.65		50.26	49.82	50.00	0.04
	03-Jul-24	7.31	78.80	13.89		50.20	49.74	49.99	0.04
	04-Jul-24	8.66	79.58	11.76		50.17	49.77	49.99	0.04
	05-Jul-24	11.70	67.66	20.64		50.24	49.69	49.99	0.06
	06-Jul-24	1.83	77.25	20.93		50.37	49.86	50.02	0.05
	07-Jul-24	7.03	81.55	11.42		50.20	49.71	49.99	0.04
	08-Jul-24	4.76	84.50	10.74		50.22	49.81	50.00	0.03
	09-Jul-24	11.92	74.46	13.62		50.25	49.64	49.99	0.05
	10-Jul-24	7.77	73.31	18.92		50.18	49.76	50.00	0.04
	11-Jul-24	3.24	83.97	12.79		50.14	49.84	50.00	0.02
	12-Jul-24	4.25	72.45	23.30		50.19	49.73	50.01	0.03
	13-Jul-24	7.25	70.95	21.81		50.30	49.79	50.01	0.05
	14-Jul-24	6.92	81.59	11.49		50.19	49.74	49.99	0.04
	15-Jul-24	8.96	71.82	19.22		50.24	49.72	49.99	0.04
	16-Jul-24	5.23	78.61	16.16		50.18	49.73	50.00	0.03
	17-Jul-24	7.11	83.90	8.99		50.10	49.82	49.99	0.03
	18-Jul-24	4.48	81.79	13.73		50.11	49.81	50.00	0.03
	19-Jul-24	3.00	88.09	8.91		50.17	49.84	50.00	0.02
	20-Jul-24	10.61	79.54	9.85		50.10	49.68	49.98	0.04
	21-Jul-24	3.03	83.73	13.24		50.14	49.82	49.99	0.02
	22-Jul-24	11.81	72.30	15.89		50.11	49.64	49.99	0.05
	23-Jul-24	3.26	80.61	16.12		50.16	49.85	50.00	0.02
	24-Jul-24	3.11	82.63	14.26		50.18	49.83	50.00	0.02
	25-Jul-24	5.79	75.13	19.09		50.21	49.64	50.00	0.04
	26-Jul-24	3.81	75.14	21.05		50.33	49.72	50.01	0.04
	27-Jul-24	5.47	86.12	8.40		50.17	49.77	49.99	0.03
	28-Jul-24	7.31	80.94	11.75		50.14	49.77	50.00	0.03
	29-Jul-24	10.65	80.21	9.14		50.18	49.72	49.98	0.05
	30-Jul-24	13.40	76.81	9.79		50.18	49.68	49.98	0.05
	31-Jul-24	1.69	73.32	24.99		50.22	49.82	50.02	0.02

6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-24	348	34	65	63	40	550
02-Jul-24	366	31	71	64	40	573
03-Jul-24	360	34	73	67	41	575
04-Jul-24	360	35	84	68	42	588
05-Jul-24	372	23	71	71	42	579
06-Jul-24	351	23	62	68	42	546
07-Jul-24	369	16	62	74	42	564
08-Jul-24	377	25	77	81	41	600
09-Jul-24	381	25	75	75	41	597
10-Jul-24	389	37	70	72	40	607
11-Jul-24	391	31	69	94	39	625
12-Jul-24	389	22	67	71	37	588
13-Jul-24	374	27	55	71	38	564
14-Jul-24	365	27	56	69	36	554
15-Jul-24	376	38	87	69	35	604
16-Jul-24	377	40	99	73	35	624
17-Jul-24	388	49	109	80	35	661
18-Jul-24	388	49	105	78	35	655
19-Jul-24	389	50	111	77	35	661
20-Jul-24	389	48	118	78	35	668
21-Jul-24	377	40	99	73	35	624
22-Jul-24	393	35	114	71	38	651
23-Jul-24	393	31	113	78	36	651
24-Jul-24	385	34	113	74	35	640
25-Jul-24	381	39	126	77	35	658
26-Jul-24	366	41	131	79	34	651
27-Jul-24	364	41	152	79	34	669
28-Jul-24	375	46	143	80	33	677
29-Jul-24	392	59	161	81	32	725
30-Jul-24	386	68	169	83	31	736
31-Jul-24	385	52	161	80	31	709
कुल TOTAL	11694	1149	3067	2317	1145	19372
उच्चतम MAXIMUM	393	68	169	94	42	736
निम्नतम MINIMUM	348	16	55	63	31	546
औसत AVERAGE	377	37	99	75	37	625
संचयी 2024-25 Cumulative 2024-25	35454	5291	7480	5215	3045	56484
अब तक का उच्चतम All Time Max.	443	167	208	157	43	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	27.06.24	30.08.22


HYDRO ENERGY DURING THE MONTH OF JULY' 2024
NR
WR
SR
ER
NER
INDIA
NR - Max: 393 MU Min: 348 MU Avg : 377 MU
SR - Max: 169 MU Min: 55 MU Avg : 99 MU
ALL INDIA - Max: 736 MU Min: 546 MU Avg : 625 MU
WR - Max: 68 MU Min: 16 MU Avg : 37 MU
ER - Max: 94 MU Min: 63 MU Avg : 75 MU
NER - Max: 42 MU Min: 31 MU Avg : 37 MU
HYDRO ENERGY (MU) -->
DATE -->

7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन

WIND GENERATION AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-24	39	176	196	----------	----------	412
02-Jul-24	45	259	220	----------	----------	524
03-Jul-24	39	231	212	----------	----------	482
04-Jul-24	35	173	184	----------	----------	392
05-Jul-24	44	149	212	----------	----------	404
06-Jul-24	67	238	235	----------	----------	540
07-Jul-24	22	213	211	----------	----------	447
08-Jul-24	7	72	194	----------	----------	274
09-Jul-24	5	66	187	----------	----------	259
10-Jul-24	16	68	188	----------	----------	272
11-Jul-24	48	170	190	----------	----------	408
12-Jul-24	79	246	200	----------	----------	525
13-Jul-24	31	234	250	----------	----------	515
14-Jul-24	28	157	291	----------	----------	476
15-Jul-24	13	69	246	----------	----------	328
16-Jul-24	15	82	306	----------	----------	403
17-Jul-24	15	79	288	----------	----------	383
18-Jul-24	12	98	311	----------	----------	421
19-Jul-24	8	119	318	----------	----------	445
20-Jul-24	6	99	304	----------	----------	408
21-Jul-24	15	82	306	----------	----------	403
22-Jul-24	16	149	280	----------	----------	445
23-Jul-24	16	215	297	----------	----------	528
24-Jul-24	22	240	287	----------	----------	548
25-Jul-24	29	243	306	----------	----------	578
26-Jul-24	30	245	323	----------	----------	598
27-Jul-24	36	212	303	----------	----------	551
28-Jul-24	14	127	310	----------	----------	451
29-Jul-24	11	109	286	----------	----------	406
30-Jul-24	13	107	247	----------	----------	367
31-Jul-24	24	146	243	----------	----------	412
कुल TOTAL	801	4877	7928	----------	----------	13606
उच्चतम MAXIMUM	79	259	323	----------	----------	598
निम्नतम MINIMUM	5	66	184	----------	----------	259
औसत AVERAGE	26	157	256	----------	----------	439
संचयी 2024-25 Cumulative 2024-25	3082	16032	17013	----------	----------	36126
अब तक का उच्चतम All Time Max.	86	310	**323**	----------	----------	619
दिनांक Date	07.08.23	28.05.24	**26.07.24**	----------	----------	28.05.24

*__Source:__ As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन

SOLAR GENERATION AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

सभी आंकड़े मिलियन यूनिट मे All figures in MU

दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Jul-24	159	52	95	2.5	0.3	309
02-Jul-24	160	41	114	2.5	0.6	318
03-Jul-24	165	70	111	1.4	1.0	348
04-Jul-24	140	61	104	2.5	1.0	309
05-Jul-24	150	59	110	2.3	0.7	321
06-Jul-24	149	69	89	2.5	0.9	310
07-Jul-24	111	64	105	2.9	0.8	284
08-Jul-24	129	68	99	3.7	0.9	301
09-Jul-24	162	74	102	3.6	0.8	341
10-Jul-24	166	82	126	3.3	0.5	377
11-Jul-24	166	78	121	3.0	0.6	369
12-Jul-24	157	70	114	1.6	1.0	344
13-Jul-24	149	74	91	2.2	1.1	317
14-Jul-24	146	76	82	7.0	1.0	313
15-Jul-24	165	66	69	2.6	0.9	304
16-Jul-24	165	91	86	2.9	0.9	345
17-Jul-24	143	69	85	2.8	1.1	302
18-Jul-24	151	71	89	2.9	1.0	316
19-Jul-24	157	75	79	2.9	1.2	315
20-Jul-24	138	68	71	2.0	0.8	281
21-Jul-24	165	91	86	2.9	0.9	345
22-Jul-24	148	58	91	1.5	0.7	299
23-Jul-24	152	50	88	1.3	1.1	292
24-Jul-24	149	42	87	1.4	1.2	281
25-Jul-24	147	45	75	1.5	1.2	271
26-Jul-24	148	48	94	2.2	1.3	293
27-Jul-24	162	37	91	1.9	1.3	294
28-Jul-24	140	48	104	2.5	1.3	296
29-Jul-24	145	35	102	2.2	0.7	285
30-Jul-24	153	55	110	2.7	0.7	322
31-Jul-24	150	62	119	2.2	1.0	335
कुल TOTAL	4687	1953	2988	79	28	9735
उच्चतम MAXIMUM	166	91	126	7.0	1.3	377
निम्नतम MINIMUM	111	35	69	1.3	0.3	271
औसत AVERAGE	151	63	96	2.6	0.9	314
संचयी 2024-25 Cumulative 2024-25	19413	9913	13981	374	114.6	43794
अब तक का उच्चतम All Time Max.	186	103	149	**7.0**	2.4	426
दिनांक Date	01.05.24	30.04.24	31.03.24	**14.07.24**	22.06.22	30.04.24

*__Source__: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति

DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: जुलाई 2024 MONTH:- JUL 2024

सभी आंकड़े मेगावाट मे All figures in MW

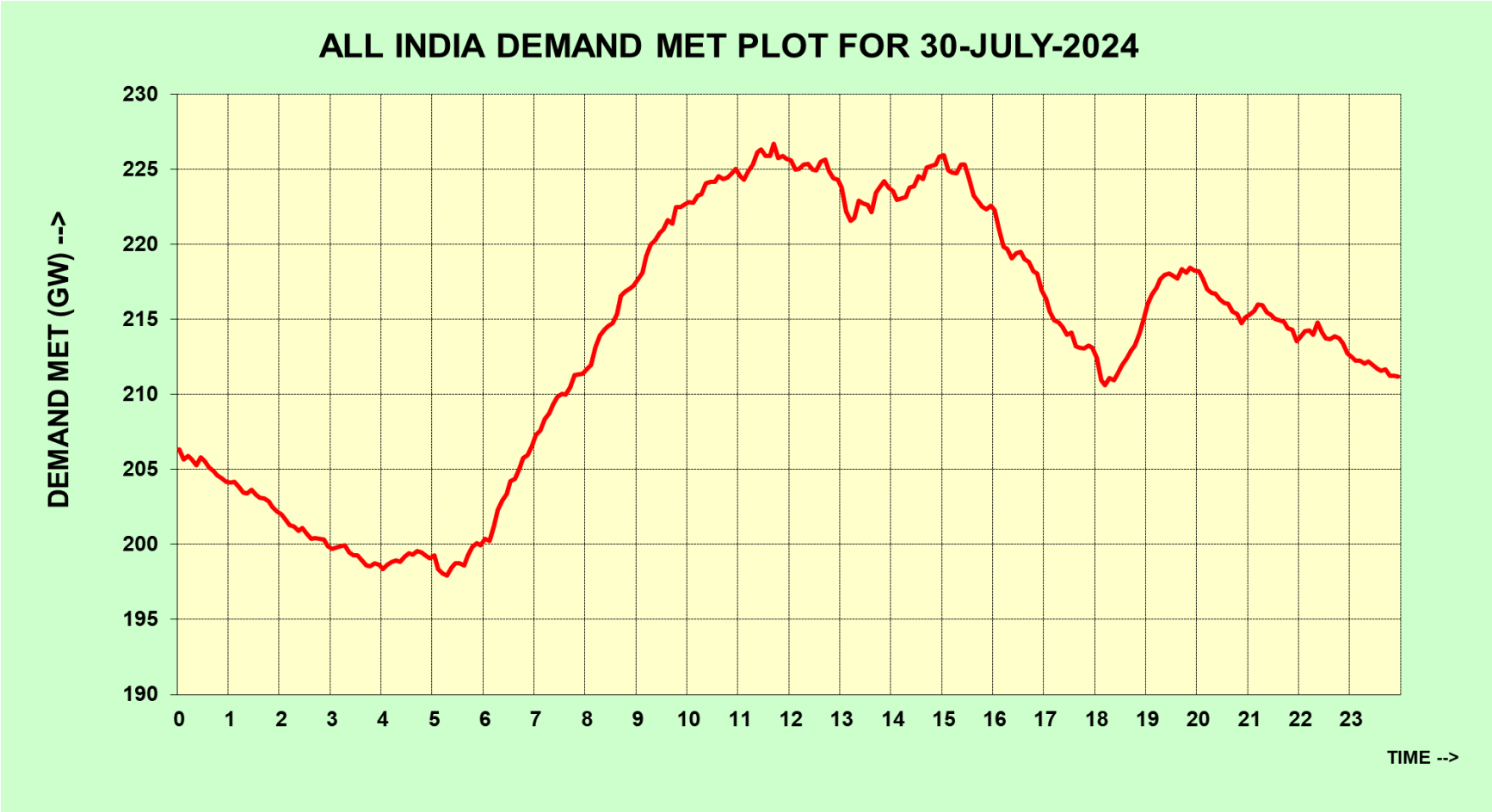
दिनांक Date	ऊत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL		विभिन्नता फ़ैक्टर Diversity Factor*
01-Jul-24	79469	56337	50120	27454	2890	208851		1.036
02-Jul-24	78609	57888	50262	26539	3079	209923		1.031
03-Jul-24	73903	59773	52762	26340	3210	206103		1.048
04-Jul-24	71450	59970	53086	26492	3206	207180		1.034
05-Jul-24	70388	58487	55726	26594	3114	208631		1.027
06-Jul-24	66270	59974	49872	27094	3256	198666		1.039
07-Jul-24	68256	57331	47323	28048	3192	197965		1.031
08-Jul-24	71404	60688	52716	29337	3291	210502		1.033
09-Jul-24	75077	60025	52184	30238	3255	214980		1.027
10-Jul-24	78441	61280	53685	30475	3323	217154		1.046
11-Jul-24	80035	60279	54073	30203	3339	221080		1.031
12-Jul-24	76811	59789	55992	28263	3352	214275		1.046
13-Jul-24	79105	59530	49609	29387	3464	213329		1.036
14-Jul-24	80067	58010	46966	28961	3431	209018		1.040
15-Jul-24	80415	61682	48945	29452	3215	214557		1.043
16-Jul-24	84969	60423	46378	29686	3486	217380		1.035
17-Jul-24	82919	59151	50712	29858	3669	219587		1.031
18-Jul-24	82990	60626	49228	30640	3580	218161		1.041
19-Jul-24	83891	60112	46745	30086	3649	215795		1.040
20-Jul-24	84969	60423	46378	29686	3486	217380		1.035
21-Jul-24	84969	60423	46378	29686	3486	217380		1.035
22-Jul-24	84969	60423	46378	29686	3486	217380		1.035
23-Jul-24	82294	55552	50174	29212	3646	212149		1.041
24-Jul-24	82294	55552	50174	29212	3646	212149		1.041
25-Jul-24	82881	55619	49670	29551	3638	212527		1.042
26-Jul-24	81661	56023	50105	29179	3732	212835		1.037
27-Jul-24	81995	56014	49464	28951	3760	211521		1.041
28-Jul-24	81995	56014	49464	28951	3760	211521		1.041
29-Jul-24	85767	57134	55487	29062	3541	223084		1.035
30-Jul-24	88085	59255	55679	29700	3620	226630		1.043
31-Jul-24	85225	57347	56170	29076	3656	225361		1.027
उच्चतम MAXIMUM	88085	61682	56170	30640	3760	226630		1.048
निम्नतम MINIMUM	66270	55552	46378	26340	2890	197965		1.027
औसत AVERAGE	79728	58746	50707	28939	3434	213647		1.037
अब तक का उच्चतम All Time Max.	91215	74974	68735	32531	**3760**	250070		
दिनांक Date	19.06.24	01.06.24	13.03.24	10.06.24	**27.07.24**	30.05.24		

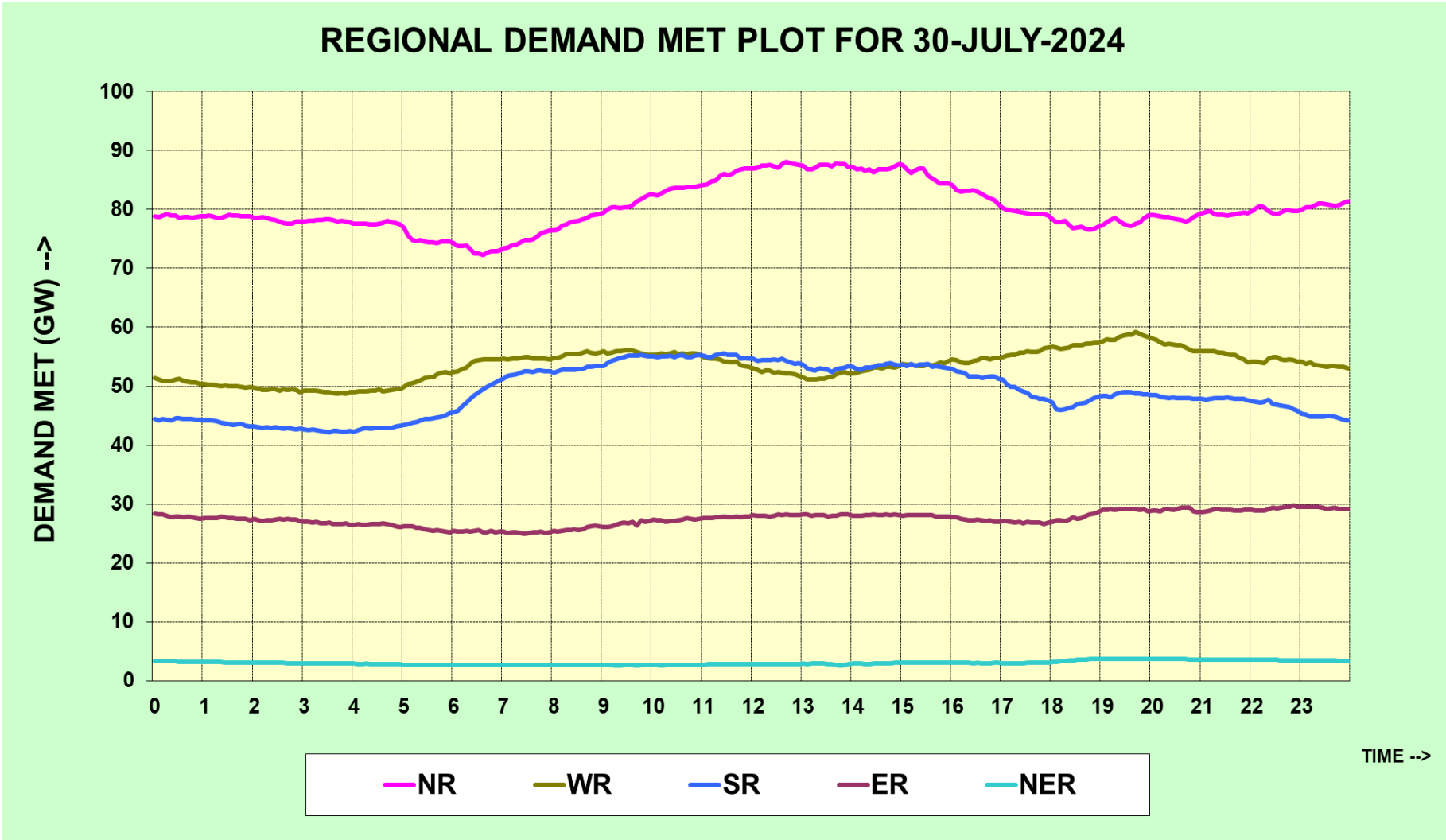
* Diversity factor = (Sum of regional max demands) / All India max demand

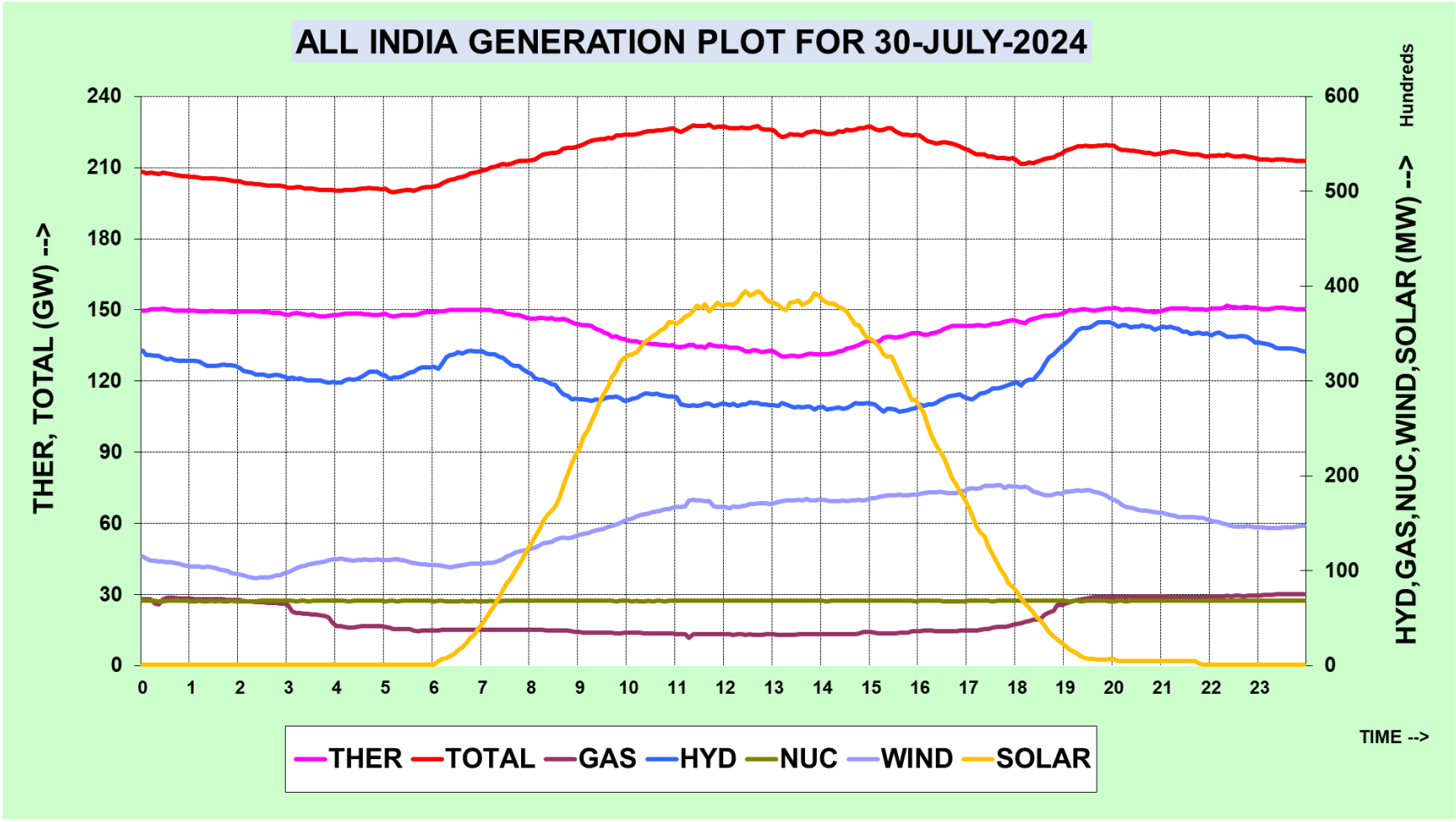
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

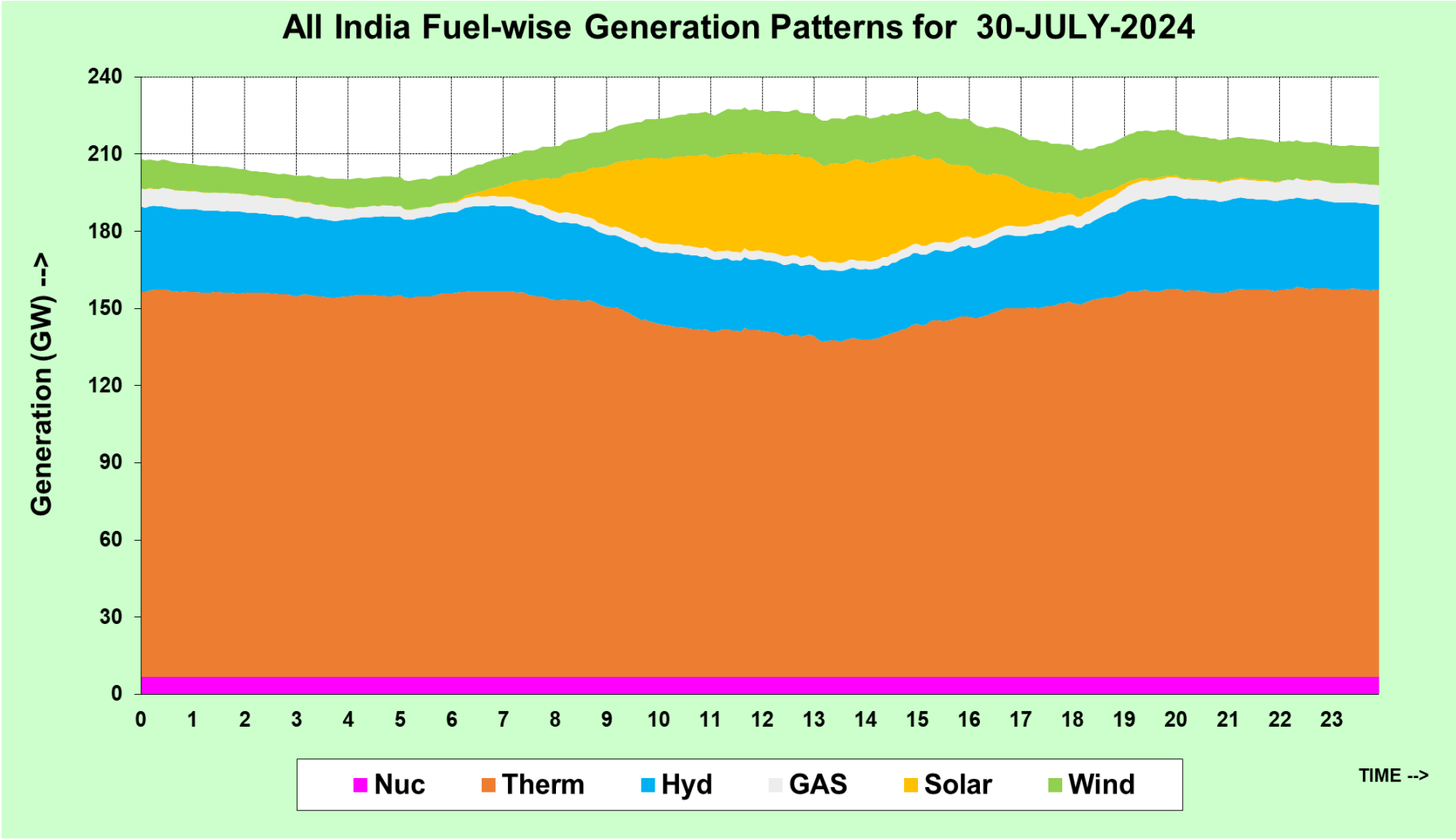
Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

10. DEMAND AND GENERATION PLOTS FOR 30 JUL (MAXIMUM DEMAND MET)




REGIONAL DEMAND MET PLOT FOR 30-JULY-2024
DEMAND MET (GW) -->
100
90
80
70
60
50
40
30
20
10
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
NR
WR
SR
ER
NER


ALL INDIA GENERATION PLOT FOR 30-JULY-2024
THER, TOTAL (GW) -->
HYD,GAS,NUC,WIND,SOLAR (MW) --> Hundreds
240
210
180
150
120
90
60
30
0
600
500
400
300
200
100
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
THER TOTAL GAS HYD NUC WIND SOLAR


All India Fuel-wise Generation Patterns for 30-JULY-2024
Generation (GW) -->
240
210
180
150
120
90
60
30
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
TIME -->
Nuc
Therm
Hyd
GAS
Solar
Wind

30 जुलाई 2024 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति

ALL INDIA DEMAND MET FOR 30 JULY 2024 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	205683		**12:05**	225014
00:20	205297		**12:20**	225370
00:35	205142		**12:35**	225534
00:50	204442		**12:50**	224401
01:05	204199		**13:05**	222226
01:20	203420		**13:20**	222939
01:35	203112		**13:35**	222138
01:50	202508		**13:50**	224207
02:05	201632		**14:05**	222988
02:20	200935		**14:20**	223774
02:35	200398		**14:35**	224394
02:50	200356		**14:50**	225310
03:05	199824		**15:05**	224948
03:20	199460		**15:20**	225342
03:35	198953		**15:35**	223275
03:50	198768		**15:50**	222337
04:05	198637		**16:05**	221010
04:20	198825		**16:20**	219085
04:35	199303		**16:35**	219045
04:50	199262		**16:50**	218076
05:05	198356		**17:05**	215549
05:20	198482		**17:20**	214500
05:35	198600		**17:35**	213212
05:50	200103		**17:50**	213243
06:05	200237		**18:05**	210980
06:20	202938		**18:20**	210962
06:35	204380		**18:35**	212397
06:50	205975		**18:50**	214015
07:05	207572		**19:05**	216696
07:20	209308		**19:20**	217971
07:35	209979		**19:35**	217713
07:50	211353		**19:50**	218471
08:05	211974		**20:05**	217736
08:20	214320		**20:20**	216729
08:35	215401		**20:35**	216059
08:50	217063		**20:50**	214769
09:05	218140		**21:05**	215596
09:20	220286		**21:20**	215478
09:35	221626		**21:35**	214971
09:50	222478		**21:50**	214304
10:05	222769		**22:05**	214244
10:20	224095		**22:20**	214804
10:35	224554		**22:35**	213677
10:50	224700		**22:50**	213424
11:05	224311		**23:05**	212255
11:20	226165		**23:20**	212207
11:35	225908		**23:35**	211601
11:50	225890		**23:50**	211250
Maximum Demand of 226630 MW met@11:40 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instanteneous demand

11. ACTUAL POWER SUPPLY POSITION - JULY 2024

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	8	8	0.0	239	239	0	0.0	434	434	0	0.0
	Delhi	141	141	0.0	4370	4370	0	0.0	8175	8175	0	0.0
	Haryana	270	269	-0.6	8366	8348	-18	-0.2	14662	14662	0	0.0
	Himachal Pradesh	38	38	0.0	1181	1181	0	0.0	1888	1888	0	0.0
	J&K(UT) and Ladakh(UT)	53	53	-0.2	1650	1643	-7	-0.4	2635	2635	0	0.0
	Punjab	335	335	0.0	10399	10399	0	0.0	16006	16006	0	0.0
	Rajasthan	313	311	-2.7	9714	9630	-84	-0.9	16371	16371	0	0.0
	Uttar Pradesh	552	551	-0.9	17118	17091	-27	-0.2	30298	30298	0	0.0
	Uttarakhand	54	54	0.0	1669	1668	-1	-0.1	2545	2545	0	0.0
	Railway	4	4	0.0	133	133	0	0.0	227	227	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	5	5	0	0.0
	TOTAL	1769	1765	-4.5	54842	54704	-138	-0.3	87468	87320	-148	-0.2
WR	Gujarat	399	399	0.0	12373	12373	0	0.0	19991	19991	0	0.0
	Madhya Pradesh	248	248	-0.8	7700	7676	-23	-0.3	12746	12735	-11	-0.1
	Chhattisgarh	116	116	-0.2	3611	3604	-7	-0.2	5948	5944	-4	-0.1
	Maharashtra	500	500	0.0	15511	15511	0	0.0	24174	24174	0	0.0
	Goa	13	13	0.0	409	409	0	0.0	697	696	-1	-0.2
	DNHDDPDCL	30	30	0.0	922	922	0	0.0	1343	1343	0	0.0
	AMNSIL	17	17	0.0	532	532	0	0.0	858	858	0	0.0
	BALCO	13	13	0.0	390	390	0	0.0	531	531	0	0.0
	TOTAL	1337	1336	-1.0	41448	41418	-30	-0.1	62168	61998	-170	-0.3
SR	Andhra Pradesh	206	205	-1.2	6391	6353	-38	-0.6	11020	11020	0	0.0
	Telangana	226	224	-1.4	6992	6949	-42	-0.6	13541	13541	0	0.0
	Karnataka	208	207	-1.5	6458	6411	-47	-0.7	13820	13820	0	0.0
	Kerala	77	76	-0.5	2378	2363	-15	-0.6	4183	4183	0	0.0
	Tamil Nadu	361	359	-1.8	11192	11135	-57	-0.5	18542	18542	0	0.0
	Puducherry	10	10	-0.1	319	318	-2	-0.5	517	517	0	0.0
	TOTAL	1088	1082	-6.5	33730	33529	-201	-0.6	56156	56134	-22	0.0
ER	Bihar	151	149	-1.3	4667	4627	-40	-0.9	7909	7879	-30	-0.4
	Jharkhand	44	44	-0.3	1374	1364	-10	-0.7	2351	2351	0	0.0
	DVC	74	74	0.0	2281	2281	0	0.0	3539	3539	0	0.0
	Odisha	119	119	0.0	3694	3694	0	0.0	6350	6350	0	0.0
	West Bengal	227	227	0.0	7026	7026	0	0.0	11672	11672	0	0.0
	Sikkim	1	1	0.0	39	39	0	0.0	94	94	0	0.0
	Railways_ER ISTS	0	0	0.0	5	5	0	0.0	25	25	0	0.0
	TOTAL	616	614	-1.6	19087	19037	-50	-0.3	31369	30622	-747	-2.4
NER	Arunachal Pradesh	3	3	0.0	89	89	0	0.0	186	186	0	0.0
	Assam	44	44	0.0	1373	1373	0	0.0	2524	2524	0	0.0
	Manipur	3	3	0.0	83	83	0	0.0	213	213	0	0.0
	Meghalaya	6	6	0.0	173	173	0	0.0	359	359	0	0.0
	Mizoram	2	2	0.0	59	59	0	0.0	130	130	0	0.0
	Nagaland	3	3	0.0	95	95	0	0.0	188	188	0	0.0
	Tripura	6	6	0.0	184	184	0	0.0	359	359	0	0.0
	TOTAL	66	66	0.0	2056	2056	0	0.0	3764	3764	0	0.0
ALL INDIA		4876	4863	-13.6	151164	150743	-420	-0.3				

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF JULY 2024 vs JULY 2023

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Jul-23	Jul-24	Difference	% Change	Average MU/day for Jul-24	Jul-23	Jul-24	Difference	% Change	Average MU/day for Jul-24
NR	Chandigarh	196	239	43	22	8	196	239	43	22	8
	Delhi	3912	4370	459	12	141	3911	4370	459	12	141
	Haryana	6737	8366	1629	24	270	6725	8348	1623	24	269
	Himachal Pradesh	1007	1181	174	17	38	1004	1181	177	18	38
	J&K(UT) and Ladakh(UT)	1591	1650	59	4	53	1574	1643	69	4	53
	Punjab	8332	10399	2067	25	335	8332	10399	2067	25	335
	Rajasthan	8375	9714	1339	16	313	8351	9630	1279	15	311
	Uttar Pradesh	15338	17118	1780	12	552	15315	17091	1775	12	551
	Uttarakhand	1375	1669	295	21	54	1361	1668	307	23	54
	Railway/NFL	108	135	27	25	4	108	135	27	25	4
	TOTAL	46971	54842	7871	17	1769	46878	54704	7826	17	1765
WR	Gujarat	11093	12373	1280	12	399	11093	12373	1280	12	399
	Madhya Pradesh	7309	7700	391	5	248	7309	7676	368	5	248
	Chhattisgarh	3362	3611	249	7	116	3356	3604	248	7	116
	Maharashtra	15308	15511	203	1	500	15304	15511	207	1	500
	Goa	379	409	30	8	13	379	409	30	8	13
	DNHDDPDCL	920	922	2	0	30	920	922	2	0	30
	AMNSIL	577	532	-45	-8	17	577	532	-45	-8	17
	BALCO	384	390	5	1	13	384	390	5	1	13
	TOTAL	39332	41448	2116	5	1337	39322	41418	2095	5	1336
SR	Andhra Pradesh	6334	6391	57	1	206	6334	6353	19	0	205
	Telangana	6407	6992	585	9	226	6489	6949	460	7	224
	Karnataka	6116	6458	341	6	208	6198	6411	213	3	207
	Kerala	2281	2378	98	4	77	2307	2363	56	2	76
	Tamil Nadu	10800	11192	392	4	361	10907	11135	228	2	359
	Puducherry	301	319	18	6	10	304	318	14	4	10
	TOTAL	32239	33730	1491	5	1088	32540	33529	990	3	1082
ER	Bihar	4362	4667	305	7	151	4318	4627	308	7	149
	Jharkhand	1158	1374	216	19	44	1149	1364	215	19	44
	DVC	2339	2281	-58	-2	74	2339	2281	-58	-2	74
	Odisha	3846	3694	-151	-4	119	3846	3694	-151	-4	119
	West Bengal	6591	7026	435	7	227	6591	7026	435	7	227
	Sikkim	37	39	2	6	1	37	39	2	6	1
	Railways_ER ISTS	5	5	0	2	0	5	5	0	2	0
	TOTAL	18339	19087	749	4	616	18286	19037	751	4	614
NER	Arunachal Pradesh	83	89	6	8	3	83	89	6	8	3
	Assam	1317	1373	55	4	44	1317	1373	55	4	44
	Manipur	82	83	1	1	3	82	83	1	1	3
	Meghalaya	182	173	-9	-5	6	159	173	14	9	6
	Mizoram	53	59	6	11	2	53	59	6	11	2
	Nagaland	89	95	6	7	3	89	95	6	7	3
	Tripura	179	184	4	2	6	179	184	4	2	6
	TOTAL	1986	2056	70	4	66	1963	2056	93	5	66
ALL INDIA		138867	151164	12296	9	4876	138988	150743	11755	8	4863

13. DEMAND COMPARISON OF JULY 2024 vs JULY 2023

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Jul-23	Jul-24	Difference	% Change	Jul-23	Jul-24	Difference	% Change
NR	Chandigarh	371	434	63	17.0	371	434	63	17.0
	Delhi	7398	8175	777	10.5	7398	8175	777	10.5
	Haryana	12227	14662	2435	19.9	12227	14662	2435	19.9
	Himachal Pradesh	1775	1888	113	6.4	1775	1888	113	6.4
	J&K(UT) and Ladakh(UT)	2628	2635	7	0.3	2543	2635	92	3.6
	Punjab	14831	16006	1175	7.9	14831	16006	1175	7.9
	Rajasthan	14204	16371	2167	15.3	14204	16371	2167	15.3
	Uttar Pradesh	28704	30298	1594	5.6	28284	30298	2014	7.1
	Uttarakhand	2298	2545	247	10.7	2223	2545	322	14.5
	Railway/NFL	203	233	30	14.7	203	233	30	14.7
WR	Gujarat	17715	19991	2276	13	17672	19991	2319	13.1
	Madhya Pradesh	12006	12746	740	6	12006	12735	729	6.1
	Chhattisgarh	5314	5948	633	12	5216	5944	728	13.9
	Maharashtra	24544	24174	-370	-2	24535	24174	-361	-1.5
	Goa	647	697	50	8	646	696	49	7.6
	DNHDDPDCL	1307	1343	36	2.8	1307	1343	36	2.8
	AMNSIL	930	858	-72	-7.7	930	858	-72	-7.7
	BALCO	523	531	7	1.3	523	531	7	1.3
SR	Andhra Pradesh	11619	11020	-599	-5.2	11584	11020	-564	-4.9
	Telangana	12629	13541	912	7.2	12610	13541	931	7.4
	Karnataka	12359	13820	1462	11.8	12359	13820	1462	11.8
	Kerala	3999	4183	184	4.6	3999	4183	184	4.6
	Tamil Nadu	17774	18542	768	4.3	17774	18542	768	4.3
	Puducherry	492	517	25	5.1	492	517	25	5.1
ER	Bihar	7578	7909	331	4.4	7578	7879	301	4.0
	Jharkhand	1857	2351	494	26.6	1857	2351	494	26.6
	DVC	3687	3539	-148	-4.0	3687	3539	-148	-4.0
	Odisha	7104	6350	-754	-10.6	7104	6350	-754	-10.6
	West Bengal	10675	11672	997	9.3	10675	11672	997	9.3
	Sikkim	96	94	-2	-2.1	96	94	-2	-2.1
	Railways_ER ISTS	26	25	-1	-3.8	26	25	-1	-3.8
NER	Arunachal Pradesh	151	186	35	22.9	151	186	35	22.9
	Assam	2390	2524	134	5.6	2390	2524	134	5.6
	Manipur	202	213	11	5.6	202	213	11	5.6
	Meghalaya	330	359	29	8.8	330	359	29	8.8
	Mizoram	122	130	8	6.8	122	130	8	6.8
	Nagaland	167	188	21	12.7	167	188	21	12.7
	Tripura	343	359	16	4.7	343	359	16	4.7

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - JULY 2024

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	237.03	242.71	5.68	2.40	7.65	7.83	0.18
	Delhi	3877.53	3883.20	5.67	0.15	125.08	125.26	0.18
	Haryana	6224.25	6205.49	-18.76	-0.30	200.78	200.18	-0.61
	Himachal Pradesh	17.65	26.83	9.18	52.02	0.57	0.87	0.30
	J&K(UT) and Ladakh(UT)	926.29	792.22	-134.07	-14.47	29.88	25.56	-4.32
	Punjab	6442.67	6349.87	-92.80	-1.44	207.83	204.83	-2.99
	Rajasthan	3185.01	3233.76	48.75	1.53	102.74	104.31	1.57
	Uttar Pradesh	8241.94	8271.83	29.89	0.36	265.87	266.83	0.96
	Uttarakhand	798.95	805.86	6.91	0.87	25.77	26.00	0.22
	Railway	115.42	133.28	17.85	15.47	3.72	4.30	0.58
	NFL	3.20	2.00	-1.21	-37.67	0.10	0.06	-0.04
	Total	30069.95	29947.05	-122.90	-0.41	970.00	966.03	-3.96
WR	Gujarat	4368.17	4238.95	-129.22	-2.96	140.91	136.74	-4.17
	Madhya Pradesh	4102.62	4026.22	-76.40	-1.86	132.34	129.88	-2.46
	Chhattisgarh	2136.00	2108.40	-27.60	-1.29	68.90	68.01	-0.89
	Maharashtra	4829.52	4651.28	-178.24	-3.69	155.79	150.04	-5.75
	Goa	392.43	402.10	9.67	2.46	12.66	12.97	0.31
	DNHDDPDCL	926.69	921.80	-4.89	-0.53	29.89	29.74	-0.16
	AMNSIL	174.26	185.94	11.68	6.70	5.62	6.00	0.38
	BALCO*	390.58	389.58	-1.00	-0.26	12.60	12.57	-0.03
	Total	17320.27	16924.27	-396.00	-2.29	558.72	545.94	-12.77
SR	Andhra Pradesh	1061.13	1021.08	-40.05	-3.77	34.23	32.94	-1.29
	Telangana	4202.40	4206.85	4.45	0.11	135.56	135.70	0.14
	Karnataka	237.89	183.29	-54.60	-22.95	7.67	5.91	-1.76
	Kerala	1366.13	1396.83	30.70	2.25	44.07	45.06	0.99
	Tamil Nadu	4636.69	4551.70	-84.99	-1.83	149.57	146.83	-2.74
	Puducherry	303.52	299.63	-3.89	-1.28	9.79	9.67	-0.13
	Goa (SR)	56.15	45.31	-10.84	-19.31	1.81	1.46	-0.35
	Total	11863.91	11704.69	-159.22	-1.34	382.71	377.57	-5.14
ER	Bihar	4302.54	4343.48	40.95	0.95	138.79	140.11	1.32
	Jharkhand	991.29	998.02	6.72	0.68	31.98	32.19	0.22
	DVC	-1140.55	-1143.43	-2.89	0.25	-36.79	-36.88	-0.09
	Odisha	1218.57	1190.43	-28.13	-2.31	39.31	38.40	-0.91
	West Bengal	2712.79	2680.20	-32.59	-1.20	87.51	86.46	-1.05
	Sikkim	40.64	39.24	-1.40	-3.45	1.31	1.27	-0.05
	Railways_ER ISTS	4.84	5.08	0.24	5.03	0.16	0.16	0.01
	Total	8130.12	8113.02	-17.10	-0.21	262.26	261.71	-0.55
NER	Arunachal Pradesh	81.78	74.50	-7.28	-8.90	2.64	2.40	-0.23
	Assam	1119.75	1133.98	14.23	1.27	36.12	36.58	0.46
	Manipur	89.33	83.04	-6.29	-7.04	2.88	2.68	-0.20
	Meghalaya	-13.59	-16.55	-2.96	21.78	-0.44	-0.53	-0.10
	Mizoram	22.15	16.04	-6.11	-27.58	0.71	0.52	-0.20
	Nagaland	85.97	83.20	-2.77	-3.22	2.77	2.68	-0.09
	Tripura	170.94	172.89	1.95	1.14	5.51	5.58	0.06
	Total	1556.33	1547.10	-9.23	-0.59	50.20	49.91	-0.30
All India Total		68940.57	68236.12	-704.45	-1.02	2223.89	2201.17	-22.72

15. INTER REGIONAL EXCHANGES 2024-25

(All figures in MU)

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
Import of NR from WR (WR-NR)					
WR - NR 765 kV Gwalior - Agra 2xS/C	529.65	1235.15	1669.21	1351.99	**4786.00**
WR-NR 765 kV Gwalior - Phagi 2xS/C	157.44	396.56	594.39	585.50	**1733.89**
WR-NR 765 kV Jabalpur-Orai D/C	304.53	976.67	1388.19	1346.63	**4016.02**
WR-NR 765 kV Satna-Orai	535.85	669.33	731.25	645.38	**2581.81**
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	0.00	**0.00**
WR - NR HVDC Champa - Kurukshetra	1525.63	2816.73	3112.34	2430.48	**9885.18**
WR - NR HVDC Vindhyachal	0.00	100.25	234.97	96.45	**431.67**
WR - NR 400 kV Zerda Kankroli	0.00	35.28	83.46	65.18	**183.92**
WR - NR 400 kV Zerda Bhinmal	0.00	1.38	0.00	4.10	**5.48**
WR - NR 400 kV RAPP C - Shujalpur D/C	0.00	60.74	171.23	154.47	**386.44**
WR - NR 220 kV Bhanpura - Ranpur	55.64	76.03	72.42	62.97	**267.06**
WR - NR 220 kV Bhanpura - Modak	69.87	76.33	55.61	69.42	**271.23**
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	3.05	7.54	1.83	**12.42**
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	0.00	**0.00**
WR - NR HVDC Mundra - Mohindergarh	717.97	1144.70	1159.61	1529.31	**4551.59**
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.67	262.23	420.74	391.70	**1075.34**
WR-NR 765 kV Varanasi-Vindhyachal	1373.47	2029.86	2265.31	2036.99	**7705.63**
WR -NR765 kV Neemach - Chittorgarh D/C	0.00	35.36	144.70	146.03	**326.09**
Total WR - NR	**5270.72**	**9919.65**	**12110.97**	**10918.43**	**38219.77**
Export of NR to WR (NR-WR)					
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior - Phagi 2xS/C	10.49	1.42	0.00	0.00	**11.91**
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Gwalior-Orai	245.97	283.52	291.19	326.07	**1146.75**
NR - WR HVDC Champa- Kurukshetra	0.00	0.00	0.00	0.00	**0.00**
NR - WR HVDC Vindhyachal	259.46	128.55	20.24	47.84	**456.09**
NR - WR 400 kV Zerda Kankroli	240.09	83.35	1.32	18.52	**343.28**
NR - WR 400 kV Zerda Bhinmal	8.05	0.00	1.50	0.00	**9.55**
NR - WR 400 kV RAPP C - Shujalpur D/C	159.07	31.59	0.21	3.55	**194.42**
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	0.00	**0.00**
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	**0.00**

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	59.69	31.06	9.61	10.21	**110.57**
NR-WR 400 kV Rihand-Vindhyachal D/C	609.68	680.87	661.37	675.02	**2626.94**
NR - WR HVDC Mundra - Mohindergarh	0.00	0.00	0.00	0.00	**0.00**
NR - WR 765 kV Banaskata-Chittorgarh D/C	463.90	132.00	0.70	25.41	**622.01**
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	0.00	**0.00**
NR -WR765 kV Chittorgarh-Neemach D/C	139.03	35.99	2.22	7.43	**184.67**
Total NR - WR	**2195.43**	**1408.35**	**988.36**	**1114.05**	**5706.19**

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
Import of NR from ER (ER-NR)					
ER - NR 765 kV Sasaram - Fatehpur	111.08	133.89	160.45	97.00	**502.42**
ER - NR 765 kV Gaya - Varanasi 2*S/C	50.67	114.34	92.48	120.98	**378.47**
ER - NR 765 kV Gaya - Balia	213.79	293.69	320.79	327.83	**1156.10**
ER - NR HVDC Alipurduar - Agra	0.00	0.00	0.00	344.72	**344.72**
ER-NR 400 kV Patna - Balia D/C	192.05	288.47	317.66	235.85	**1034.03**
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	3.14	135.70	339.63	320.97	**799.44**
ER - NR 400 kV Biharshariff - Balia D/C	0.00	10.98	26.88	8.44	**46.30**
ER - NR 400 kV Motihari - Gorakhpur D/C	63.12	164.75	249.69	181.64	**659.20**
ER - NR 400 kV Biharshariff - Varanasi D/C	3.31	1.28	6.08	0.00	**10.67**
ER - NR 400 kV Sasaram -Varanasi	41.13	16.74	11.73	41.85	**111.45**
ER - NR 400 kV Sasaram -Allahabad	0.13	1.97	22.79	26.07	**50.96**
ER - NR 220 kV Sahupuri-Karamnasa	26.02	26.80	20.84	18.12	**91.78**
ER - NR 132 kV Sahupuri - Karamnasa	2.88	2.88	1.01	1.44	**8.21**
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.66	0.00	0.00	**0.66**
ER - NR 132 kV Rihand - Garhwa	0.00	0.72	0.00	0.23	**0.95**
ER-NR 400 kV Naubatpur - Balia D/C	44.01	81.16	88.27	54.33	**267.77**
ER-NR 400 kV Biharshriff - Sahupuri D/C	2.89	29.05	73.37	83.48	**188.79**
Total ER-NR	**754.22**	**1303.08**	**1731.67**	**1862.95**	**5651.92**
Import of NR from NER (NER-NR)					
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	0.00	371.98	**371.98**
Total NER - NR	**0.00**	**0.00**	**0.00**	**371.98**	**371.98**

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
Export of NR to ER (NR-ER)					
NR - ER 765 kV Sasaram - Fatehpur	0.00	0.46	0.00	0.40	**0.86**
NR - ER 765 kV Gaya - Varanasi 2*S/C	65.96	56.17	23.53	10.09	**155.75**
NR - ER 765 kV Gaya - Balia	0.00	0.00	0.00	0.00	**0.00**
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	0.00	**0.00**
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	0.00	0.00	**0.00**
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	87.39	32.41	0.00	0.00	**119.80**
NR - ER 400 kV Biharshariff - Balia D/C	148.68	86.56	34.72	43.49	**313.45**
ER - NR 400 kV Motihari - Gorakhpur D/C	0.29	1.01	0.00	0.00	**1.30**
NR - ER 400 kV Biharshariff - Varanasi D/C	19.06	0.00	0.00	0.00	**19.06**
NR - ER 400 kV Sasaram -Varanasi	0.00	2.63	4.50	0.00	**7.13**
NR - ER 400 kV Sasaram -Allahabad	10.41	12.82	11.40	0.10	**34.73**
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	1.38	**1.38**
NR - ER 132 kV Sahupuri - Karamnasa	1.44	1.23	0.00	0.00	**2.67**
NR - ER 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.92	0.33	**1.25**
ER - NR 132 kV Rihand - Garhwa	18.99	19.86	16.50	16.68	**72.03**
NR-ER 400 kV Balia- Naubatpur D/C	0.00	1.31	0.00	0.00	**1.31**
NR-ER 400 kV Sahupuri- Biharshariff D/C	0.00	1.60	2.33	1.30	**5.23**
Total NR - ER	**352.22**	**216.06**	**93.90**	**73.77**	**735.95**
Export of NR to NER (NER-NR)					
NR-NER HVDC Biswanath Chariali - Agra	433.33	466.77	450.41	15.02	**1365.53**
Total NR - NER	**433.33**	**466.77**	**450.41**	**15.02**	**1365.53**

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
Export of WR to ER (WR-ER)					
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	388.72	751.76	1168.90	1091.31	**3400.69**
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	721.67	524.74	158.97	147.48	**1552.85**
WR - ER 765 kV Durg - Jharsuguda D/C	0.05	1.25	9.28	8.12	**18.70**
WR - ER 400 kV Sipat - Ranchi D/C	51.26	216.17	285.29	228.90	**781.61**
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	3.12	46.71	321.05	240.61	**611.49**
WR - ER 220 kV Korba - Budhipadar D/C	8.86	26.39	12.71	1.56	**49.53**
WR - ER 220 kV Raigarh - Budhipadar	7.90	2.55	1.47	0.43	**12.34**
Total WR - ER	**1181.58**	**1569.56**	**1957.67**	**1718.40**	**6427.21**
Import of WR from ER (ER - WR)					
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	48.43	15.64	0.11	0.81	**64.99**
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	2.75	169.79	403.73	491.76	**1068.04**
ER - WR 765 kV Durg - Jharsuguda D/C	376.32	377.94	175.66	129.22	**1059.14**
ER - WR 400 kV Sipat - Ranchi D/C	22.01	4.57	0.14	0.92	**27.64**
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	195.37	120.88	235.08	300.81	**852.15**
ER - WR 220 kV Korba - Budhipadar D/C	1.27	8.92	26.67	47.83	**84.69**
ER - WR 220 kV Raigarh - Budhipadar	15.69	25.32	47.74	55.61	**144.37**
Total ER - WR	**661.85**	**723.06**	**889.13**	**1026.96**	**3301.01**
Export of ER to NER (ER - NER)					
ER - NER 400 kV Binaguri - Bongaigaon D/C	20.06	23.09	0.13	105.85	**149.13**
ER - NER 400 kV Alipurduar - Bongaigaon D/C	16.74	47.75	0.06	174.13	**238.69**
ER - NER 220 kV Birpara - Salakati D/C	2.74	6.73	0.18	40.66	**50.31**
Total ER - NER	**39.54**	**77.57**	**0.38**	**320.64**	**438.13**
Import of ER from NER (NER - ER)					
NER - ER 400 kV Binaguri - Bongaigaon D/C	79.80	122.93	220.43	26.32	**449.48**
NER - ER 400 kV Alipurduar - Bongaigaon D/C	164.15	208.72	302.14	30.00	**705.01**
NER - ER 220 kV Birpara - Salakati D/C	18.58	38.63	53.31	5.56	**116.08**
Total NER - ER	**262.52**	**370.28**	**575.88**	**61.88**	**1270.57**

	Apr'24	May'24	June'24	July'24	Fin. Year 2024-25
Name of Line					
Export of ER to SR (ER - SR)					
ER - SR HVDC Gazuwaka	413.69	249.41	22.79	26.22	**712.10**
ER - SR HVDC Talchar - Kolar	1069.04	1104.35	1220.43	999.65	**4393.48**
ER - SR 765 kV Angul- Srikakulam D/C	1640.93	1313.30	1089.30	1030.39	**5073.91**
Total ER - SR	**3123.66**	**2667.05**	**2332.52**	**2056.26**	**10179.49**
Import of ER from SR (SR - ER)					
SR - ER HVDC Gazuwaka	0.00	16.28	175.68	241.91	**433.87**
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	0.00	**0.00**
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	0.00	**0.00**
Total SR - ER	**0.00**	**16.28**	**175.68**	**241.91**	**433.87**
Export of WR to SR (WR-SR)					
WR - SR HVDC Bhadrawati	601.92	221.20	27.24	0.00	**850.36**
WR - SR 765 kV Sholapur - Raichur 2xS/C	802.05	292.07	41.83	26.71	**1162.66**
WR - SR 765 kV Wardha - Nizamabad D/C	1407.32	763.52	504.48	602.22	**3277.54**
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.02	0.00	0.00	**0.02**
WR-SR HVDC Raigarh-Pugalur	2668.37	1464.59	692.13	375.04	**5200.13**
WR-SR 220kV Xeldem - Ambewadi S/C	73.01	75.81	61.15	50.36	**260.34**
WR-SR 220kV Ponda - Ambewadi S/C	43.06	54.26	44.46	5.32	**147.10**
WR-SR 765 kV Warora - Warangal D/C	1603.13	954.21	557.06	549.21	**3663.61**
Total WR - SR	**7198.86**	**3825.69**	**1928.36**	**1608.85**	**14561.76**
Import of WR from SR (SR - WR)					
SR - WR HVDC Bhadrawati	2.36	182.40	422.51	566.72	**1173.98**
SR - WR 765 kV Raichur - Sholapur 2xS/C	0.55	262.88	727.33	826.50	**1817.27**
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	6.86	32.86	33.19	**72.91**
SR - WR 400KV Kolhapur-Kudgi D/C	429.02	646.95	994.86	1212.56	**3283.39**
SR-WR HVDC Pugalur-Raigarh	0.00	0.00	22.86	291.64	**314.50**
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	0.00	**0.00**
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.00	0.00	**0.00**
SR-WR 765 kV Warangal - Warora D/C	0.00	6.90	27.23	50.82	**84.95**
Total SR - WR	**431.93**	**1105.99**	**2227.65**	**2981.43**	**6747.00**
TOTAL ALL INDIA	**21906**	**23669**	**25463**	**24373**	**95410**

***In case of mutiple ckt / DC pole year corres. to the commissioning of final element**

15.1 Import-Export of NR with WR during July 2024

Import of NR from WR (WR-NR)

Date	WR - NR 765 kV Gwalior - Agra 2xS/C	WR-NR 765 kV Gwalior-Phagi 2xS/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa-Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda-Kankroli	WR - NR 400 kV Zerda-Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Neemach - Chittorgarh D/C	Total WR - NR
1-Jul-24	49.76	22.24	42.85	22.88	0.00	71.20	12.03	0.32		5.56	2.05	1.06	0.65	0.00	36.28	9.01	51.62	4.99	**332.50**
2-Jul-24	49.22	20.23	39.54	22.33	0.00	70.05	12.00	5.12		6.05	2.37	1.07	0.70	0.00	37.16	29.21	52.57	4.07	**351.69**
3-Jul-24	38.70	18.30	34.80	20.65	0.00	60.88	9.56	3.44		3.75	2.55	1.19	0.09	0.00	37.50	21.72	52.74	1.56	**307.43**
4-Jul-24	28.23	15.10	30.13	17.97	0.00	26.95	4.50	0.00		0.39	2.55	1.12	0.00	0.00	48.35	0.00	47.20	0.00	**222.49**
5-Jul-24	29.55	15.28	29.89	19.52	0.00	23.00	4.56	0.00		0.09	2.39	1.07	0.00	0.00	48.34	0.00	54.21	0.25	**228.15**
6-Jul-24	27.95	4.50	24.05	17.26	0.00	22.94	0.03	0.00		0.00	2.61	2.01	0.00	0.00	49.67	7.97	56.88	0.00	**215.87**
7-Jul-24	32.64	18.33	34.15	17.85	0.00	9.45	0.00	2.89		2.72	2.77	2.54	0.09	0.00	48.29	17.28	52.81	0.24	**242.05**
8-Jul-24	31.19	27.51	36.31	20.34	0.00	36.23	0.00	0.00		3.84	2.42	2.38	0.00	0.00	48.33	0.00	48.08	2.22	**258.85**
9-Jul-24	34.85	25.36	39.56	21.11	0.00	49.98	1.13	0.00		4.41	2.01	2.51	0.00	0.00	48.33	0.00	56.67	3.53	**289.45**
10-Jul-24	39.37	23.92	42.12	21.91	0.00	72.78	0.00	0.00		5.32	2.37	3.27	0.00	0.00	44.53	0.00	62.12	4.41	**322.12**
11-Jul-24	35.66	12.97	34.61	20.28	0.00	91.88	0.00	2.49		1.95	2.20	3.56	0.00	0.00	36.29	18.84	56.18	0.90	**317.81**
12-Jul-24	31.96	2.24	28.83	18.35	0.00	86.95	0.00	3.09		0.00	2.11	3.73	0.00	0.00	42.93	23.16	57.30	0.00	**300.65**
13-Jul-24	45.41	14.30	41.75	21.86	0.00	64.22	1.10	5.30		4.43	2.10	3.43	0.00	0.00	53.24	28.83	68.50	2.90	**357.37**
14-Jul-24	44.92	20.25	36.82	22.85	0.00	78.79	6.69		3.00	6.24	2.26	3.44	0.00	0.00	60.41	17.24	74.22	5.34	**382.47**
15-Jul-24	42.72	23.48	46.98	21.08	0.00	95.45	2.45	0.00		4.88	2.12	3.31	0.00	0.00	60.40	0.00	67.12	4.80	**374.79**
16-Jul-24	49.72	24.65	46.78	21.95	0.00	110.92	0.00	1.81		7.31	2.16	3.37	0.00	0.00	60.28	8.01	73.12	6.62	**416.70**
17-Jul-24	50.39	25.87	55.27	22.88	0.00	102.27	0.00	2.32		8.35	1.98	3.55	0.00	0.00	60.41	7.79	69.71	9.14	**419.93**
18-Jul-24	53.63	25.87	53.25	23.33	0.00	80.89	2.15	3.12		8.84	2.58	2.55	0.00	0.00	54.84	10.50	76.96	9.39	**407.90**
19-Jul-24	46.75	20.71	52.18	20.87	0.00	106.23	5.08	1.45		6.13	1.94	1.97	0.00	0.00	60.10	7.50	72.80	6.37	**410.08**
20-Jul-24	52.13	31.51	57.33	23.08	0.00	85.64	2.41	2.05		9.30	1.94	2.04	0.00	0.00	60.40	7.10	76.14	9.63	**420.70**
21-Jul-24	50.35	23.96	53.35	20.40	0.00	101.54	1.69	1.55	1.10	7.89	1.94	2.05	0.00	0.00	49.15	9.70	78.01	8.01	**410.69**
22-Jul-24	40.78	22.62	48.20	19.79	0.00	108.00	1.99	0.99		5.65	2.06	1.99	0.00	0.00	60.40	5.92	65.99	5.40	**389.78**
23-Jul-24	44.02	17.78	45.01	20.23	0.00	91.01	6.96	1.50		5.13	2.06	2.03	0.00	0.00	48.92	15.80	69.92	4.46	**374.83**
24-Jul-24	47.84	14.65	46.25	20.90	0.00	101.64	1.31	5.62		5.38	2.06	1.85	0.10	0.00	23.41	29.80	74.01	4.02	**378.84**
25-Jul-24	48.23	13.12	45.23	20.34	0.00	94.50	1.58	4.52		4.85	1.53	1.64	0.10	0.00	39.97	28.42	77.25	5.35	**386.63**
26-Jul-24	48.18	12.35	44.10	19.92	0.00	104.15	7.21	5.35		4.86	1.44	1.79	0.00	0.00	54.33	25.91	75.11	5.09	**409.79**
27-Jul-24	49.50	10.79	48.10	20.20	0.00	113.25	4.27	2.88		4.63	1.04	1.65	0.10	0.00	43.43	16.63	76.90	5.41	**398.78**
28-Jul-24	50.85	19.32	48.73	21.17	0.00	109.89	6.54	2.99		6.67	0.95	1.72	0.00	0.00	39.22	10.48	78.11	8.18	**404.82**
29-Jul-24	52.28	23.71	51.57	21.96	0.00	87.98	0.00	1.21		7.16	1.20	1.78	0.00	0.00	60.38	0.68	75.10	9.13	**394.14**
30-Jul-24	51.97	20.66	60.83	20.70	0.00	93.00	0.00	2.63		7.47	1.49	1.83	0.00	0.00	60.39	8.28	70.40	9.20	**408.85**
31-Jul-24	53.24	13.92	48.06	21.42	0.00	78.82	1.21	2.54		5.22	1.72	1.92	0.00	0.00	53.63	25.92	69.24	5.42	**382.28**
Total	**1351.99**	**585.50**	**1346.63**	**645.38**	**0.00**	**2430.48**	**96.45**	**65.18**	**4.10**	**154.47**	**62.97**	**69.42**	**1.83**	**0.00**	**1529.31**	**391.70**	**2036.99**	**146.03**	**10918.43**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.1 Import-Export of NR with WR during July 2024

Export of NR to WR (NR- WR)

Date	NR - WR 765 kV Gwalior - Agra 2xS/C	NR-WR 765 kV Gwalior-Phagi 2xS/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa-Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda-Kankroli	NR - WR 400 kV Zerda-Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand-Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi-Vindhyachal	WR -NR 765 kV Chittorgarh-Neemach D/C	Total NR - WR
1-Jul-24	0.00	0.00	0.00	0.00	9.95	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.88	0.00	0.00	0.00	0.00	**31.83**
2-Jul-24	0.00	0.00	0.00	0.00	8.69	0.00	0.00	0.00		0.00	0.00	0.00	0.00	20.48	0.00	0.00	0.00	0.00	**29.17**
3-Jul-24	0.00	0.00	0.00	0.00	8.34	0.00	0.00	0.00		0.00	0.00	0.00	0.00	20.92	0.00	0.00	0.00	0.00	**29.26**
4-Jul-24	0.00	0.00	0.00	0.00	9.39	0.00	1.52	3.01		0.00	0.00	0.00	0.85	22.38	0.00	1.20	0.00	0.50	**38.85**
5-Jul-24	0.00	0.00	0.00	0.00	8.89	0.00	0.32	5.11		0.00	0.00	0.00	0.20	22.06	0.00	6.06	0.00	0.00	**42.64**
6-Jul-24	0.00	0.00	0.00	0.00	5.39	0.00	1.19	2.20		2.33	0.00	0.00	0.19	20.30	0.00	0.00	0.00	3.82	**35.42**
7-Jul-24	0.00	0.00	0.00	0.00	9.13	0.00	1.24	0.00		0.00	0.00	0.00	0.00	19.99	0.00	0.00	0.00	0.00	**30.36**
8-Jul-24	0.00	0.00	0.00	0.00	13.00	0.00	3.69	2.60		0.00	0.00	0.00	0.85	22.39	0.00	6.93	0.00	0.00	**49.46**
9-Jul-24	0.00	0.00	0.00	0.00	12.72	0.00	2.68	2.84		0.00	0.00	0.00	0.77	22.14	0.00	8.32	0.00	0.00	**49.47**
10-Jul-24	0.00	0.00	0.00	0.00	11.98	0.00	6.77	0.68		0.00	0.00	0.00	0.68	22.29	0.00	0.42	0.00	0.00	**42.82**
11-Jul-24	0.00	0.00	0.00	0.00	9.07	0.00	2.43	0.00		0.00	0.00	0.00	0.83	22.28	0.00	0.00	0.00	0.00	**34.61**
12-Jul-24	0.00	0.00	0.00	0.00	5.98	0.00	2.44	0.00		1.22	0.00	0.00	0.18	21.01	0.00	0.00	0.00	3.11	**33.94**
13-Jul-24	0.00	0.00	0.00	0.00	9.32	0.00	2.20	0.00		0.00	0.00	0.00	0.12	21.39	0.00	0.00	0.00	0.00	**33.03**
14-Jul-24	0.00	0.00	0.00	0.00	8.66	0.00	0.00		0.00	0.00	0.00	0.00	0.36	21.93	0.00	0.00	0.00	0.00	**30.95**
15-Jul-24	0.00	0.00	0.00	0.00	13.15	0.00	3.58	2.08		0.00	0.00	0.00	0.85	22.27	0.00	2.48	0.00	0.00	**44.41**
16-Jul-24	0.00	0.00	0.00	0.00	12.98	0.00	6.09	0.00		0.00	0.00	0.00	0.51	22.25	0.00	0.00	0.00	0.00	**41.83**
17-Jul-24	0.00	0.00	0.00	0.00	12.98	0.00	6.81	0.00		0.00	0.00	0.00	0.29	22.47	0.00	0.00	0.00	0.00	**42.55**
18-Jul-24	0.00	0.00	0.00	0.00	12.41	0.00	3.88	0.00		0.00	0.00	0.00	0.14	22.39	0.00	0.00	0.00	0.00	**38.82**
19-Jul-24	0.00	0.00	0.00	0.00	11.70	0.00	0.00	0.00		0.00	0.00	0.00	0.19	22.31	0.00	0.00	0.00	0.00	**34.20**
20-Jul-24	0.00	0.00	0.00	0.00	14.91	0.00	0.00	0.00		0.00	0.00	0.00	0.27	22.09	0.00	0.00	0.00	0.00	**37.27**
21-Jul-24	0.00	0.00	0.00	0.00	13.16	0.00	0.36	0.00	0.00	0.00	0.00	0.00	0.26	21.79	0.00	0.00	0.00	0.00	**35.57**
22-Jul-24	0.00	0.00	0.00	0.00	13.25	0.00	0.00	0.00		0.00	0.00	0.00	0.12	21.36	0.00	0.00	0.00	0.00	**34.73**
23-Jul-24	0.00	0.00	0.00	0.00	10.90	0.00	0.00	0.00		0.00	0.00	0.00	0.61	22.26	0.00	0.00	0.00	0.00	**33.77**
24-Jul-24	0.00	0.00	0.00	0.00	8.43	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.73	0.00	0.00	0.00	0.00	**30.16**
25-Jul-24	0.00	0.00	0.00	0.00	9.54	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.89	0.00	0.00	0.00	0.00	**31.43**
26-Jul-24	0.00	0.00	0.00	0.00	9.47	0.00	0.00	0.00		0.00	0.00	0.00	0.52	21.85	0.00	0.00	0.00	0.00	**31.84**
27-Jul-24	0.00	0.00	0.00	0.00	8.26	0.00	0.00	0.00		0.00	0.00	0.00	0.00	21.44	0.00	0.00	0.00	0.00	**29.70**
28-Jul-24	0.00	0.00	0.00	0.00	11.19	0.00	0.18	0.00		0.00	0.00	0.00	0.15	21.22	0.00	0.00	0.00	0.00	**32.74**
29-Jul-24	0.00	0.00	0.00	0.00	12.17	0.00	1.22	0.00		0.00	0.00	0.00	0.56	22.44	0.00	0.00	0.00	0.00	**36.39**
30-Jul-24	0.00	0.00	0.00	0.00	11.20	0.00	1.24	0.00		0.00	0.00	0.00	0.59	22.30	0.00	0.00	0.00	0.00	**35.33**
31-Jul-24	0.00	0.00	0.00	0.00	9.86	0.00	0.00	0.00		0.00	0.00	0.00	0.12	21.52	0.00	0.00	0.00	0.00	**31.50**
Total	**0.00**	**0.00**	**0.00**	**0.00**	**326.07**	**0.00**	**47.84**	**18.52**	**0.00**	**3.55**	**0.00**	**0.00**	**10.21**	**675.02**	**0.00**	**25.41**	**0.00**	**7.43**	**1114.05**

Disclaimer:- Blank entry if the line under outage/shutdown on corrosponding day

15.2 Import-Export of NR with ER & NER during July 2024

Date	Import of NR from ER (ER-NR)																		Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*S/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri-Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	ER-NR 400 kV Biharshriff - Sahupuri D/C	Total ER -NR	NER-NR HVDC Biswanath Chariali - Agra	Total NER-NR
1-Jul-24	8.17	9.49	9.46		12.18	18.30	3.39	11.64		0.86	1.51	0.99			0.00	3.85	3.85	**83.69**	0.00	**0.00**
2-Jul-24	6.85	10.65	9.26	2.68	12.30	18.24	3.13	9.94		0.51	1.53	0.58			0.00	3.46	5.26	**84.39**	0.00	**0.00**
3-Jul-24	4.53	5.53	7.35	8.35	6.35	10.50	0.00	6.35		1.10	1.11	0.23	0.00		0.00	1.37	2.52	**55.29**	0.00	**0.00**
4-Jul-24	2.96	1.45	5.61	8.53	4.75	9.20	0.00	5.45		1.66	0.55	0.08			0.00	0.73	1.20	**42.17**	13.46	**13.46**
5-Jul-24	3.42	0.00	4.26	8.20	5.10	8.28	0.00	4.47		1.44	0.87	0.60			0.23	1.05	1.01	**38.93**	16.60	**16.60**
6-Jul-24	0.00	0.00	3.67	8.64	1.11	3.21	0.00	2.84		1.68	0.44	0.48			0.00	0.56	0.00	**22.63**	17.04	**17.04**
7-Jul-24	2.05	0.12	3.91	8.75	4.48	6.96	0.00	4.42		1.22	0.84	0.44	0.00		0.00	0.61	1.11	**34.91**	16.89	**16.89**
8-Jul-24	4.84	6.05	6.37	8.62	7.78	11.38	0.00	7.30		1.47	0.67	0.36	0.00		0.00	1.66	2.81	**59.31**	16.84	**16.84**
9-Jul-24	3.91	4.59	7.73	7.82	8.56	12.72	0.00	6.31		1.70	0.45	0.22		0.00	0.00	1.85	1.95	**57.81**	15.92	**15.92**
10-Jul-24	4.40	6.16	9.82	7.85	12.31	12.27	0.00	8.18		1.18	0.83	0.85			0.00	3.22	2.12	**69.19**	15.72	**15.72**
11-Jul-24	3.78	3.32	8.42	8.88	9.76	11.76	0.00	7.12		2.10	0.81	0.25			0.00	2.51	2.10	**60.81**	16.93	**16.93**
12-Jul-24	2.06	0.55	7.48	8.72	6.80	11.42	0.00	6.11		2.15	0.00	0.02			0.00	1.45	1.35	**48.11**	17.15	**17.15**
13-Jul-24	2.13	0.08	7.97	8.21	6.20	10.65	0.00	5.80		1.39	0.84	0.29			0.00	1.72	1.19	**46.47**	16.86	**16.86**
14-Jul-24	1.48	0.00	9.60	8.54	5.64	9.73	0.00	5.31		1.67	0.64	0.00			0.00	0.97	0.35	**43.93**	16.83	**16.83**
15-Jul-24	1.98	3.34	11.29	7.93	9.28	10.86	0.00	4.89	0.00	1.45	0.62	0.00	0.00		0.00	2.11	1.72	**55.47**	15.02	**15.02**
16-Jul-24	3.51	3.01	12.16	8.98	8.98	10.42	0.00	6.82	0.00	1.01	0.64	0.11			0.00	2.11	2.32	**60.07**	15.92	**15.92**
17-Jul-24	6.40	11.51	13.05	8.22	12.05	14.45	0.55	9.02		0.27	1.87	0.51			0.00	3.07	4.92	**85.89**	16.44	**16.44**
18-Jul-24	3.22	7.34	13.72	8.12	8.98	11.64	0.00	6.92		0.42	1.62	0.58			0.00	1.95	5.61	**70.12**	16.29	**16.29**
19-Jul-24	4.35	2.15	12.12	10.48	4.01	8.05	0.00	4.32		1.20	1.15	0.09			0.00	0.25	2.95	**51.12**	14.68	**14.68**
20-Jul-24	2.58	5.08	15.43	16.96	7.05	11.26	0.00	5.26		0.79	1.31	0.42	0.00		0.00	1.46	3.99	**71.59**	11.15	**11.15**
21-Jul-24	1.71	3.31	14.05	18.38	5.17	12.42	0.00	5.38		1.31	1.02	0.61	0.00		0.00	1.01	3.14	**67.51**	12.16	**12.16**
22-Jul-24	2.90	6.62	13.22	15.81	11.40	10.91	0.94	7.24		1.44	1.08	1.60	0.00		0.00	3.02	3.71	**79.89**	12.12	**12.12**
23-Jul-24	2.00	5.04	13.25	17.11	10.80	11.78	0.43	5.98		1.44	0.83	1.40	0.00		0.00	2.49	3.64	**76.19**	12.43	**12.43**
24-Jul-24	0.82	1.41	12.20	15.80	8.83	8.44	0.00	4.92	0.00	1.44	0.77	0.35		0.00	0.00	1.73	2.85	**59.56**	12.06	**12.06**
25-Jul-24	0.32	1.85	13.85	16.62	9.51	8.15	0.00	4.52		1.35	0.50	0.19		0.00	0.00	2.19	1.72	**60.77**	9.66	**9.66**
26-Jul-24	0.47	0.10	13.01	16.50	6.65	6.97	0.00	3.83		1.79	0.40	1.42		0.00	0.00	2.61	2.29	**56.04**	7.07	**7.07**
27-Jul-24	3.89	1.75	13.65	16.26	7.45	8.44	0.00	4.46		1.50	0.71	1.31	0.00		0.00	2.55	3.01	**64.98**	7.51	**7.51**
28-Jul-24	4.62	2.40	14.03	16.46	5.04	9.75	0.00	4.49		1.72	0.49	1.43	0.00		0.00	0.42	2.97	**63.82**	7.34	**7.34**
29-Jul-24	2.63	6.28	15.09	16.75	6.64	8.50	0.00	4.20		1.53	0.75	0.27	0.96	0.00	0.00	0.95	4.50	**69.05**	7.43	**7.43**
30-Jul-24	2.54	7.28	15.00	14.75	5.89	8.11	0.00	4.43		1.72	0.61	1.40	0.00	0.00	0.00	0.66	4.00	**66.39**	7.34	**7.34**
31-Jul-24	2.48	4.52	11.80	15.80	4.80	6.20	0.00	3.72		1.34	0.61	1.04	0.48		0.00	0.74	3.32	**56.85**	7.12	**7.12**
Total	**97.00**	**120.98**	**327.83**	**344.72**	**235.85**	**320.97**	**8.44**	**181.64**	**0.00**	**41.85**	**26.07**	**18.12**	**1.44**	**0.00**	**0.23**	**54.33**	**83.48**	**1862.95**	**371.98**	**371.98**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.2 Import-Export of NR with ER & NER during July 2024

Date	Export of NR to ER (NR- ER)																		Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*S/C	NR - ER 765 kV Gaya Balia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Balia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Balia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharshariff - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri-Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Balia-Naubatpur D/C	NR-ER 400 kV Sahupuri-Biharshariff D/C	Total NR -ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-Jul-24	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.53	0.00	0.00	**0.53**	11.95	**11.95**
2-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.52	0.00	0.00	**0.52**	3.07	**3.07**
3-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.72	0.00		0.00	0.00	0.00	0.00		0.45	0.00	0.00	**1.17**	0.00	**0.00**
4-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.22	0.00		0.00	0.00	0.00			0.49	0.00	0.00	**2.71**	0.00	**0.00**
5-Jul-24	0.00	0.53	0.00	0.00	0.00	0.00	1.38	0.00		0.00	0.00	0.00			0.00	0.00	0.00	**1.91**	0.00	**0.00**
6-Jul-24	0.40	8.01	0.00	0.00	0.00	0.00	4.78	0.00		0.00	0.00	0.00			0.64	0.00	1.30	**15.13**	0.00	**0.00**
7-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.84	0.00		0.00	0.00	0.00	0.00		0.61	0.00	0.00	**3.45**	0.00	**0.00**
8-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.07	0.00		0.00	0.00	0.00	0.00		0.55	0.00	0.00	**2.62**	0.00	**0.00**
9-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.36	0.00		0.00	0.00	0.00		0.07	0.65	0.00	0.00	**3.08**	0.00	**0.00**
10-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00		0.00	0.00	0.00			0.52	0.00	0.00	**1.20**	0.00	**0.00**
11-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.37	0.00		0.00	0.00	0.00			0.47	0.00	0.00	**1.84**	0.00	**0.00**
12-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.58	0.00		0.00	0.10	0.00			0.48	0.00	0.00	**2.16**	0.00	**0.00**
13-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.35	0.00		0.00	0.00	0.00			0.48	0.00	0.00	**2.83**	0.00	**0.00**
14-Jul-24	0.00	1.55	0.00	0.00	0.00	0.00	2.98	0.00		0.00	0.00	0.70			0.58	0.00	0.00	**5.81**	0.00	**0.00**
15-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.52	0.00	0.00	0.00	0.00	0.68	0.00		0.54	0.00	0.00	**2.74**	0.00	**0.00**
16-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.28	0.00	0.00	0.00	0.00	0.00			0.48	0.00	0.00	**1.76**	0.00	**0.00**
17-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00			0.39	0.00	0.00	**0.39**	0.00	**0.00**
18-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.55	0.00		0.00	0.00	0.00			0.58	0.00	0.00	**1.13**	0.00	**0.00**
19-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.33	0.00		0.00	0.00	0.00			0.68	0.00	0.00	**3.01**	0.00	**0.00**
20-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.39	0.00		0.00	0.00	0.00	0.00		0.66	0.00	0.00	**1.05**	0.00	**0.00**
21-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00		0.00	0.00	0.00	0.00		0.61	0.00	0.00	**0.79**	0.00	**0.00**
22-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.46	0.00	0.00	**0.46**	0.00	**0.00**
23-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.00	0.00	0.00	0.00		0.59	0.00	0.00	**0.59**	0.00	**0.00**
24-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.39	0.00	0.00	0.00	0.00	0.00		0.04	0.50	0.00	0.00	**1.93**	0.00	**0.00**
25-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.90	0.00		0.00	0.00	0.00		0.09	0.52	0.00	0.00	**1.51**	0.00	**0.00**
26-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.28	0.00		0.00	0.00	0.00		0.13	0.60	0.00	0.00	**2.01**	0.00	**0.00**
27-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.51	0.00		0.00	0.00	0.00	0.00		0.52	0.00	0.00	**2.03**	0.00	**0.00**
28-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.51	0.00		0.00	0.00	0.00	0.00		0.68	0.00	0.00	**2.19**	0.00	**0.00**
29-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.23	0.00		0.00	0.00	0.00	0.00	0.00	0.72	0.00	0.00	**1.95**	0.00	**0.00**
30-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	1.25	0.00		0.00	0.00	0.00	0.00	0.00	0.63	0.00	0.00	**1.88**	0.00	**0.00**
31-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	2.84	0.00		0.00	0.00	0.00	0.00		0.55	0.00	0.00	**3.39**	0.00	**0.00**
Total	**0.40**	**10.09**	**0.00**	**0.00**	**0.00**	**0.00**	**43.49**	**0.00**	**0.00**	**0.00**	**0.10**	**1.38**	**0.00**	**0.33**	**16.68**	**0.00**	**1.30**	**73.77**	**15.02**	**15.02**

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

15.3 Import and Export of ER with WR & NER during July 2024

Date	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2xS/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda- 2xD/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2xS/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda- 2xD/C	ER - WR 220 kV Korba - Budhipadar D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2xD/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Jul-24	24.2	0.6	0.0	5.2	9.4	0.0	0.0	**39.34**	0.1	12.8	9.7	0.1	11.1	1.4	2.5	**37.57**	0.00	0.00	0.00	**0.00**	12.50	15.51	3.00	**31.01**
2-Jul-24	25.0	1.4	0.0	5.6	9.9	0.0	0.0	**41.92**	0.2	13.8	5.8	0.1	11.8	2.8	2.7	**37.06**	0.01	0.10	0.04	**0.15**	7.86	8.64	1.67	**18.17**
3-Jul-24	24.4	4.0	0.0	5.4	8.9	0.0	0.0	**42.72**	0.3	11.3	5.7	0.2	11.4	2.8	2.4	**34.18**	1.43	3.22	0.66	**5.31**	1.71	2.15	0.45	**4.32**
4-Jul-24	28.4	11.0	0.0	5.5	10.3	0.0	0.0	**55.29**	0.0	8.2	6.5	0.0	13.6	2.2	2.8	**33.34**	1.30	2.47	0.80	**4.57**	0.29	0.14	0.02	**0.44**
5-Jul-24	32.2	13.4	0.0	7.0	12.0	0.0	0.0	**64.61**	0.0	9.5	5.3	0.0	13.7	2.6	3.3	**34.48**	0.55	1.87	0.54	**2.95**	0.98	0.78	0.14	**1.91**
6-Jul-24	38.2	12.4	0.0	9.1	15.4	0.0	0.0	**75.10**	0.0	13.8	4.3	0.0	14.5	2.3	3.5	**38.46**	1.58	3.98	1.14	**6.71**	0.23	0.05	0.00	**0.28**
7-Jul-24	36.1	16.4	0.2	8.1	13.4	0.0	0.0	**74.27**	0.0	11.4	3.4	0.0	14.9	2.3	2.7	**34.75**	1.44	3.94	1.08	**6.46**	0.06	0.00	0.00	**0.06**
8-Jul-24	29.4	14.8	0.0	5.7	10.0	0.0	0.0	**59.91**	0.0	6.1	3.5	0.0	13.6	2.6	2.8	**28.64**	1.63	3.93	1.20	**6.76**	0.34	0.17	0.00	**0.50**
9-Jul-24	35.1	12.6	0.0	7.4	12.7	0.0	0.0	**67.78**	0.0	11.1	3.2	0.0	14.6	2.7	3.0	**34.52**	1.45	3.28	1.21	**5.94**	0.41	0.16	0.00	**0.57**
10-Jul-24	36.8	9.5	0.0	5.9	14.4	0.0	0.0	**66.55**	0.0	11.7	2.6	0.0	14.1	2.3	3.0	**33.79**	1.55	3.38	0.91	**5.84**	0.67	0.56	0.09	**1.33**
11-Jul-24	31.9	9.8	0.1	7.3	13.0	0.0	0.0	**62.00**	0.0	6.5	1.4	0.0	11.8	1.6	2.6	**23.90**	1.56	4.16	0.80	**6.52**	0.39	0.23	0.04	**0.66**
12-Jul-24	30.2	15.3	0.4	7.0	12.9	0.0	0.0	**65.75**	0.2	3.9	1.2	0.1	11.8	2.3	2.4	**21.97**	2.52	6.11	1.18	**9.80**	0.27	0.09	0.01	**0.37**
13-Jul-24	42.4	10.2	0.3	10.4	16.7	0.0	0.0	**80.05**	0.0	16.2	1.6	0.0	15.8	1.7	2.6	**37.96**	3.65	7.62	1.64	**12.91**	0.05	0.00	0.00	**0.05**
14-Jul-24	49.8	2.5	0.0	12.0	17.6	0.0	0.0	**81.86**	0.0	22.2	2.4	0.0	17.8	1.8	0.9	**45.06**	5.75	9.65	1.90	**17.30**	0.00	0.00	0.00	**0.00**
15-Jul-24	41.2	0.2	0.0	9.2	14.2	0.0	0.0	**64.77**	0.0	21.0	7.8	0.0	15.0	1.9	2.6	**48.25**	4.59	7.44	1.51	**13.54**	0.00	0.00	0.00	**0.00**
16-Jul-24	46.3	0.0	0.0	10.4	15.0	0.0	0.0	**71.68**	0.0	28.4	7.7	0.0	15.2	1.7	2.9	**55.91**	5.57	9.71	1.98	**17.27**	0.00	0.00	0.00	**0.00**
17-Jul-24	33.2	0.0	0.0	7.2	8.6	0.0	0.0	**48.99**	0.0	29.3	7.2	0.0	10.8	1.8	2.5	**51.56**	5.51	10.15	1.88	**17.54**	0.00	0.00	0.00	**0.00**
18-Jul-24	44.5	0.0	0.2	8.8	8.9	0.0	0.0	**62.31**	0.0	21.8	5.3	0.0	11.0	1.6	1.6	**41.27**	5.93	11.23	2.21	**19.37**	0.00	0.00	0.00	**0.00**
19-Jul-24	38.1	0.0	0.8	7.5	3.0	0.0	0.0	**49.40**	0.0	23.8	2.3	0.0	8.1	1.6	0.8	**36.52**	6.26	10.94	2.38	**19.58**	0.00	0.00	0.00	**0.00**
20-Jul-24	36.6	0.0	0.8	7.6	0.0	0.0	0.0	**45.05**	0.0	26.3	1.6	0.0	7.1	1.6	0.7	**37.31**	4.61	6.02	1.51	**12.14**	0.00	0.03	0.00	**0.03**
21-Jul-24	36.0	0.0	0.7	7.0	0.1	0.0	0.0	**43.76**	0.0	24.7	2.9	0.0	7.8	1.7	1.0	**38.19**	3.94	5.00	1.28	**10.23**	0.00	0.01	0.00	**0.01**
22-Jul-24	27.3	0.3	0.4	5.3	0.1	0.0	0.0	**33.42**	0.0	19.2	4.2	0.2	7.6	1.3	0.9	**33.42**	1.23	2.51	0.81	**4.55**	0.47	1.02	0.12	**1.61**
23-Jul-24	30.8	1.6	0.2	5.7	0.3	0.0	0.0	**38.68**	0.0	11.0	3.7	0.0	5.8	0.5	0.4	**21.34**	2.64	5.46	1.30	**9.41**	0.05	0.13	0.01	**0.19**
24-Jul-24	35.0	2.8	0.3	7.1	1.5	0.1	0.1	**46.91**	0.0	13.0	4.1	0.0	3.5	0.3	0.2	**21.17**	5.96	7.27	1.73	**14.96**	0.00	0.02	0.00	**0.02**
25-Jul-24	38.0	0.9	0.6	7.4	1.8	0.3	0.1	**49.01**	0.0	16.4	4.1	0.1	3.0	0.1	0.5	**24.07**	5.21	6.89	1.72	**13.82**	0.00	0.00	0.00	**0.00**
26-Jul-24	37.7	0.8	0.8	7.4	2.1	0.2	0.1	**49.00**	0.0	19.7	3.9	0.0	2.8	0.4	0.7	**27.44**	5.88	7.55	1.95	**15.37**	0.00	0.00	0.00	**0.00**
27-Jul-24	35.1	0.4	0.8	7.4	2.2	0.4	0.1	**46.47**	0.0	22.0	4.2	0.0	2.4	0.2	0.7	**29.43**	6.07	8.06	1.95	**16.08**	0.00	0.00	0.00	**0.00**
28-Jul-24	37.0	1.0	0.6	7.1	1.5	0.1	0.0	**47.17**	0.0	18.9	3.6	0.0	2.6	0.3	0.7	**25.99**	4.46	5.56	1.44	**11.46**	0.00	0.06	0.00	**0.07**
29-Jul-24	36.7	1.1	0.0	6.5	0.6	0.0	0.0	**45.01**	0.0	12.3	4.4	0.0	3.5	0.7	0.8	**21.59**	3.90	4.62	1.33	**9.85**	0.03	0.19	0.00	**0.22**
30-Jul-24	38.8	0.8	0.3	8.1	1.7	0.2	0.0	**49.91**	0.0	16.0	2.8	0.0	2.3	0.6	0.9	**22.53**	4.43	5.27	1.24	**10.94**	0.00	0.06	0.00	**0.06**
31-Jul-24	34.8	3.8	0.6	7.6	2.5	0.3	0.0	**49.68**	0.0	9.6	2.8	0.0	1.9	0.3	0.6	**15.32**	5.25	6.75	1.30	**13.31**	0.00	0.00	0.00	**0.00**
Total	**1091.31**	**147.48**	**8.12**	**228.90**	**240.61**	**1.56**	**0.43**	**1718.40**	**0.81**	**491.76**	**129.22**	**0.92**	**300.81**	**47.83**	**55.61**	**1026.96**	**105.85**	**174.13**	**40.66**	**320.64**	**26.32**	**30.00**	**5.56**	**61.88**

15.4 Import and Export of SR with ER & WR during July 2024

Date	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)									Import of WR from SR (SR - WR)								
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar -Kolar	ER-SR 765 kV Angul-Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul-Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadrawati	WR - SR 765 kV Sholapur - Raichur 2xS/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400KV Kolhapur-Kudgi D/C	WR-SR HVDC Raigarh-Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	WR-SR 765 kV Warora - Warangal D/C	Total WR - SR	SR - WR HVDC Bhadrawati	SR - WR 765 kV Raichur - Sholapur 2xS/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400KV Kolhapur-Kudgi D/C	SR-WR HVDC Raigarh-Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	SR-WR 765 kV Warangal - Warora D/C	Total SR - WR
1-Jul-24	0.00	40.98	47.17	**88.16**	4.81	0.00	0.00	**4.81**	0.00	0.17	27.62	0.00	31.31	0.00	0.00	27.33	**86.43**	10.61	15.95	0.00	35.39	0.00	0.00	0.00	0.00	**61.95**
2-Jul-24	0.00	41.00	48.56	**89.56**	7.59	0.00	0.00	**7.59**	0.00	1.67	30.06	0.00	24.29	0.00	0.00	28.37	**84.39**	12.74	13.72	0.00	37.19	0.00	0.00	0.00	0.00	**63.65**
3-Jul-24	2.49	41.00	47.67	**91.15**	3.95	0.00	0.00	**3.95**	0.00	4.22	31.62	0.00	40.92	0.45	0.00	30.40	**107.61**	8.70	13.78	0.00	32.39	0.00	0.00	0.00	0.00	**54.87**
4-Jul-24	3.78	45.35	54.37	**103.49**	0.00	0.00	0.00	**0.00**	0.00	2.61	40.55	0.00	24.82	2.30	0.00	40.57	**110.86**	7.18	5.09	0.00	28.70	0.00	0.00	0.00	0.00	**40.97**
5-Jul-24	3.79	48.13	47.68	**99.59**	0.00	0.00	0.00	**0.00**	0.00	2.27	35.68	0.00	20.86	2.11	0.01	36.89	**97.82**	7.18	7.06	0.00	30.68	0.00	0.00	0.00	0.00	**44.92**
6-Jul-24	2.47	48.12	41.10	**91.69**	1.22	0.00	0.00	**1.22**	0.00	1.32	32.13	0.00	48.91	2.16	0.01	31.17	**115.69**	7.18	17.62	0.00	34.60	0.00	0.00	0.00	0.00	**59.40**
7-Jul-24	0.00	48.13	37.30	**85.43**	3.56	0.00	0.00	**3.56**	0.00	0.80	23.91	0.00	14.61	2.09	0.01	22.79	**64.21**	13.40	24.65	0.11	34.24	0.00	0.00	0.00	0.33	**72.74**
8-Jul-24	1.77	48.16	37.78	**87.71**	1.15	0.00	0.00	**1.15**	0.00	0.10	24.90	0.00	8.76	2.09	0.00	24.10	**59.95**	11.86	22.43	0.00	32.43	0.00	0.00	0.00	0.00	**66.73**
9-Jul-24	2.63	48.16	29.38	**80.18**	0.00	0.00	0.00	**0.00**	0.00	0.01	15.09	0.00	26.32	2.10	0.00	14.86	**58.38**	9.57	27.88	0.09	35.27	0.00	0.00	0.00	0.06	**72.86**
10-Jul-24	2.63	48.16	30.34	**81.13**	0.00	0.00	0.00	**0.00**	0.00	0.42	19.39	0.00	24.25	1.98	0.00	17.42	**63.46**	23.81	26.59	0.00	30.92	0.00	0.00	0.00	0.00	**81.33**
11-Jul-24	2.63	48.17	30.87	**81.66**	0.00	0.00	0.00	**0.00**	0.00	0.72	25.41	0.00	39.05	1.99	0.00	24.45	**91.64**	23.83	14.62	0.00	27.52	0.00	0.00	0.00	0.01	**65.97**
12-Jul-24	2.63	48.15	36.28	**87.05**	0.00	0.00	0.00	**0.00**	0.00	4.50	34.17	0.00	25.17	1.98	0.00	32.11	**97.94**	15.48	10.95	0.00	24.46	0.00	0.00	0.00	0.00	**50.89**
13-Jul-24	1.40	48.15	27.53	**77.08**	1.15	0.00	0.00	**1.15**	0.00	0.39	19.91	0.00	19.92	1.71	0.01	16.13	**58.07**	11.99	29.20	0.61	37.02	0.00	0.00	0.00	1.48	**80.31**
14-Jul-24	0.00	48.15	23.35	**71.50**	6.91	0.00	0.00	**6.91**	0.00	0.00	8.52	0.00	7.76	1.08	0.01	6.02	**23.39**	17.80	43.64	1.71	46.72	14.92	0.00	0.00	3.59	**128.39**
15-Jul-24	0.00	48.15	35.25	**83.40**	17.79	0.00	0.00	**17.79**	0.00	0.00	17.69	0.00	0.00	1.46	0.01	15.55	**34.70**	23.85	35.48	1.54	41.32	17.17	0.00	0.00	2.77	**122.14**
16-Jul-24	0.00	39.94	29.22	**69.16**	17.85	0.00	0.00	**17.85**	0.00	0.00	8.98	0.00	3.29	1.96	0.00	6.69	**20.92**	23.85	48.43	2.18	49.91	15.68	0.00	0.00	4.72	**144.76**
17-Jul-24	0.00	30.13	28.34	**58.47**	17.90	0.00	0.00	**17.90**	0.00	0.00	9.42	0.00	11.29	2.20	0.01	6.56	**29.48**	23.84	45.93	2.55	47.23	0.00	0.00	0.00	4.77	**124.33**
18-Jul-24	0.00	21.56	25.97	**47.53**	16.77	0.00	0.00	**16.77**	0.00	0.00	7.12	0.00	3.51	2.12	0.01	4.14	**16.90**	23.15	51.05	3.62	50.77	7.30	0.00	0.00	6.97	**142.86**
19-Jul-24	0.00	9.79	22.13	**31.92**	12.27	0.00	0.00	**12.27**	0.00	0.00	7.64	0.00	0.00	1.23	1.06	6.21	**16.13**	18.53	42.28	4.76	46.93	18.40	0.00	0.00	6.26	**137.16**
20-Jul-24	0.00	8.55	21.41	**29.96**	13.37	0.00	0.00	**13.37**	0.00	0.00	5.46	0.00	0.00	1.87	1.42	4.44	**13.19**	22.75	44.13	4.86	48.28	13.97	0.00	0.00	5.87	**139.86**
21-Jul-24	0.00	7.34	22.20	**29.54**	18.05	0.00	0.00	**18.05**	0.00	0.00	4.10	0.00	0.00	1.05	1.58	3.68	**10.42**	23.76	47.02	5.68	48.52	19.76	0.00	0.00	5.92	**150.67**
22-Jul-24	0.00	12.92	25.88	**38.79**	8.64	0.00	0.00	**8.64**	0.00	0.00	9.35	0.00	0.00	0.59	1.13	9.91	**20.98**	23.77	33.20	2.16	41.02	12.49	0.00	0.00	2.05	**114.68**
23-Jul-24	0.00	19.33	31.50	**50.84**	2.48	0.00	0.00	**2.48**	0.00	0.82	20.91	0.00	0.00	1.48	0.01	18.15	**41.37**	23.81	21.81	0.08	36.35	11.52	0.00	0.00	0.24	**93.82**
24-Jul-24	0.00	15.41	33.54	**48.95**	6.44	0.00	0.00	**6.44**	0.00	2.06	23.11	0.00	0.00	0.89	0.00	19.94	**46.00**	23.81	16.59	0.00	39.52	13.22	0.00	0.00	0.08	**93.23**
25-Jul-24	0.00	17.03	30.30	**47.33**	12.27	0.00	0.00	**12.27**	0.00	1.02	14.87	0.00	0.00	1.99	0.00	12.45	**30.33**	18.91	21.84	0.50	42.83	20.51	0.00	0.00	0.53	**105.12**
26-Jul-24	0.00	18.56	29.53	**48.09**	14.27	0.00	0.00	**14.27**	0.00	0.57	11.21	0.00	0.00	0.86	0.01	10.01	**22.64**	23.80	25.77	0.42	45.32	28.49	0.00	0.00	0.67	**124.47**
27-Jul-24	0.00	19.60	27.49	**47.09**	11.01	0.00	0.00	**11.01**	0.00	0.18	13.11	0.00	0.00	2.17	0.00	11.44	**26.90**	23.68	25.14	0.62	43.99	28.57	0.00	0.00	1.08	**123.08**
28-Jul-24	0.00	14.68	26.24	**40.92**	14.63	0.00	0.00	**14.63**	0.00	0.00	8.88	0.00	0.00	2.10	0.00	6.76	**17.74**	20.79	34.09	1.51	46.76	32.43	0.00	0.00	2.76	**138.34**
29-Jul-24	0.00	16.94	34.51	**51.45**	13.49	0.00	0.00	**13.49**	0.00	0.08	19.77	0.00	0.00	2.14	0.01	15.86	**37.86**	23.87	21.38	0.19	39.67	18.89	0.00	0.00	0.63	**104.62**
30-Jul-24	0.00	16.94	31.08	**48.03**	7.16	0.00	0.00	**7.16**	0.00	0.43	19.34	0.00	0.00	2.14	0.01	14.28	**36.19**	23.87	23.54	0.00	39.83	9.16	0.00	0.00	0.00	**96.40**
31-Jul-24	0.00	32.97	36.42	**69.38**	7.16	0.00	0.00	**7.16**	0.00	2.35	32.29	0.00	0.00	2.06	0.01	30.55	**67.24**	19.35	15.63	0.00	36.76	9.15	0.00	0.00	0.03	**80.92**
TOTAL	**26.22**	**999.65**	**1030.39**	**2056.26**	**241.91**	**0.00**	**0.00**	**241.91**	**0.00**	**26.71**	**602.22**	**0.00**	**375.04**	**50.36**	**5.32**	**549.21**	**1608.85**	**566.72**	**826.50**	**33.19**	**1212.56**	**291.64**	**0.00**	**0.00**	**50.82**	**2981.43**

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2024 से मार्च 2025 April 2024 to March 2025

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/ 'को आयात']

Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2024 APR'24	86.47	18.58	346.08	0.00	694.34	0.00	0.76	0.00
मई 2024 MAY'24	51.37	68.17	271.84	3.92	655.55	0.00	0.74	0.00
जून 2024 JUN'24	0.00	520.23	20.51	152.32	709.14	0.00	0.87	0.00
जुलाई 2024 JUL'24	0.00	1225.84	0.00	476.74	747.37	0.00	0.91	0.00
कुल Total	**137.84**	**1832.82**	**638.42**	**632.97**	**2806.40**	**0.00**	**3.27**	**0.00**

* Based on daily operational data

16.1 Import from neighbouring countries during Jul 2024

(All figures in MU)

Date	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jul-24	13.97	3.62	0.53	0.00	6.56	9.85	0.94	0.21	0.00	0.00	1.05	12.77	0.03	0.00
2-Jul-24	13.78	3.51	0.80	0.00	6.77	10.19	0.50	0.31	0.00	0.00	0.99	13.95	0.09	0.00
3-Jul-24	13.12	3.11	0.69	0.17	4.83	12.40	1.25	0.65	0.00	0.00	0.94	14.30	0.04	0.00
4-Jul-24	13.16	3.15	0.49	0.27	1.63	12.89	1.25	0.56	0.00	0.00	0.86	13.42	0.03	0.00
5-Jul-24	13.37	3.29	0.50	0.00	6.74	10.21	1.31	0.72	0.00	0.00	0.98	13.69	0.03	0.00
6-Jul-24	13.26	3.33	0.44	0.00	6.79	10.09	1.41	0.76	0.00	0.00	0.27	8.27	0.02	0.00
7-Jul-24	16.38	4.17	0.60	0.25	6.79	10.28	1.31	0.81	0.00	0.00	0.75	15.17	0.00	0.00
8-Jul-24	16.64	4.13	0.72	0.40	6.89	10.45	1.33	0.20	0.00	0.00	0.66	14.80	0.00	0.00
9-Jul-24	16.61	4.13	0.68	0.46	6.61	10.01	1.23	0.10	0.00	0.00	0.57	15.27	0.00	0.00
10-Jul-24	16.60	4.09	0.66	0.49	6.62	10.01	1.06	0.79	0.00	0.00	0.98	15.86	0.00	0.00
11-Jul-24	16.05	3.76	0.00	0.15	7.48	11.32	1.15	0.50	0.00	0.00	0.85	16.18	0.00	0.00
12-Jul-24	16.82	4.27	1.47	0.40	7.05	10.66	1.23	0.87	0.00	0.00	1.00	16.55	0.00	0.00
13-Jul-24	17.04	4.33	2.07	0.54	6.86	10.35	1.21	0.86	0.00	0.00	1.32	16.52	0.00	0.00
14-Jul-24	16.48	4.00	1.32	0.59	5.42	8.15	1.34	0.75	0.00	0.00	1.27	15.90	0.02	0.00
15-Jul-24	16.35	4.00	1.35	0.60	5.05	7.60	1.26	0.66	0.00	0.00	1.15	14.69	0.02	0.00
16-Jul-24	15.34	4.99	1.94	0.81	5.71	8.61	1.16	0.79	0.00	0.00	1.00	15.29	0.03	0.00
17-Jul-24	21.75	0.44	1.84	0.00	6.72	10.16	1.35	0.90	0.00	0.00	0.53	15.75	0.06	0.00
18-Jul-24	15.64	4.82	2.21	0.86	6.93	10.48	0.57	1.20	0.00	0.00	0.24	15.16	0.02	0.00
19-Jul-24	16.66	4.17	2.39	0.85	6.97	10.56	0.60	1.24	0.00	0.00	0.97	14.66	0.02	0.00
20-Jul-24	16.47	4.07	2.11	0.80	6.76	10.23	1.27	0.83	0.00	0.00	1.05	15.96	0.02	0.00
21-Jul-24	16.60	4.15	2.17	0.76	6.79	10.24	1.25	0.80	0.00	0.00	1.09	14.25	0.02	0.00
22-Jul-24	16.74	4.24	2.04	0.58	6.82	10.30	1.21	0.72	0.00	0.00	1.22	15.28	0.02	0.00
23-Jul-24	16.63	4.17	2.17	0.75	6.83	10.33	1.39	0.79	0.00	0.00	1.34	15.23	0.01	0.00
24-Jul-24	16.33	3.99	2.03	0.89	5.72	8.64	1.30	0.69	0.00	0.00	1.05	14.22	0.02	0.00
25-Jul-24	17.76	2.25	2.05	0.98	6.71	10.13	1.35	0.82	0.00	0.00	1.02	14.34	0.01	0.00
26-Jul-24	16.42	4.04	2.06	0.83	6.49	9.25	1.11	0.92	0.00	0.00	1.14	13.74	0.01	0.00
27-Jul-24	16.39	4.03	1.97	0.81	6.65	10.03	1.43	0.83	0.00	0.00	1.27	14.24	0.02	0.00
28-Jul-24	16.49	4.09	2.02	0.77	6.58	9.95	1.31	0.76	0.00	0.00	1.01	13.13	0.02	0.00
29-Jul-24	16.42	4.04	1.93	0.95	6.53	9.88	1.34	0.73	0.00	0.00	0.73	12.58	0.01	0.00
30-Jul-24	22.23	0.00	1.59	0.00	6.47	9.81	1.32	0.73	0.00	0.00	0.92	13.37	0.00	0.00
31-Jul-24	23.22	0.00	1.21	0.00	6.30	9.54	1.37	0.73	0.00	0.00	1.31	12.84	0.00	0.00
Total	**510.69**	**110.41**	**44.03**	**14.98**	**197.05**	**312.60**	**37.11**	**22.23**	**0.00**	**0.00**	**29.52**	**447.38**	**0.56**	**0.00**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

16.2 Export to neighbouring countries during Jul 2024

(All figures in MU)

Date	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Jul-24	0.00	0.00	0.00	0.59	0.00	0.00	0.00	0.00	22.57	1.81	0.00	0.00	0.00	0.03
2-Jul-24	0.00	0.00	0.00	0.31	0.00	0.00	0.00	0.00	22.56	1.81	0.00	0.00	0.00	0.03
3-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	17.02	1.78	0.00	0.00	0.00	0.03
4-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.09	1.81	0.00	0.00	0.00	0.03
5-Jul-24	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	22.24	1.54	0.00	0.00	0.00	0.03
6-Jul-24	0.00	0.00	0.00	0.09	0.00	0.00	0.00	0.00	22.43	1.86	0.00	0.00	0.00	0.02
7-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.44	2.10	0.00	0.00	0.03	0.03
8-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.10	2.03	0.00	0.00	0.05	0.03
9-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.39	1.90	0.00	0.00	0.03	0.03
10-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.24	2.01	0.00	0.00	0.00	0.03
11-Jul-24	0.00	0.00	0.31	0.00	0.00	0.00	0.00	0.00	22.32	2.10	0.00	0.00	0.01	0.03
12-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.50	1.73	0.00	0.00	0.03	0.03
13-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.22	1.96	0.00	0.00	0.02	0.03
14-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.27	2.01	0.00	0.00	0.00	0.03
15-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.96	2.06	0.00	0.00	0.00	0.03
16-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.23	2.10	0.00	0.00	0.00	0.03
17-Jul-24	0.00	0.00	0.00	0.03	0.00	0.00	0.00	0.00	22.27	2.14	0.00	0.00	0.00	0.03
18-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.03	2.33	0.00	0.00	0.00	0.04
19-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.30	1.95	0.00	0.00	0.00	0.02
20-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.34	1.81	0.00	0.00	0.00	0.03
21-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.34	1.89	0.00	0.00	0.00	0.03
22-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.17	1.81	0.00	0.00	0.00	0.03
23-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.07	1.72	0.00	0.00	0.00	0.03
24-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.10	1.83	0.00	0.00	0.00	0.03
25-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.99	2.01	0.00	0.00	0.00	0.03
26-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.89	2.09	0.00	0.00	0.00	0.03
27-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.79	2.22	0.00	0.00	0.00	0.03
28-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.08	2.29	0.00	0.00	0.00	0.03
29-Jul-24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	22.04	2.06	0.00	0.00	0.00	0.03
30-Jul-24	0.00	0.75	0.00	0.26	0.00	0.00	0.00	0.00	21.76	2.26	0.00	0.00	0.01	0.03
31-Jul-24	0.00	3.83	0.00	1.18	0.00	0.00	0.00	0.00	21.38	2.24	0.00	0.00	0.01	0.03
Total	**0.00**	**4.58**	**0.31**	**2.48**	**0.00**	**0.00**	**0.00**	**0.00**	**682.13**	**61.25**	**0.00**	**0.00**	**0.20**	**0.91**

Based on SEM/Energy meter data for links where available (*Mangdechu generation reciept at APD through a bypassed arrangement at 400kV Punatsanchu station)

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – जुलाई 2024

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - JULY 2024

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

जुलाई 2024 JULY 2024			Apr'24- Jul'24	
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	848	6007	3218	13323
प.क्षे. WR	352	542	1473	2691
द.क्षे. SR	238	311	1440	6910
पू.क्षे. ER	388	1665	1519	6707
पूर्वोत्तर क्षे. NER	65	101	393	248
कुल TOTAL	1891	8626	8043	29879

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA		द्विपक्षीय एसटीओए Bilateral STOA	
माह MONTH	क्रेताओं / विक्रेताओं की स. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2024 Apr'24	9860	7128	1565	5921
मई 2024 May'24	497	8268	2258	7240
जून 2024 Jun'24	10971	8732	2329	8093
जुलाई 2024 Jul'24	11338	9378	1891	8626
कुल TOTAL	32666	33506	8043	29879

मासिक डी एस एम बिलिंग का ब्योरा* 2024-25
MONTHLY DSM BILLING DETAILS* 2024-25

अनंतिम आँकड़े
Provisional data
subject to change

करोड़ रु. मे (RS. IN CRORES)

क्षेत्र REGION → / सप्ताह WEEK ↓	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
01.07.24 to 07.07.24	115.39	281.91	423.74	245.01	32.89
08.07.24 to 14.07.24	159.15	377.17	541.87	356.11	70.97
15.07.24 to 21.07.24	170.76	334.13	376.77	258.67	70.09
22.07.24 to 28.07.24	120.41	234.76	268.20	170.57	40.00

* Amount shown is Payble to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
POWER MARKET INFORMATION (Source IEX & PXIL)
पावर एक्स्चेंज के माध्यम से विनिमय – माह:- जुलाई 2024
EXCHANGES THROUGH POWER EXCHANGES -JULY 2024

क्र. स. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
1	AD HYDRO POWER LIMITED	उत्तरी क्षेत्र NR	114.39	0.00
2	ADANI SOLAR ENERGY JAISALMER TWO PRIVATE LIMITED(Project-2)		32.59	0.00
3	ADANI SOLAR ENERGY JODHPUR TWO LIMITED		11.25	0.00
4	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED		45.13	0.00
5	ADANI SOLAR ENERGY RJ TWO PRIVATE LIMITED (infirm)		4.02	0.00
6	Adept Renewable Technologies Private Limited		29.67	0.00
7	Amp Energy Green Six Private Limited		26.90	0.00
8	Amplus Ages Private Limited		13.19	0.00
9	Ayana Renewable Power Three Private Limited Infirm		19.45	0.00
10	Budhil HEP (Greenko Budhil Hydro Power Pvt. Ltd.)		9.59	0.00
11	Chandigarh (UT)		16.19	8.74
12	Delhi		369.52	103.97
13	Grian Energy Private Limited		25.24	0.00
14	Haryana		243.75	217.61
15	Himachal Pradesh		281.24	130.41
16	IGSTPS JHAJJAR		8.55	0.00
17	Jammu Kashmir		51.91	0.57
18	KARCHAM WANGTOO HYDRO ELECTRIC PLANT.		124.44	0.00
19	NEA Nepal Upper Chameliya Hydropower Project		26.09	0.00
20	NEA Nepal Upper Kalangagad Hydropower Project new		8.50	0.00
21	NTPC Dadri Stage I		6.78	0.00
22	NTPC Dadri Stage II		8.65	0.00
23	NTPC Rihand stage I		9.27	0.00
24	NTPC Rihand stage II		8.69	0.00
25	NTPC Rihand stage III		10.51	0.00
26	NTPC Singrauli		9.02	0.00
27	NTPC Tanda Stage II		10.04	0.00
28	NTPC Unchahar Stage I		2.93	0.00
29	NTPC Unchahar Stage II		5.02	0.00
30	NTPC Unchahar Stage III		2.91	0.00
31	NTPC Unchahar Stage IV		6.15	0.00
32	North Central Railway Prayagraj		0.00	45.47
33	Onevolt Energy Private Limited		22.11	0.00
34	PARBATI II HE PROJECT		5.39	0.00
35	Punjab		30.37	1433.22
36	RENEW SURYA PRATAP PRIVATE LIMITED		44.41	0.00
37	RENEW SURYA ROSHNI PRIVATE LIMITED-Fatehgarh-III PS		0.58	0.00
38	Rajasthan		104.31	780.11
39	ReNew Surya Ravi Private Limited		45.33	0.00
40	SHREE CEMENT LIMITED TPS		42.79	0.00
41	Sainj HEP		39.65	0.88
42	Serentica Renewables India 4 Private Limited		4.35	0.00
43	Singoli Bhatwari HEP		17.50	0.00
44	Sorang HEP (Himachal Sorang Power Pvt. Ltd.))		47.42	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एक्स. एवं पी.एक्स.आई.एल.)
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			(DAM+HP DAM+RTM)	
45	TPSL 200MW TPTCL Banderwala		23.42	0.00
46	TPSL Banderwala SECI 100_INF		13.63	0.00
47	Transition Energy Services Private Limited		19.68	0.00
48	Transition Green Energy Private Limited		24.74	0.00
49	Uttar Pradesh		373.26	508.57
50	Uttarakhand		48.89	146.35

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			(DAM+HP DAM+RTM)	
51	ACB (INDIA) LIMITED	पश्चिमी क्षेत्र WR	41.84	0.51
52	ADANI GREEN ENERGY TWENTY SIX A LIMITED_infirm		0.49	0.00
53	ADANI RENEWABLE ENERGY FIFTY FIVE LIMITED_Infirm		15.82	0.00
54	ADANI RENEWABLE ENERGY FORTY ONE LIMITED_infirm		31.69	0.00
55	ADANI RENEWABLE ENERGY HOLDING FOUR LIMITED		148.95	0.00
56	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana		94.21	0.00
57	ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhatrana (infirm)		12.50	0.00
58	Adani Green Energy Twenty Five A Limited_Infirm		9.38	0.00
59	Adani Green Energy Twenty Five B Limited_infirm		83.40	0.00
60	Adani Green Energy Twenty Six B Limited_infirm		12.95	0.00
61	Adani Power Limited - Raigarh TPP		85.91	2.70
62	Adani Power Limited-Raipur TPP		120.10	0.00
63	ArcelorMittal Nippon Steel India Limited		0.00	85.34
64	BHARAT ALUMINIUM COMPANY LTD		27.45	0.00
65	BHARAT ALUMINIUM COMPANY LTD (Bulk Consumer)		0.00	7.21
66	CONTINUUM POWER TRADING (TN) PRIVATE LIMITED		7.70	0.00
67	Chhattisgarh		202.94	140.59
68	D B Power Limited		114.88	0.00
69	DGEN MEGA POWER PROJECT		1.90	0.00
70	DHARIWAL STU OTH		1.61	0.00
71	Daman and Diu - Dadra and Nagar Haveli		1.26	278.48
72	Dhariwal ISTS		1.48	0.00
73	GMR Warora Energy Limited		4.70	0.00
74	Goa WR		11.12	9.82
75	Gujarat		129.74	637.89
76	Jaypee Nigrie Super Thermal Power Plant		114.10	0.00
77	Jhabua Power Limited		1.11	1.20
78	Jindal Power Limited, Stage-1		145.45	0.05
79	Jindal Power Limited, Stage-2		271.76	0.00
80	Jindal Steel & Power Ltd , DCPP		73.91	2.70
81	MAHAN ENERGEN LIMITED U#1		157.16	7.78
82	MAHAN ENERGEN LIMITED U#2		61.97	0.00
83	MB POWER (MADHYA PRADESH) LIMITED		10.78	3.58
84	Madhya Pradesh		187.55	114.37
85	Maharashtra		152.98	268.93
86	Maruti Clean Coal and Power Limited		21.28	0.00
87	NREL-Dayapar Wind		10.15	0.00
88	NTPC Gadarwara		4.53	0.00
89	NTPC Jhanor Gandhar GPS		0.00	0.49
90	NTPC Kawas GPS		0.00	0.73
91	NTPC Korba Stage I &II		7.11	0.00
92	NTPC Korba Stage III		0.83	0.00
93	NTPC Lara Stage I		6.14	0.00
94	NTPC Mouda Stage I		3.39	0.00

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			(DAM+HP DAM+RTM)	
95	NTPC Mouda Stage II		7.65	0.00
96	NTPC SAIL POWER COMPANY LIMITED		0.60	0.00
97	NTPC Sipat Stage I		13.88	0.00
98	NTPC Sipat Stage II		6.94	0.00
99	NTPC Solapur		4.06	0.00
100	NTPC Vindhyachal Stage I		5.53	0.00
101	NTPC Vindhyachal Stage II		4.59	0.00
102	NTPC Vindhyachal Stage III		6.99	0.00
103	NTPC Vindhyachal Stage IV		9.11	0.00
104	NTPC Vindhyachal Stage V		5.43	0.00
105	NTPC khargone		2.90	0.00
106	Nani Virani Wind Energy Private Limited		4.42	0.00
107	R.K.M POWERGEN PRIVATE LIMITED		44.45	0.00
108	Ratnagiri Gas & Power Private Limited		0.00	0.60
109	SKS Power Generation Chhattisgarh Limited		81.08	5.32
110	Sasan Power Limited		8.11	0.00
111	TRN ENERGY PRIVATE LIMITED		2.84	0.00
112	The Tata Power Co Ltd (MTPS)		27.20	0.00
113	WIND FIVE RENERGY LIMITED		12.16	0.00

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			(DAM+HP DAM+RTM)	
114	Andhra Pradesh	दक्षिणी क्षेत्र SR	254.49	250.31
115	COASTAL ENERGEN PRIVATE LIMITED		0.11	0.00
116	Goa SR		0.06	7.83
117	IL&FS TAMIL NADU POWER COMPANY LIMITED		5.91	0.00
118	JINDAL POWER LIMITED SIMHAPURI		12.47	8.13
119	JSW RENEW ENERGY LIMITED INFIRM		4.29	0.00
120	JSW RENEW ENERGY TWO LIMITED INFIRM		2.67	0.00
121	Karnataka		1035.17	63.45
122	Kerala		25.82	216.36
123	MEENAKSHI ENERGY LIMITED		14.87	0.00
124	NLC INDIA LIMITED NEYVELI NEW THERMAL POWER STATION		7.13	0.00
125	NLC INDIA LIMITED THERMAL POWER STATION I EXPANSION		5.70	0.00
126	NLC INDIA LIMITED THERMAL POWER STATION II EXPANSION		0.33	0.00
127	NLC INDIA LIMITED THERMAL POWER STATION II STAGE I		0.77	0.00
128	NLC INDIA LIMITED THERMAL POWER STATION II STAGE II		1.08	0.00
129	NLC Tamilnadu Power Limited		6.71	0.00
130	NTECL VALLUR		5.99	0.00
131	NTPC KUDGI		18.10	0.00
132	NTPC Ramagundam Stage I &II		7.02	0.00
133	NTPC Ramagundam Stage III		1.75	0.00
134	NTPC Simhadri Stage I		6.79	0.00
135	NTPC Simhadri Stage II		3.82	0.00
136	NTPC Talcher Super Thermal Power Station Stage II		6.97	0.00
137	NTPC Telangana		10.29	0.00
138	OSTRO KANNADA POWER PRIVATE LIMITED		45.86	0.00
139	Pondicherry UT		19.28	0.58
140	RENEW SURYA OJAS PRIVATE LIMITED-Wind		149.60	0.00
141	RENEW SURYA OJAS PRIVATE LIMITED_INFIRM		13.18	0.00
142	RENEW SURYA ROSHNI PRIVATE LIMITED Koppal PS		16.80	0.00
143	Ramagundam Floating solar		6.11	0.00
144	SEIL ENERGY INDIA LIMITED		79.37	0.00
145	SEIL Energy India Limited Project II		0.57	0.00
146	Serentica Renewables India 1 Private Limited- Koppal (Infirm)		17.58	0.00
147	Simhadri FSP 15 MW		1.68	0.00
148	Simhadri Floating solar (10 MW)		0.58	0.00
149	Tamil Nadu		456.12	354.64
150	Telangana		81.32	2362.32

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			(DAM+HP DAM+RTM)	
151	ADHUNIK POWER & NATURAL RESOURCES LIMITED	पूर्वी क्षेत्र ER	16.50	0.00
152	Basochhu Hydropower Plant Bhutan		39.16	0.00
153	Bihar		242.53	218.55
154	Chilime Hydro Power Project NEPAL ELECTRICITY AUTHORITY		15.86	0.00
155	Chuzachen HEP		0.13	0.00
156	DEVIGHAT NEPAL ELECTRICITY AUTHORITY		10.75	0.00
157	Damodar Valley Corporation		45.20	64.92
158	GMR KAMALANGA ENERGY LTD-CTU		0.20	0.00
159	IND BARATH ENERGY UTKAL LIMITED		34.15	0.00
160	JORETHANG LOOP HEP, DANS ENERGY PRIVATE LIMITED		1.61	0.00
161	Jharkhand		32.45	68.04
162	Jindal India Thermal Power Limited		24.93	0.00
163	KALI GANDAKI NEPAL ELECTRICITY AUTHORITY		88.76	0.00
164	KANTI BIJLEE UTPADAN NIGAM LIMITED		21.65	0.00
165	Kabeli B-1 Hydro Power Project NEPAL ELECTRICITY AUTHORITY		17.32	0.00
166	LIKHU-iV NEA		37.74	0.00
167	Lower Modi Hydro Power Project NEPAL ELECTRICITY AUTHORITY		14.04	0.00
168	MAITHON POWER LIMITED		2.47	0.00
169	MARSYANGDI NEPAL ELECTRICITY AUTHORITY		49.71	0.00
170	NABINAGAR POWER GENERATING COMPANY LIMITED		78.73	0.00
171	NEA Nepal Upper Balephi A Hydropower Project1		18.50	0.00
172	NEA Nepal Upper Dordi A Hydropower Project		17.56	0.00
173	NEPAL ELECTRICITY AUTHORITY-MIDDLE MARSYANGDI		49.78	0.00
174	NTPC BARH Stage I		10.89	0.00
175	NTPC BARH Stage II		23.02	0.00
176	NTPC Darlipali		3.07	0.00
177	NTPC Farakka stage I		54.77	0.00
178	NTPC Farakka stage III		20.02	0.00
179	NTPC Kahalgaon stage I		10.63	0.00
180	NTPC Kahalgaon stage II		21.76	0.00
181	NTPC North Karanpura STPS		24.61	0.00
182	NTPC Talcher Stage I		37.60	0.00
183	NTPP BRBCL		10.69	0.00
184	Nikachhu Hydropower Project		21.45	0.00
185	Odisha		345.57	148.23
186	Sikkim		16.13	11.60
187	Solu Hydropower Project NEPAL ELECTRICITY AUTHORITY		16.90	0.00
188	TRISHULI NEPAL ELECTRICITY AUTHORITY		17.20	0.00
189	Tashiding HEP, Shiga Energy Private Limited		0.73	0.00
190	West Bengal		73.67	543.02

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191	AGARTALA GAS BASED POWER STATION	पूर्वोत्तर क्षेत्र NER	2.09	0.00
192	ASSAM GAS BASED POWER STATION		2.64	0.00
193	Arunachal Pradesh		46.88	0.02
194	Assam		223.64	83.67
195	Bongaigaon Thermal Power Station NTPC		13.06	0.00
196	KAMENG HYDRO POWER STATION		0.03	0.00
197	Kopili Power Station NEEPCO		0.02	0.00
198	Manipur		10.85	6.34
199	Meghalaya		47.35	5.22
200	Mizoram		25.68	0.00
201	Nagaland		6.57	0.00
202	Palatana Plant		8.19	0.00
203	Tripura		6.50	20.50
Total			9378	9378

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JULY 2024

RE Source & Unit wise break up (01.07.2024-31.07.2024)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	6	2	12	1
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	63	4	9	2
5	Small Hydro	0	0	0	0
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**69**	**6**	**21**	**3**

RECs Issued (01.07.2024-31.07.2024)

Sr.No.	Non Solar	Solar	Total
1	492563	59129	551692

Redemption of REC (01.07.2024-31.07.2024)

Sr.No.	Non Solar	Solar	Total
1	5273724	56342	5330066

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JULY 2024

RE Source & Unit wise break up (Apr'24-Jul'24)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	19	4	14	3
2	Urban or Municipal Waste	0	0	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	91	13	17	6
5	Small Hydro	24	1	24	1
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	**134**	**18**	**67**	**11**

RECs Issued (Apr'24-Jul'24)

Sr.No.	Non Solar	Solar	Total
1	8501315	260621	8761936

Redemption of REC (Apr'24-Jul'24)

Sr.No.	Non Solar	Solar	Total
1	8673871	130630	8804501

19. INFORMATION ABOUT RENEWABLE ENERGY CERTIFICATE MECHANISM

MONTH : JULY 2024

RE Source & Unit wise break up Since Inception to Jul'24

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3152	534	2991	534
2	Urban or Municipal Waste	12	1	12	1
3	Solar Thermal	0	0	0	0
4	Solar PV	1727	587	1489	486
5	Small Hydro	869	43	881	45
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	823	91	383	55
	Total	**6989**	**1295**	**6137**	**1157**

RECs Issued since Inception to Jul'24

Sr.No.	Non Solar	Solar	Total
1	123606353	13800999	137407352

Redemption of REC since Inception to Jul'24

Sr.No.	Non Solar	Solar	Total
1	87178200	12579887	99758087

REC Closing balance as on 31.07.2024

Sr.No.	Non Solar	Solar	Total
1	36428153	1221112	37649265

20.Details of Grid Events during the Month of July 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Uttar Pradesh	01-07-2024 21:37	01-07-2024 23:44	02:07	0	80	0.000	0.102	55304	78768	i) 220kV Baghpat(UP) has main and transfer bus scheme at 220kV level. ii) During antecedent condition, incoming power at Baghpat(UP) was approx. 80 MW through 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C, 220/132kV 160MVA ICT-1 and 220/132kV 100MVA ICT-2 were connected to 220kV main bus. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C is on the same towers. iii) As per SCADA SOE, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped at 21:37:59.415hrs and 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 tripped at 21:37:59.535 hrs. iv) As reported, at 21:37 hrs, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped on Y-B-N double phase to earth fault and line tripped on zone-1 distance protection operation from both ends. As per Baghpat(UP) reporting, initially a B-ph fault occurred on 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 with fault current - 10.39kA in zone-1. At the same time, jumper at tower no 45 of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 got broken which created line-line (Y-B) fault due to which line got tripped from both ends. v) Further 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 also tripped on B-N phase to earth fault with fault current of ~9.7kA and fault distance of 11.8km from Baghpat(PG) end. As per DR of Baghpat(PG) end, B-N phase to earth fault with no A/R operation is observed and line tripped on zone-1 distance protection operation. Since there was no source remaining at 220kV Baghpat(UP) hence 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 didn't trip from Baghpat(UP) end. vi) As per PMU at Meerut(PG), Y-B phase to phase fault with fault clearance time of 80ms is observed. vii) Due to tripping of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2, Baghpat(UP) lost its connectivity from the grid and 220kV Baghpat(UP) S/s became dead. viii) As per SCADA, change in demand of approx. 68 MW in UP control area. However, SLDC-UP reported 80MW load loss.	1) 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 2) 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2
2	GD-1	Uttar Pradesh	01-07-2024 00:15	01-07-2024 00:28	00:13	0	195	0.000	0.267	53003	72935	i) 220kV Chinhat(UP) has main and transfer bus scheme at 220kV level. ii) During antecedent condition, incoming power at Chinhat(UP) was through 220kV Satrikh ckt (~100MW), Kursi Road ckt (~30MW) and Lucknow(PG) ckt (~80MW) and outgoing power was through 220kV Gomatinagar ckt (~30MW) and load at Chinhat(UP) S/s (~90MW). All 220kV lines and ICTs connected to 220kV main bus at Chinhat(UP) S/s. 220kV Chinhat-LMRC D/C is radial line from Chinhat(UP) S/s. iii) As reported, at 00:15 hrs, LA of 220 KV Chinhat-Satrikh Road (UP) Ckt bay burst at Chinhat(UP) S/s which caused R-N phase to earth fault. iv) On this fault 220kV lines from Chinhat(UP) to Satrikh Road (UP), Gomtinagar (UP), Kursi Road (UP) & Lucknow_1(PG) tripped (Reason of tripping and type of protection operated for all elements yet to receive). v) Due to these trippings at Chinhat(UP) S/s, 220kV Chinhat-LMRC D/C, 220/132kV ICT-1 & 2 became dead and blackout occurred at 220kV Chinhat(UP) S/s. vi) As per PMU at Lucknow(PG), R-N phase to earth fault with delayed fault clearance of 440msec is observed (reason for delayed fault clearance yet to receive). vii) As per SCADA, change in demand of approx. 195 MW in UP control area.	1) 220 KV Chinhat-Satrikh Road (UP) Ckt 2) 220 KV Chinhat-Gomtinagar (UP) Ckt 3) 220 KV Chinhat-Kursi Road (UP) Ckt 4) 220 KV Chinhat(UP)-Lucknow_1(PG) (UP) Ckt
3	GD-1	Delhi	04-07-2024 14:21	04-07-2024 15:00	00:39	0	106	0.000	0.160	59455	66088	i) 220kV Vasant Kunj(DTL) has double main Bus arrangement at 220kV side. ii) During antecedent condition, 220 KV Vasant Kunj-RK Puram(DTL) Ckt-1, 220 KV Vasant Kunj-Mehrauli(DTL) Ckt-1, 220/66kV 100MVA ICT-1 & 2 were connected to 220kV Bus-1 and 220 KV Vasant Kunj-RK Puram(DTL) Ckt-2 & 220/66kV 160MVA ICT-3 connected to 220kV Bus-2 at Vasant Kunj(DTL) S/s. 220 KV Vasant Kunj-Mehrauli(DTL) Ckt-1 & 2 were not in service (ckt-2 opened from Mehrauli end). iii) As reported, at 14:21 hrs, R-phase jumper of 220kV bus coupler connected to 220kV bus-2 got damaged which cause bus fault on both 220kV buses which led to bus bar protection operation on both 220kV buses at Vasant Kunj(DTL) S/s. iv) As per PMU at Dadri Thermal(NTPC), R-B phase to phase fault with fault clearing time of 80ms is observed. v) Due to bus bar protection operation at Vasant Kunj(DTL), all elements connected to 220kV bus-1 & 2 got tripped and blackout of 220kV Vasant Kunj(DTL) S/s occurred. vi) As per SCADA, change in demand of approx. 153 MW in Delhi control area (as per SCADA). However, 106 MW load loss is reported by SLDC-Delhi.	1) 220 KV Vasant Kunj-RK Puram(DTL) Ckt-1 2) 220 KV Vasant Kunj-RK Puram(DTL) Ckt-2 3) 220 KV Vasant Kunj-Mehrauli(DTL) Ckt-1 4) 220/66kV 100MVA ICT-1 at Vasant Kunj(DTL) 5) 220/66kV 100MVA ICT-2 at Vasant Kunj(DTL) 6) 220/66kV 160MVA ICT-3 at Vasant Kunj(DTL)
4	GI-1	Rajasthan	06-07-2024 05:26	06-07-2024 06:30	01:04	650	0	1.362	0.000	47708	54877	i) 400/220kV Akal(RS) has one and half breaker scheme at 400kV level and double main and transfer bus scheme at 220kV level. ii) During antecedent condition, incoming power at Akal(RS) S/s through 220 KV Akal-Akal(Suzlon) (RS) D/C and 220 KV Akal-Mulana (RS) Ckt were approx. 235 MW and 125 MW respectively. iii) As reported, at 05:26 hrs, R-phase conductor of 220 KV Akal-Akal(Suzlon) (RS) ckt-2 broke at a distance of approx. 160m from Akal(RS) S/s which caused R-N phase to earth fault and subsequently 220 KV Akal-Akal(Suzlon) (RS) ckt-2 tripped on zone-1 distance protection from Akal(RS) end. iv) As per PMU at ASPS1(IP), R-Y phase to phase fault followed by R-N phase to earth fault with fault clearance time of 80msec and 80msec respectively are observed. v) At the same time, 220 KV Akal-Akal(Suzlon) (RS) Ckt-1 and 220 KV Akal-Mulana (RS) Ckt also tripped from Akal(RS) end (Reason of tripping yet to be received). vi) During this event, dip in Rajasthan wind generation of approx. 1800 MW is observed out of which approx. 1150 MW recovered within 10 minutes. (As per SCADA). vii) As per SCADA, no change in demand is observed in Rajasthan control area. viii) As per SCADA, change in Rajasthan wind generation of approx. 168MW is observed.	1) 220 KV Akal-Akal(Suzlon) (RS) Ckt-2 2) 220 KV Akal-Akal(Suzlon) (RS) Ckt-1 3) 220 KV Akal-Mulana (RS) Ckt
5	GI-2	Uttar Pradesh	07-07-2024 11:44	07-07-2024 14:01	02:17	0	60	0.000	0.097	52587	61926	i) 220kV Mau(UP) has double main and transfer bus scheme at 400kV level. ii) During antecedent condition, 400 KV Azamgarh-Mau (UP) Ckt, 400 KV Mau(UP)-Balia(PG) (PG) Ckt & 400/132 kV 200 MVA ICT-3 connected to 400kV bus-1 and 400kV Mau-Rasra (UP) ckt, 400/132/33kV 200MVA ICT-1 & 2 connected to 400kV bus-2. 400 KV Anpara_B(UPUN)-Mau(UP) (UP) Ckt was not in service during the event. iii) As reported, at 11:44 hrs, B-phase CT of 400 KV Azamgarh-Mau (UP) Ckt burst which caused bus fault on 400kV bus-1 which led to bus bar protection operation on 400kV bus-1 at Mau(UP) S/s (Reason for delayed operation of bus bar protection yet to be received). iv) As per PMU at Azamgarh(UP), B-N phase to earth fault converted into Y-B phase to phase fault with delayed fault clearance time of 560ms is observed (Reason for delayed fault clearance is yet to receive). v) Due to bus bar protection operation, all elements connected to 400kV bus-1 (400kV Azamgarh(UP) ckt, Balia(PG) ckt and 400/132 kV 200 MVA ICT-3) tripped at 400kV Mau(UP) S/s. vi) As per SCADA, change in demand of approx. 60 MW in UP control area.	1) 400 KV Azamgarh-Mau (UP) Ckt 2) 400 KV Mau(UP)-Balia(PG) (PG) Ckt 3) 400/132 kV 200 MVA ICT 3 at Mau(UP)

20.Details of Grid Events during the Month of July 2024 in Northern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
6	GI-2	Uttar Pradesh	10-07-2024 15:54	10-07-2024 17:53	01:59	0	100	0.000	0.137	60415	72949	i)400/220kV Sahupuri(UP) has double main double scheme at 400kV and 220kV level. ii)During antecedent condition at 15:52 hrs, 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-2, 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-1 and 400/220 kV 500 MVA ICT-2 were connected to 400kV bus-1 and 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 and 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-2 were connected to 400kV bus-2 at 400kV Sahupuri(UP) S/s. 400/220 kV 500 MVA ICT-1 at Sahupuri(UP) is under installation (commissioning) process. iii)As reported, at 15:54 hrs, Y-N phase to earth fault occurred in GIS compartment at 400kV Sahupuri(UP) (exact location of fault is yet to be received). It is suspected that fault location was in the bay of 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-2 in GIS compartment at 400kV Sahupuri(UP). iv)On this fault, 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-1 & 2 (fault current Iy=~1.1kA from Sahupuri end) tripped only from Sahupuri(UP) end on zone-4 distance protection. 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 & 2 (fault current Iy=~3.3kA from Varanasi end, Iy=~3.6kA from Sahupuri end) tripped only from Varanasi(PG) end on zone-2 distance protection and fault sensed in zone-4 from Sahupuri(UP) end. 400/220 kV 500 MVA ICT 2 at Sahupuri(UP) and also tripped (details of protection operation is yet to receive). v)As reported, from 15:27 hrs to 17:49 hrs, multiple 220 & 132kV line also tripped at 220kV Sahupuri S/s i.e. 220kV Sahupuri-Churk ckt, 220kV Sahupuri-Raja Talab ckt, 220kV Sahupuri(400kV)-Sahupuri(200kV) interconnector, 132kV Sahupuri-Aliapur ckt, 132kV Sahupuri-Karamnasha ckt, 132kV Sahupuri-Chandauli ckt, 132kV Sahupuri-Sadat ckt and 132kV Sahupuri-Dhanapur ckt. Reason of tripping of these lines are yet to be received. vi)As per PMU at Varanasi(PG), at 15:54 hrs, Y-N phase to earth fault with delayed fault clearance time of 400msec is observed (Reason for delayed fault clearance is yet to receive). vii)As per SCADA, at 15:54 hrs, change in demand of approx. 100 MW in UP control area. viii)By 18:29 hrs, 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) D/C and 400/220 kV 500 MVA ICT-2 were charged.	1)400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 2)400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-2 3)400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-1 4)400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-2 5)400/220 kV 500 MVA ICT 2 at Sahupuri(UP) 6)132 KV Sahupuri(UP)-Karamnasa(BS) (UP) Ckt-1
7	GI-2	Uttar Pradesh	10-07-2024 18:37	10-07-2024 22:59	04:22	0	60	0.000	0.085	53214	70406	i)400/220kV Sahupuri(UP) has double main double scheme at 400kV and 220kV level. ii)During antecedent condition at 18:35 hrs, 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-1 and 400/220 kV 500 MVA ICT-2 were connected to 400kV bus-1 and 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-2 were connected to 400kV bus-2 at 400kV Sahupuri(UP) S/s. 400/220 kV 500 MVA ICT-1 at Sahupuri(UP) is under installation (commissioning) process. 400 KV Varanasi(PG)-Sahupuri(UP) (PG) D/C was not in service. iii)As reported at 18:37 hrs, Y-N phase to earth fault occurred in GIS compartment at 400kV Sahupuri(UP) (exact location of fault is yet to received). iv)On this fault, 400KV Sahupuri(UP)-Biharshariff(PG) (PG) D/C tripped (fault current Iy=~ 1.3kA from Sahupuri end) only from Sahupuri(UP) end on zone-4 distance protection. 400/220kV 500 MVA ICT-2 at Sahupuri(UP) also tripped (details of protection operation is yet to receive). v)As per PMU at Varanasi(PG), at 18:37 hrs, Y-N phase to earth fault with delayed fault clearance time of 240msec is observed (Reason for delayed fault clearance is yet to receive). vi)As per SCADA, at 18:37 hrs, change in demand of approx. 60 MW in UP control area. vii)At 18:56 hrs 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 was charged at Sahupuri(UP) S/s. Charging attempt of 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-2 was not taken because it was suspected that fault at 15:54 hrs was in the bay of this line.	1)400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-1 2)400 KV Sahupuri(UP)-Biharshariff(PG) (PG) Ckt-2 3)400/220 kV 500 MVA ICT 2 at Sahupuri(UP)
8	GI-2	Uttar Pradesh	10-07-2024 19:25	10-07-2024 22:57	03:32	0	0	0.000	0.000	55759	72471	i)400/220kV Sahupuri(UP) has double main double scheme at 400kV and 220kV level. ii)During antecedent condition at 19:23 hrs, 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 was connected to 400kV bus-2 at 400kV Sahupuri(UP) S/s. 400/220 kV 500 MVA ICT-1 at Sahupuri(UP) is under installation (commissioning) process. 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) D/C and 400/220 kV 500 MVA ICT-2 were not in service. iii)As reported, at 19:25 hrs, Y-N phase to earth fault occurred in GIS compartment at 400kV Sahupuri(UP) (exact location of fault is yet to receive). iv)On this fault 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 tripped (fault current Iy=~ 4.7kA from Varanasi end) only from Varanasi end on zone-2 distance protection. v)As per PMU at Varanasi(PG), at 19:25 hrs, Y-N phase to earth fault with delayed fault clearance time of 400msec is observed (Reason for delayed fault clearance is yet to receive). vi)As per SCADA, at 19:25 hrs, no change in demand in UP control area. vii)During these three tripping events, 400/220kV Sahupuri(UP) S/s connected with grid through 220kV Sahupuri-Bhelpur (UP) ckt and interconnector of 220kV Sahupuri(UP) S/s. viii)Restoration of 400 KV Sahupuri(UP)-Biharshariff(PG) (PG) D/C, 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1 and 400/220 kV 500 MVA ICT-2 at Sahupuri(UP) started from 22:57 hrs by taking all these elements on 400kV bus-1 at Sahupuri(UP) and 400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-2 is still not charged.	1)400 KV Varanasi(PG)-Sahupuri(UP) (PG) Ckt-1
9	GI-2	Delhi	12-07-2024 11:55	12-07-2024 12:32	00:37	0	0	0.000	0.000	65549	73566	i)765/400kV Jhatikara(PG) has one and half bus arrangement at 400kV side. ii)During antecedent condition, 400 KV Jhatikara-Dwarka (PG) ckt was not in service and due to high loading of 400 KV Bamnoli(DV)-Jhatikara(PG) (DTL) Ckt (~1570MW at 10:15 hrs), 400 KV Jhatikara-Dwarka (PG) ckt taken into service at 10:16 hrs. iii)At 10:20 hrs, 765/400 kV 1500 MVA ICT-1 & 2 were taken into emergency outage due hotspot/sparking in their isolators by opening their tie breakers. This led to reversal of power flow in 400 KV Bamnoli(DV)-Jhatikara(PG) (DTL) Ckt (i.e. Bamnoli to Jhatikara) of approx. 145 MW which was evacuating through 400 KV Jhatikara-Dwarka (PG) ckt. iv)At 11:37 hrs, 400 KV Jhatikara(PG)-Mundka(DV) (DTL) Ckt-1 (~1075MW) hand tripped (emergency shutdown) from Mundka(DV) end due to hotspot on the bay of same line at Mundka(DV) end. v)Due to tripping of 400 KV Jhatikara(PG)-Mundka(DV) (DTL) Ckt-1, the load of ckt-1 shifted on 400 KV Jhatikara(PG)-Mundka(DV) (DTL) Ckt-2 and due to overloading, 400 KV Jhatikara(PG)-Mundka(DV) (DTL) Ckt-2 (~1925 MW) also tripped at 11:55 hrs. vi)As per PMU at Jhatikara(PG), fluctuation in voltage and no fault in system is observed. vii)As per SCADA, no change in demand of Delhi control area.	1)765/400 kV 1500 MVA ICT 1 at Jhatikara(PG) 2)765/400 kV 1500 MVA ICT 2 at Jhatikara(PG) 3)400 KV Jhatikara(PG)-Mundka(DV) (DTL) Ckt-1 4)400 KV Jhatikara(PG)-Mundka(DV) (PG) Ckt-2
10	GI-2	Rajasthan	13-07-2024 04:33	13-07-2024 07:13	02:40	0	0	0.000	0.000	50388	66264	i)765/400/220kV Fatehgarh-2(PG) has one and half bus arrangement at 765kV side. ii)During antecedent condition, power flow from Fatehgarh-2 to Bhadla-2 through 765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PBTL) Ckt-3 & 765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PFTL) Ckt-1 were 17 MW and 16 MW respectively. iii)At 04:33:41.642 hrs, 765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PBTL) Ckt-3 tripped on R-N phase to earth fault (as reported). iv)As per PMU at Fatehgarh-2(PG), R-N phase to earth fault with fault clearing time of 120msec is observed. v)At 04:33:47.041 hrs, 765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PFTL) Ckt-1 tripped on overvoltage protection. As per DR, voltage shoot up to 810kV (~1.06 pu) on 765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PFTL) Ckt-1. vi)As per SCADA, no change in demand of Rajasthan control area and no change in RE generation.	1)765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PBTL) Ckt-3 2)765 KV Bhadla_2 (PG)-Fatehgarh_II(PG) (PFTL) Ckt-1

20.Details of Grid Events during the Month of July 2024 in Northern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
11	GI-2	Uttar Pradesh	14-07-2024 15:53	14-07-2024 16:08	00:15	0	250	0.000	0.342	56035	73115	i) 220kV Lucknow(UP) has double main and transfer bus scheme at 220kV level. ii) During antecedent condition, 400/220kV 500 MVA ICT-1 & 2, 220/132kV 200 MVA ICT-1 & 2, 220kV Lucknow-Hardoi Road (UP) ckt & 220kV Lucknow-Unnao (UP) ckt were connected to 220kV bus-1 and 220kV lines from Lucknow(UP) to Bachrawan, Gomatinagar, Kanpur Road & 220/132kV 200MVA ICT-1 & 2 connected to 220kV bus-2 at 220kV Lucknow(UP) S/s. 220kV Lucknow-Kanpur Road (UP) ckt was not in service during the event. iii) As reported, at 15:53 hrs, R-N phase to earth fault occurred on 220kV bus-1 which led to tripping of all elements connected to 220kV bus-1 at 220kV Lucknow(UP). Bus bar protection failed to operate and 400/220 kV 500 MVA ICT-1 & 2 tripped on LBB protection (Type of protection operated in tripping of other elements is yet to receive) iv) As per PMU at Lucknow(PG), R-N phase to earth fault with delayed fault clearance time of 880ms is observed (Reason for delayed fault clearance is yet to receive). v) As per SCADA, change in demand of approx. 280 MW in UP control area. However, approx. 250 MW load loss in UP control area as per SLDC-UP.	1) 400/220 kV 500 MVA ICT 1 at Lucknow(UP) 2) 400/220 kV 500 MVA ICT 2 at Lucknow(UP) 3) 220kV Lucknow-Hardoi Road (UP) ckt 4) 220kV Lucknow-Unnao (UP) ckt 5) 220/132 kV 200 MVA ICT 1 at Lucknow(UP) 6) 220/132 kV 200 MVA ICT 2 at Lucknow(UP)
12	GD-1	Haryana and Delhi	16-07-2024 22:10	17-07-2024 00:30	02:20	0	1580	0.000	1.956	56799	80778	i) During antecedent condition, 220 kV Palli S/S importing load from 220 KV Samaypur (BB)-Palli (HV) (HVPNL) Ckt-1 & Ckt-2, 220 KV Badshahpur (HV)-Palli (HV) (HVPNL) Ckt-1 & ckt-2 and 220 KV Sector-56 (Gurgaon) -Palli (HV) (HVPNL) Ckt-1 & Ckt-2 and feeding that load to 220 KV Palla (HV) (Sec-46) & 220 KV Palli (2*100MVA+1*160MVA) S/S. ii) As reported, to manage the line loading on sector-72 Gurgaon ckt, 220 KV Sector 52 (HV) (Sec-56 Gurgaon)-Palli (HV) (HVPNL) Ckt-1 was opened at 22:10 hrs on the instruction of SLDC-Haryana. This led to sparking on the 220 KV Sector 52 (HV) (Sec-56 Gurgaon)-Palli (HV) (HVPNL) Ckt-2 at Palli S/S end. iii) At the same time, busbar protection operated at 220kV Palli(HV) due to which all the elements connected to 220kV Bus-1 and 2 at Palli(HV) tripped and complete blackout occurred at Palli(HV) S/s. iv) As per PMU, R-Y phase to phase fault with delayed fault clearing time of 880 ms was observed. v) As per SCADA, change in demand of approx. 600 MW and 980 MW in Delhi and Haryana control area respectively were observed. However, as reported, approx. 400 MW load loss occurred at Palli & Sec-46 (Faridabad). Rest of the change in demand is suspected due to stalling of induction motor.	1) 220 KV Samaypur (BB)-Palli (HV) (HVPNL) Ckt-1 2) 220 KV Samaypur (BB)-Palli (HV) (HVPNL) Ckt-2 3) 220 KV Badshahpur (HV)-Palli (HV) (HVPNL) Ckt-1 4) 220 KV Badshahpur (HV)-Palli (HV) (HVPNL) Ckt-2 5) 220 KV Palla (HV) (Sec-46) -Palli (HV) (HVPNL) Ckt-1 6) 220 KV Palla (HV) (Sec-46) -Palli (HV) (HVPNL) Ckt-2 7) 220 KV Sector 52 (HV) (Sec-56 Gurgoan)-Palli (HV) (HVPNL) Ckt-1 8) 220 KV Sector 52 (HV) (Sec-56 Gurgoan)-Palli (HV) (HVPNL) Ckt-2
13	GI-1	Jammu and Kashmir	18-07-2024 11:01	18-07-2024 12:51	01:50	0	210	0.000	0.257	69460	81592	i) 220/132kV Ziankote S/s have two bus at 220kV side i.e., main bus & reserve bus. 220kV Amargarh-Ziankote ckt-1&2 are on the same tower (D/C tower) and line length is ~21.4km. ii) During antecedent condition, 220kV Amargarh(INDIGRID) –Ziankote(JK) D/C was carrying 109 MW each and feeding Ziankote load. iii) As reported, at 11:01 hrs, 220 KV Amargarh(INDIGRID)-Ziankote(JK) (PDD JK) D/C tripped from both ends on B-N phase to earth fault. During patrolling it was found that fault occurred due to vegetation fire in bottom of the line. iv) As per DR of both the circuits of Amargarh end, Amargarh end distance protection relay sensed B-N fault in Z-1 (15.66km) in line-1 and in Z-2(21.69km) in line-2. Both the lines tripped instantaneously from Amargarh end. Fault current was ~4kA. v) As confirmed by Amargarh(INDIGRID), in view of non-availability of carrier communication and A/R scheme at Ziankote end, A/R has been kept disabled at Amargarh end and time delay of Z-2 also kept as instantaneous at Amargarh end. vi) As per PMU at Amargarh(PG), B-N phase to earth fault which cleared within 120 msec is observed. vii) As per SCADA, change in demand of approx. 210MW is observed in J&K control area.	1) 220 KV Amargarh (INDIGRID)-Ziankote(JK) (PDD JK) Ckt-1 2) 220 KV Amargarh (INDIGRID)-Ziankote(JK) (PDD JK) Ckt-2
14	GD-1	Uttarakhand	19-07-2024 21:31	19-07-2024 22:03	00:32	300	30	0.526	0.037	57033	80484	i) During antecedent condition, all the four 30MW units of Khodri and 60 MW units of Chhibro were running and total active power generation of Khodri and Chhibro was approx. 89 MW and 196 MW (as per SCADA). Total generation of Chhibro was evacuating through 220 KV Khodri-Chhibro (UK) Ckt-1 & 2. ii) As reported, at 21:31 hrs, while taking out 30MW Khodri Unit-2, B-phase pole of CB of Unit-2 did not open. This led to LBB protection operation which further resulted in tripping of all the elements connected to both the buses at 220kV Khodri(UK) and complete blackout occurred at 220kV Khodri(UK) S/s. iii) Due to tripping of 220 KV Khodri-Chhibro (UK) Ckt-1 & 2, 60 MW Chhibro Unit-1, 2, 3 & 4 also tripped due to loss of evacuation path and complete blackout occurred at 220kV Chhibro(UK) S/s. iv) As per PMU, no fault was observed in the system. v) As per SCADA, change in demand and generation of approx. 30 MW and 300 MW respectively in Uttarakhand control area were observed. vi) As remedial action taken, over hauling & testing of generator CB has been performed and found satisfactory.	1) 220 KV Khodri(UK)-Majri(HP) (UK) Ckt-1 2) 220 KV Khodri(UK)-Majri(HP) (UK) Ckt-2 3) 220 KV Khodri(UK)-Sarsawan(UP) (UP) Ckt 4) 220 KV Khodri(UK)-Saharanpur(UP) (UP) Ckt 5) 220 KV Khodri-Chhibro (UK) Ckt-1 6) 220 KV Khodri-Chhibro (UK) Ckt-2 7) 30 MW Khodri Unit-1, 2, 3 & 4 8) 60 MW Chhibro Unit-1, 2, 3 & 4
15	GI-2	Punjab	19-07-2024 18:50	19-07-2024 20:27	01:37	0	245	0.000	0.322	57860	76105	i) 400/220kV Patiala(PG) has one and half bus scheme at 400kV level and double main & transfer bus scheme at 220kV level. ii) During antecedent condition, 400/220kV 315 MVA ICT-1 & 500 MVA ICT-3, 220kV Bahadurgarh-I, Nabha-I, Ablowal-I were connected at 220kV Bus-1 and 400/220kV 315 MVA ICT-2 & 500 MVA ICT-4, 220kV Bahadurgarh-II, Nabha-II, Ablowal-II were connected at 220kV Bus-2. 400/220kV ICT-1,2,3 & 4 were carrying approx. 156MW, 153MW, 243MW & 238 MW respectively. 220kV D/C to Nabha, Bahadurgarh & Ablowal were carrying approx. 171MW, 98MW & 127MW respectively per circuit. iii) As reported at 18:50 hrs, B-N phase to earth fault occurred on 220 KV Patiala(PG)-Nabha(PS) (PSTCL) Ckt-1. Fault location was ~7.3km from Nabha end. Distance protection at Patiala end sensed fault in Z-2 and initiated tripping command however, breaker at Patiala end failed to open. This further led to the operation of LBB protection of Nabha-I bay at Patiala(PG). iv) On the result of LBB protection operation, 400/220kV 315 MVA ICT-1, Ablowal-I, bus coupler tripped however, 400/220kV 500 MVA ICT-3 & 220kV Bahadurgarh-I didn't trip. v) Further, 400/220kV 500 MVA ICT-3 tripped on over current earth fault protection operation and 220kV Bahadurgarh-I tripped from Bahadurgarh end only. vi) Further, at the same time, Nabha-II, Ablowal-II also tripped due to overloading. vii) At 18:50:33 hrs, 220 KV Bahadurgarh(PS)-Patiala(PG) (PSTCL) Ckt-2 tripped on another B-N fault. As reported, fault occurred due to conductor snapping at distance ~1.5km from Bahadurgarh end. viii) As per PMU at Patiala(PG), B-N phase to earth fault at 18:50:15 hrs & 18:50:33 hrs with fault clearance time of 2400 msec at 18:50:15 hrs and 120 msec at 18:50:33 hrs is observed. ix) As per SCADA, change in demand of approx. 245MW is observed in Punjab control area. x) As reported by POWERGRID(NR-2), CB operating mechanism problem of 220kV Nabha-I line has been rectified and reason of non-tripping of 400/220kV 500 MVA ICT-3 & 220kV Bahadurgarh-I at Patiala(PG) on LBB operation is under investigation.	1) 220KV Bus 1 at Patiala(PG) 2) 400/220 kV 315 MVA ICT 1 at Patiala(PG) 3) 400/220 kV 315 MVA ICT 3 at Patiala(PG) 4) 220 KV Bahadurgarh(PS)-Patiala(PG) (PSTCL) Ckt-1 5) 220 KV Patiala(PG)-Ablowal(PS) (PSTCL) Ckt-1 6) 220 KV Patiala(PG)-Nabha(PS) (PSTCL) Ckt-1 7) 220 KV Bahadurgarh(PS)-Patiala(PG) (PSTCL) Ckt-2 8) 220 KV Patiala(PG)-Ablowal(PS) (PSTCL) Ckt-2 9) 220 KV Patiala(PG)-Nabha(PS) (PSTCL) Ckt-2

20.Detailds of Grid Events during the Month of July 2024 in Northern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
16	GI-1	Delhi	20-07-2024 10:46	20-07-2024 11:02	00:16	0	227	0.000	0.283	68287	80318	i)During antecedent condition, 220/66 kV 160 MVA ICT-4 at Mehrauli(DTL) was under shutdown (as informed by SLDC Delhi). ii)As reported, at 10:45 hrs, 66 kV incomers of 200/66 kV 100 MVA ICT-1, 2 & 3 at Mehrauli(DTL) tripped on over-current protection operation (exact reason, location and nature of fault yet to be shared). iii)As per PMU, B-N phase to earth fault with delayed fault clearing time of 400 ms followed by Y-N phase to earth fault with fault clearing time of 120 ms was observed. iv)As per SCADA, change in demand of approx. 290 MW was observed in Delhi. However, as reported by SLDC-Delhi, load loss of approx. 227 MW occurred in Delhi.	1) 66 kV incomer of 200/66 kV 100 MVA ICT-1 at Mehrauli(DTL) 2) 66 kV incomer of 200/66 kV 100 MVA ICT-2 at Mehrauli(DTL) 3) 66 kV incomer of 200/66 kV 100 MVA ICT-3 at Mehrauli(DTL)
17	GI-2	Delhi	28-07-2024 18:24	28-07-2024 18:46	00:22	0	95	0.000	0.134	53018	70818	i)400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) D/C and 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) D/C are on same towers. ii)During antecedent condition, incoming power at Maharani Bagh(PG) through 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) D/C and 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) Ckt-1 was approx. 295 MW and 292 MW respectively (as per SCADA). iii)As reported, at 18:24 hrs, 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) Ckt-1 and 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) Ckt-2 tripped on Y-B phase to phase fault and at the same time 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) Ckt-1 also tripped from Mandaula(PG) end (reason of tripping is yet to be received). iv)During patrolling of Ckts, it was found that Y-B phase to phase fault occurred on 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) Ckt-1 and 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) Ckt-2 due to kite thread. v)As per PMU at Maharani Bagh(PG), Y-B followed by Y-B phase to phase fault with fault clearing time of 120msec & 120msec is observed. vi)As per DR of Bawana(DTL) end of 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) Ckt-1, Y-B phase to phase fault (Iy=~9.8kA & Ib=~10.3kA) sensed in zone-2 with carrier signal received is observed. Fault distance was 39.58km from Bawana(DTL) end (as reported). vii)As per SCADA, change in demand of approx. 95 MW in Delhi control area.	1) 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) Ckt-1 2) 400 KV Mandaula(PG)-Maharani Bagh(PG) (DTL) Ckt-2 3) 400 KV Bawana(DTL)-Maharani Bagh(PG) (DTL) Ckt-1
18	GI-1	Delhi	28-07-2024 21:53	28-07-2024 22:14	00:21	0	110	0.000	0.136	56168	81095	i)During antecedent condition, 200/66 kV 100 MVA ICT-1 & 3 at Mehrauli(DTL) were connected to 220kV Bus-1 at Mehrauli(DTL) and 220/66 kV 100MVA ICT-3 and 160 MVA ICT-4 at Mehrauli(DTL) were connected to 220kV Bus-2 at Mehrauli(DTL). 220kV Bus coupler was in ON position whereas 66kV Bus coupler was in OFF position (as informed by SLDC Delhi). ii)As reported, at 21:53 hrs, 200/66 kV 100 MVA ICT-1 at Mehrauli(DTL) tripped on Restricted earth fault protection operation (exact nature, location and reason of fault yet to be shared). iii)During the same time, 66 kV incomer of 200/66 kV 100 MVA ICT-3 at Mehrauli(DTL) tripped without any relay indication (exact reason of tripping yet to be shared). iv)As per PMU, no fault is observed in the system at 21:53 hrs. v)As reported by SLDC Delhi, load loss of approx. 110 MW occurred at 21:53hrs. Major affected load areas were Vasant Kunj C Block, C Dot, Fatehpur Beri, Caffims and 220kV Mehrauli.	1) 200/66 kV 100 MVA ICT-1 at Mehrauli(DTL) 2) 66 kV incomer of 200/66 kV 100 MVA ICT-3 at Mehrauli(DTL)
19	GI-1	Delhi	28-07-2024 22:05	28-07-2024 22:14	00:09	0	198	0.000	0.242	56370	81704	i)As reported, at 22:05 hrs, 66 kV incomers of 200/66 kV 100 MVA ICT- 2 at Mehrauli(DTL) tripped on over-current (B-ph) protection operation. ii)During the same time, 66 kV incomers of 200/66 kV 160 MVA ICT- 4 at Mehrauli(DTL) tripped without any relay indication (exact reason of tripping yet to be shared). iii)As per PMU, no fault is observed in the system at 22:05 hrs. iv)As per SCADA, change in demand of approx. 240 MW is observed in Delhi control area. However, as reported by SLDC Delhi, load loss of approx. 198 MW occurred at 21:53hrs. Major affected load areas were Vasant Kunj C Block, Malviya Nagar, Shivalik, TKD, DC Saket, IGNOU, C Dot, Fatehpur Beri, Caffims, Bijwashan and 220kV Mehrauli.	1) 66 kV incomer of 200/66 kV 100 MVA ICT-2 at Mehrauli(DTL) 2) 66 kV incomer of 200/66 kV 160 MVA ICT-4 at Mehrauli(DTL)
20	GI-1	Delhi	29-07-2024 14:40	29-07-2024 15:03	00:23	0	125	0.000	0.146	70093	85659	i)220/66/33kV Shalimarbagh(DTL) has double main Bus arrangement at 220kV side. ii)During antecedent condition, 220 kV Shalimarbagh-Rohini Ckt-1 (No load), 220kV Shalimargarh-SGTN Ckt-1 & 2, 220kV Shalimarbagh-DMRC Ckt-1, 220kV Shalimarbagh-Wazirpur Ckt-1 & 2 and 220 kV Bawana- Shalimarbagh Ckt-1 were connected to 220kV Bus-1 and 100 MVA 220/33kV ICT-1 & 3 and 220/66kV ICT-2 at Shalimarbagh(DTL), 220 kV Shalimarbagh-Rohini Ckt-2 (No load), 220kV Shalimarbagh-DMRC Ckt-2 and 220 kV Bawana- Shalimarbagh Ckt-2 were connected to 220kV Bus-2 at Shalimarbagh(DTL) S/s. 220kV bus coupler was in ON position (as informed by SLDC Delhi). iii)As reported, at 14:40 hrs, heavy flashover was observed in R-ph line isolator at Bawana end of 220 KV Bawana- Shalimarbagh (DTL) Ckt-2 and line tripped from Shalimarbagh end. As reported and as per SCADA SOE, line was manually opened from Bawana end at 14:42:17hrs. iv)During the same time, 100 MVA 220/33kV ICT-1 & 3 and 220/66kV ICT-2 at Shalimarbagh(DTL) also tripped (exact nature of protection operated yet to be shared). v)As per PMU at Abdullapur(PG), no fault is observed in the system. vi)As per SCADA, change in demand of approx. 185 MW in Delhi control area. However, as reported by SLDC-Delhi, load loss of approx. 125 MW occurred. Major load affected area were Tigipur, Haiderpur, Pitampura-III & I, Rohini-I, Wazirpur-II, Ranibagh, SMB-FC and SMB-Khosla.	1) 220 KV Bawana- Shalimarbagh (DTL) Ckt-2 2) 220/33kV 100 MVA ICT-1 at Shalimarbagh(DTL) 3) 220/66kV 100 MVA ICT-2 at Shalimarbagh(DTL) 4) 220/33kV 100 MVA ICT-3 at Shalimarbagh(DTL)

20.Detailds of Grid Events during the Month of July 2024 in Northern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
21	GD-1	Uttarakhand	29-07-2024 13:56	29-07-2024 14:28	00:32	108	0	0.152	0.000	70868	84775	i) During antecedent condition, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were generating approx. 36MW each. 220 KV Singoli Bhatwari (Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-2 was under planned outage. Total generation of 108MW of Singoli Bhatwari was evacuating through 220 KV Singoli Bhatwari (Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1. ii) As reported, at 13:56 hrs, 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 tripped on R-N phase to ground fault (exact reason of fault yet to be shared). iii) As per DR, fault current was ~2.44kA and fault distance was 53.2 km (69.1%) from Srinagar(UK) end; fault sensed in zone-1 and fault clearing time was ~50 ms.. iv) Due to tripping of 220 KV Singoli Bhatwari(Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP tripped due to loss of evacuation path and blackout occurred at 220kV Singoli Bhatwari HEP. v) As per PMU at Roorkee(PG), R-N phase to ground fault is observed with fault clearing time of 80 ms. vi) As per SCADA, generation loss of approx. 108MW at Singoli Bhatwari HEP is observed.	1) 220 KV Singoli Bhatwari (Singoli(LTUHP))-Srinagar(UK) (PTCUL) Ckt-1 2) 33MW Unit-1 at Singoli Bhatwari HEP 3) 33MW Unit-2 at Singoli Bhatwari HEP 4) 33MW Unit-3 at Singoli Bhatwari HEP
22	GI-1	Rajasthan	30-07-2024 11:38	30-07-2024 12:52	01:14	370	0	0.524	0.000	70619	86135	i) 400/220kV Bhadla(RS) has double main and transfer bus arrangement at 220kV side. ii) During antecedent condition, 220 kV Bhadla(RS)-Saurya Urja-2 and 220kV Bhadla(RS)-RSDCL I Ckt-2 were carrying approx. 242 MW & 128 MW respectively (reported data). iii) As reported, at 11:38hrs, B-ph jumper of 220kV Bhadla(RS)-Saurya Urja Ckt-2 snapped from Main Bus at Bhadla(RS) which led to tripping of 220kV Bhadla(RS)-Saurya Urja Ckt-2. iv) During the same time, 220 kV Bus sectionalizer-I (Bay no. 09) and 220 kV Bus Coupler-I (Bay no. 13) at Bhadla(RS) also tripped due to B-N phase to ground fault (As per PMU, Y-N fault; phase sequence issue is observed). v) Further as reported, 220kV Bhadla(RS)-RSDCL I Ckt-2 also tripped from RSDCL I end only due to LBB operation at the same time (exact reason of LBB operation yet to be shared). vi) As per PMU at Bhadla(PG), Y-N phase to ground fault is observed with delayed fault clearing time of 160 ms. vii) As per SCADA, change in solar generation of approx. 905MW is observed in Rajasthan control area. viii) As reported by SLDC Rajasthan, approx. 370 MW of solar generation loss occurred in Rajasthan control area and there is total approx. 730 MW reduction in solar generation by RE plants connected at Bhadla(RS).	1) 220kV Bhadla(RS)-Saurya Urja Ckt-2 2) 220 kV Bus sectionalizer-I (Bay no. 09) 3) 220 kV Bus Coupler-I (Bay no. 13) 4) 220kV Bhadla(RS)-RSDCL I Ckt-2
23	GD-1	Delhi	30-07-2024 14:55	30-07-2024 15:05	00:10	0	304	0.000	0.347	71027	87542	i) 220kV Najafgarh(DTL) has double main bus arrangement at 220kV level. ii) During antecedent condition, incoming power at Najafgarh(DTL) S/s was approx. 335 MW through 220KV Bamanauli-Najafgarh (DTL) D/C. 220kV Najafgarh-Mundka (DTL) Ckt & 220kV Najafgarh-Kanjhawala (DTL) Ckt were not in service. iii) As reported, at 14:55 hrs, 220KV Bamanauli-Najafgarh (DTL) Ckt-2 tripped on B-N phase to earth fault and 220KV Bamanauli-Najafgarh (DTL) Ckt-1 tripped on O/C protection. iv) As reported, 220KV Bamanauli-Najafgarh (DTL) Ckt-2 tripped on zone-1 distance protection on B-N fault with fault distance of 6.3km from Najafgarh(DTL) end and on differential protection from Bamanauli(DTL) end. v) As 220KV Bamanauli-Najafgarh (DTL) Ckt-2 tripped, complete load shifted on 220KV Bamanauli-Najafgarh (DTL) Ckt-1 and Ckt-1 also tripped on overcurrent protection from Najafgarh(DTL) end. vi) As per PMU at Dwarka(PG), B-N phase to earth fault with fault clearing time of 120msec is observed. vii) As 220kV Najafgarh-Mundka (DTL) Ckt & 220kV Najafgarh-Kanjhawala (DTL) Ckt were not in service and 220KV Bamanauli-Najafgarh (DTL) D/C also tripped, Najafgarh (DTL) lost its connectivity from Grid and blackout occurred at 220kV Najafgarh(DTL) S/s. viii) As per SCADA, change in demand of approx. 285 MW in Delhi control area. However, SLDC-Delhi reported load loss of approx. 304 MW in Delhi control area.	1) 220KV Bamanauli-Najafgarh (DTL) Ckt-1 2) 220KV Bamanauli-Najafgarh (DTL) Ckt-2
24	GI-1	Himachal Pradesh	31-07-2024 22:40	31-07-2024 23:33	00:53	0	140	0.000	0.203	54106	68837	i) 220kV Giri(HP) S/s has double main bus arrangement at 220kV level. ii) During antecedent condition, incoming power at Giri(HP) S/s through 220kV Khodri(UK)-Giri(HP) (UK) D/C was 105MW. iii) As reported, at 22:40 hrs, Y-phase LA of 220/132kV 100 MVA ICT-2 at Giri(HP) S/s blasted which caused Y-N phase to earth fault. On this fault 220/132kV 100 MVA ICT-1 & 2 tripped on overcurrent earth fault protection (ICT-1 at Giri(HP) should not trip on this fault, reason for the same is yet to be received). iv) On the same fault, 220kV Khodri(UK)-Giri(HP) (UK) Ckt-2 tripped only from Khodri(UK) end on Y-N phase to earth fault. Fault sensed in zone-3 with fault current of Iy=~2.4kA from Khodri(UK) end. v) As per PMU at Saharanpur(PG), Y-N phase to earth fault with fault clearing time of 120msec is observed. vi) As per SCADA, change in demand of approx. 110 MW in HP control area. However, SLDC-HP reported load loss of approx. 140 MW in HP control area.	1) 220 KV Khodri(UK)-Majri/Giri(HP) (UK) ckt-2 2) 220/132kV 100 MVA ICT-1 at Giri(HP) 3) 220/132kV 100 MVA ICT-2 at Giri(HP)

20.Details of Grid Events during the Month of July 2024 in Western Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	WR	02-07-2024 07:33	02-07-2024 15:56	08:23	282.07	-	0.42%	-	67515	52976	i)220kV Baghpat(UP) has main and transfer bus scheme at 220kV level. ii)During antecedent condition, incoming power at Baghpat(UP) was approx. 80 MW through 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C, 220/132kV 160MVA ICT-1 and 220/132kV 100MVA ICT-2 were connected to 220kV main bus. 220 KV Baghpat(PG)-Baghpat(UP) (UP)	Tripping of following Elements: 220 kV Bhuj-II-Nakhatrana-1
2	GD-1	WR	05-07-2024 06:25	05-07-2024 07:39	01:14	24	-	0.04%	-	66396	55434	At 06:25 Hrs/05-07-2024, 220kV Bhuj- Kotda Madh (Netra) line auto reclosed successfully from Bhuj end and tripped at Kotda madh end due to Y-N fault. Due to loss of evacuation path, wind generation loss of 24MW occurred at Kotda Madh (Netra)	Tripping of following Elements: 220 kV Bhuj-Kotda Madh-1
3	GD-1	WR	08-07-2024 00:02	08-07-2024 04:34	04:32	120	-	0.18%	-	67421	56031	At 00:02 Hrs/08-07-2024, 220kV Bhuj- Gadhsisa line tripped on R-Y fault due to falling of Nirona-Junacay line conductor. Due to loss of evacuation path, generation loss of 120MW occurred at Gadhsisa WPP.	Tripping of following Elements: 220 kV Bhuj-Gadhsisa-1
4	GD-1	WR	09-07-2024 15:47	09-07-2024 16:39	00:52	218.5	-	0.31%	-	70415	58699	At 15:47 Hrs/09-07-2024, 220 kV Bhuj- Kotda Madh tripped on B-N fault at Bhuj end & on DT receipt at Kotda Madh end. At the same time, 220 kV Bhuj- Nanavalka tripped at Bhuj end only on R-N fault. At the same time, 220kV Bhuj- Gadhsisa tripped at Gadhsisa end only due to maloperation. Due to loss of evacuation path, generation loss of 111 MW, 37.5 MW & 70 MW occurred at Gadhsisa, Nanavalka & Kotda Madh WPPs respectively.	Tripping of following Elements: 220 kV Bhuj-Gadhsisa-1 220 kV Bhuj-Kotda Madh-1 220 kV Bhuj-Nanavalka
5	GD-1	WR	15-07-2024 13:26	15-07-2024 17:10	03:44	-	-	-	-	66921	57735	At 13:26 Hrs/15-07-2024, 400KV-Jabalpur-PS-Jhabua-1 tripped on R-N fault from Jhabua end, A/R successful from Jabalpur end. Ckt-2, Bus-2 and Jhabua Unit-1 were already under planned outage due to annual overhauling work. Therefore, 400 kV Jhabua S/S became dead. No generation loss reported as the Unit was already out.	Tripping of following Elements: 400 kV Jabalpur(PS) - Jhabua - I Jhabua - 400KV - Bus 1
6	GD-1	WR	15-07-2024 20:03	15-07-2024 22:03	02:00	22	-	0.03%	-	73890	58501	At 20:03 Hrs/15-07-2024, 220kV Baranda Bhuj tripped on R-N fault from Baranda end, A/R successful from Bhuj end. Due to loss of evcuation path, Baranda Wind RE station went to blackout leading to generation loss of 22 MW.	Tripping of following Elements: 220 kV Bhuj-Baranda-1
7	GD-1	WR	23-07-2024 07:23	23-07-2024 16:56	09:33	120	-	0.17%	-	69157	55551	At 07:23 Hrs/23-07-2024, 220/33 kV Gadhsisa ICT-1 tripped due to operation of Restricted earth fault (REF) Relay, which resulted in loss of evacuation path from 33 kV to 220 kV side as 220/33 kV Gadhsisa ICT-2 was already in tripped condition since 21:15 Hrs/22.07.2024 due to sparking in 33 KV side isolator. About 120 MW wind generation loss occurred at 220 KV Gadhsisa WPP.	Tripping of following Elements: 220KV/33KV Gadhsisa-ICT-1
8	GD-1	WR	25-07-2024 09:17	25-07-2024 17:32	08:15	253.16	-	0.39%	-	64438	54172	At 09:17 Hrs 220kV Nakhatrana-Bhuj-II-1 tripped on Y-N fault due to Y-Phase Disc Insulator failure on Tower No.22 from Nakhatrana side. Generation loss of 253.16 MW occurred due to loss of evacuation path.	Tripping of following Elements: 220 kV Bhuj-II-Nakhatrana-1
9	GD-1	WR	25-07-2024 11:33	25-07-2024 12:52	01:19	95	-	0.15%	-	61920	52889	At 11:33 Hrs/25.07.2024, 220kV Srijan Morjhar-Bhuj-II-1 tripped on Z-2, R-N fault at Srijan end only. No indications received. Generation loss of 95 MW occurred due to loss of evacuation path.	Tripping of following Elements: 220kV Srijan Morjhar-Bhuj-II-1
10	GD-1	WR	25-07-2024 17:33	25-07-2024 19:56	02:23	113.7	-	0.18%	-	64524	52709	At 17:33 Hrs/25.07.2024 220kV Srijan Morjhar – Bhuj-II-1 tripped on R-N fault at Srijan end. During inspection by Srijan, main 2, R phase voltage was not available as loose connection identified in Terminal Box, which was tightened. Generation loss of 113.7 MW occurred due to loss of evacuation path.	Tripping of following Elements: 220kV Srijan Morjhar-Bhuj-II-1
11	GD-1	WR	29-07-2024 16:46	30-07-2024 00:01	07:15	-	131	-	0.25%	66129	53429	At 16:46 Hrs/29.07.2024, 220kV Vaghchhipa-Sayali Ckt & 220kV Khadoli-Sayali Ckt tripped on Zone-2, Y-B Fault and 220kV Sayali Switching Substation became dead. After Line patrolling by DNH, Y-phase jumper found broken of 220kV Sayli-Khadoli line at tower location no.28. This resulted in complete load loss of 220kV Alok Industries Ltd. Total Load loss of around 131MW reported.	Tripping of following Elements: 220kV Vaghchhipa-Sayali Ckt 220kV Khadoli-Sayali Ckt 220kV Sayali- Alok Industries Ltd.

20.Details of Grid Events during the Month of July 2024 in Southern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD - 1	KARNATAKA	01-07-2024 02:53	01-07-2024 12:01	09:08	32	152	0.09%	0.41%	35141	37322	i)220kV Baghpat(UP) has main and transfer bus scheme at 220kV level. ii) During antecedent condition, incoming power at Baghpat(UP) was approx. 80 MW through 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C, 220/132kV 160MVA ICT-1 and 220/132kV 100MVA ICT-2 were connected to 220kV main bus. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C is on the same towers. iii) As per SCADA SOE, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped at 21:37:59.415hrs and 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 tripped at 21:37:59.535 hrs. iv) As reported, at 21:37 hrs, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped on Y-B-N double phase to earth fault and line tripped on zone-1 distance protection operation from both ends. As per Baghpat(UP) reporting, initially a B-ph fault occurred on 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 with fault current - 10.39kA in zone-1. At the same time, jumper at tower no 45 of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 got broken which created line-line (Y-B) fault due to which line got tripped from both ends. v) Further 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 also tripped on B-N phase to earth fault with fault current of ~9.7kA and fault distance of 11.8km from Baghpat(PG) end. As per DR of Baghpat(PG) end, B-N phase to earth fault with no A/R operation is observed and line tripped on zone-1 distance protection operation. Since there was no source remaining at 220kV Baghpat(UP) hence 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 didn't trip from Baghpat(UP) end. vi)As per PMU at Meerut(PG), Y-B phase to phase fault with fault clearance time of 80ms is observed. vii) Due to tripping of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2, Baghpat(UP) lost its connectivity from the grid and 220kV Baghpat(UP) S/s became dead. viii)As per SCADA, change in demand of approx. 68 MW in UP control area. However, SLDC-UP reported 80MW load loss.	220KV-KADAKOLA-KANIYAMPETTA , 220KV-MYSORE-HUNSUR-1 , 220KV-MYSORE-KADAKOLA-1
2	GD - 1	TAMILNADU	03-07-2024 20:53	03-07-2024 21:29	0:36	0	79	0.00%	0.17%	45332	46298	Complete Outage of 230kV Kilpauk SS: During antecedent conditions, 230kV Koratur-Kilpauk was in idle charged condition and 230kV Kilpauk was radially connected to 230kV NCTPS. The triggering incident is the fault in 230kV NCTPS-NCTPSST3 line near to 230kV NCTPS. The fault was cleared within time, however 230kV Kilpauk-NCTPS line tripped at Kilpauk end on Z1 protection in only Main-1. Tripping of the only line connected to Kilpauk led to the complete outage of 230kV Kilpauk station.	230KV-NCTPS-KILPAUK-1 , 230KV-NCTPS-NCTPS STG3-1
3	GD - 1	KARNATAKA	11-07-2024 11:18	11-07-2024 11:45	0:27	104	0	0.20%	0.00%	50990	53603	Complete Outage of 220kV/66kV Tallak SS, 230kV/33kV Sagitur SS and 220kV BARC SS. As per the reports submitted, the triggering incident was 96 relay maloperation in 220kV Bus-1 of 220kV/66kV Tallak SS tripping all elements connected to the Bus-1. Since, 230kV/33kV Sagitur SS and 220kV BARC SS are being radially connected to 220kV Bus-1 of 220kV/66kV Tallak SS, this resulted in complete outage of 230kV/33kV Sagitur SS and 220kV BARC SS	220KV-Chitradurga-TALLAK-1
4	GD - 1	KARNATAKA	12-07-2024 11:54	12-07-2024 12:06	0:12	0	360	0.00%	0.65%	51678.2	55761.41	Complete Outage of 220kV/66kV Manyatha Tech SS, 220kV/66kV ITI, and 220kV/66kV HBR Layout: Triggering incident was tripping of 220kV Yelahanka-Yelahanka PG line-2 at Yelahanka end due to suspected maloperation of 96 trip relay. This led to the overloading of 220kV Yelahanka-Yelahanka PG line-1. Immediately SPS operated in two stages to reduce the flow in 220kV Yelahanka-Yelahanka PG line-1 resulting in the tripping of 150MVA PTRs and 220kV Yelahanka Sahakarnagar line-2 at 220kV Yelahanka SS. This further resulted in overlaoding of 220kV Yelahanka Sahakarnagar line-1 and tripping of 220kV Sahakarinagar-Manyatha lines on operation of SPS St-2 protection at 220kV Sahakarinagar SS. Due to tripping of these 220kV lines, 220kV/66kV Manyatha Tech SS, 220kV/66kV ITI, and 220kV/66kV HBR Layout became radial on Hoody source, which eventually tripped due to overload, leading to a complete outage of these substations.	220KV-YELHANKA-YELAHANKA DG-2 , 220KV-HOODY-HBR_LAYOUT-1 , 220KV-HOODY-ITI-1 , 220KV-SAHKARI_NAGAR-MANYATA-1 , 220KV-SAHKARI_NAGAR-MANYATA-2 , 220KV-YELHANKA-SAHKARI_NAGAR-2
5	GD - 1	KARNATAKA	17-07-2024 14:55	17-07-2024 15:40	0:45	0	45	0.00%	0.00%	52920.42	48007.53	Complete Outage of 220kV/66kV Chikkamangalore SS of KPTCL: As per the reports submitted, the triggering incident was Y-N fault in 220kV Shimoga Chikmangalore Line. At Chikkamangalore end, the fault was sensed in zone-1 and Y-pole opened. however no protection operated at Chikkamangalore end during subsequent fault in dead time. Fault was cleared by tripping of 220kV Chikkamangalore Hassan line on zone-3 protection at Hassan end with a delay of 800ms. This led to the complete outage of 220kV/66kV Chikkamangalore SS.	220KV-CHIKKAMAGALURU-MRS SHIMOGA-1 , 220KV-HASSAN-CHIKKAMAGALURU-1
6	GD - 1	TAMILNADU	18-07-2024 12:14	18-07-2024 12:30	0:16	40	0	0.00%	0.00%	54290.55	47128.61	Complete Outage of 230kV JSW_Vilathikulam Wind Station: As per the reports submitted, the 230kV TTGS-JSW_Vilathikulam line-1 tripped only at JSW_Vilathikulam end on operation of Y-Ph under voltage stage -1 protection while changing settings in P546 relay. Tripping of the only connected line resulted in the complete outage of 230kV JSW_Vilathikulam Wind Station.	230KV-TTGS-JSW_Vilathikulam-1
7	GD - 1	KARNATAKA	22-07-2024 11:47	22-07-2024 12:09	0:22	950	170	2.00%	0.00%	51797.49	46718.3	Tripping of 220kV Bus-1&2 of 400kV/220kV Jagalur SS and Complete Outage of 220kV/66kV Chitradurga SS, 220kV/66kV Tallak SS, 220kV/66kV Kudligi SS of KPTCL and 220kV Barc SS, 220kV Sagitur SS of KPTCL :As per the reports submitted, the triggering incident was cascaded tripping of 220kV Chitradurga Hiriyur, 220kV Guttur Chitradurga Lines and 400kV/220kV Jagalur ICTS on operation of over current protection. At the same time, all 220kV lines connected to 220kV Tallak SS tripped due to suspected maoperation of 96 trip relay. Tripping of all these elements led to de-energisaiton of 220kV Bus-1& 2 of 400kV/220kV Jagalur SS and complete outage of 220kV/66kV Chitradurga SS, 220kV/66kV Tallak SS, 220kV/66kV Kudligi SS, 220kV Barc SS, 220kV Sagitur SS.	220KV-HIRIYUR-Chitradurga-1 , 400KV/220KV JAGALUR-ICT-1 , 220KV-Chitradurga-TALLAK-1 , 220KV-GUTTUR-Chitradurga-1 , 400KV/220KV JAGALUR-ICT-2
8	GD - 1	KERALA	24-07-2024 09:46	24-07-2024 10:31	0:45	0	36	0.00%	0.00%	49280.56	49903.61	Complete Outage of 400kV/220kV Kottayam SS and 220kV/110kV Ettumanoor SS of KSEB: As per the reports submitted, the triggering incident was high resistance B-N fault in 400kV Kottayam Tirunelveli Line-2. At Tirunelveli end, the fault was sensed in zone-1 and the line tripped only at Tirunelveli end. Due to non-operation of protection at Kottayam end, 400kV Kottayam Tirunelveli Line-1 tripped at Tirunelveli end on zone-2 and all other connected 400kV lines and 220kV lines tripped on operation of DEF protection at remote ends. Tripping of all lines led to complete outage of 400kV/220kV Kottayam SS and 220kV/110kV Ettumanoor SS.	400KV-KOTTAYAM-KOCHI-1 , 400KV-KOTTAYAM-KOCHI-2 , 400KV-TIRUNELVELI-KOTTAYAM-1 , 400KV-TIRUNELVELI-KOTTAYAM-2 , 400KV/220KV KOTTAYAM-ICT-1 , 400KV/220KV KOTTAYAM-ICT-2, 220KV KOTTAYAM-ETTAMANUR, 220KV KOTTAYAM-PALLOM
9	GD - 1	KARNATAKA	27-07-2024 12:49	27-07-2024 15:00	2:11	112	113	0.00%	0.00%	53879.28	45339.17	Complete Outage of 220kV Kodasalli PH of KPCL: 220kV Kodsally PH is operating with single bus with transfer bus configuration. . As per the reports submitted, the triggering incident was B-N fault in 220kV Bus-1 of 220kV Kodsally PH. Immediately BBP operated and all elements connected to the bus tripped. This resulted in complete outage of 220kV Kodsally PH.	220KV KODASALLI-KAIGA 220KV KODASALLI-KADRA 220KV KODASALLI-NAGHJHERI-1 220KV KODASALLI-NAGHJHERI-2
10	GI-1	ANDHRA PRADESH	18-07-2024 17:26	18-07-2024 18:26	01:00	0	0	0.00%	0.00%	49602	43030	Tripping of 220kV Bus-1 of 400kV/220kV KV Kota SS: As per the reports furnished, while test charging the tie feeder-1 (line from 400kV/220kV KV Kota SS to 220kV/132kV KV Kota SS) from 400kV/220kV KV Kota SS end, line tripped on operation of SOTF protection due to YN fault in the line because of insulator failure. Since Y-pole breaker didn't open at 400kV/220kV KV Kota SS, LBB operated resulted in the tripping of all the elements connected to 220kV Bus-1 of 400kV/220kV KV Kota SS.	400KV/220KV KV_KOTA-ICT-1
11	GI-1	ANDHRA PRADESH	19-07-2024 18:46	19-07-2024 20:21	1:35	0	0	0.00%	0.00%	48238	42062	Tripping of 220kV Bus-2 of 400kV/220kV KV Kota SS of APTRANSCO : As per the reports furnished, the triggering incident was YN fault in tie feeder-2 (line from 400kV/220kV KV Kota SS to 220kV/132kV KV Kota SS) from 400kV/220kV KV Kota SS end. Since Y-pole breaker didn't open 400kV/220kV KV Kota SS, LBB operated resulted in the tripping of all the elements connected to 220kV Bus-2 of 400kV/220kV KV Kota SS.	400KV/220KV KV_KOTA-ICT-2

20.Details of Grid Events during the Month of July 2024 in Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Chatra	08.07.2024 13:10	08.07.2024 14:25	01:15	0	36	0.00%	0.14%	28189	26236	i)220kV Baghpat(UP) has main and transfer bus scheme at 220kV level. ii) During antecedent condition, incoming power at Baghpat(UP) was approx. 80 MW through 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C, 220/132kV 160MVA ICT-1 and 220/132kV 100MVA ICT-2 were connected to 220kV main bus. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C is on the same towers. iii) As per SCADA SOE, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped at 21:37:59.415hrs and 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 tripped at 21:37:59.535 hrs. iv) As reported, at 21:37 hrs, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped on Y-B-N double phase to earth fault and line tripped on zone-1 distance protection operation from both ends. As per Baghpat(UP) reporting, initially a B-ph fault	220kV Latehar-Chatra 220kV Daltongunj-Chatra
2	GD-1	Bokaro	20.07.2024 19:38	20.07.2024 22:18	02:40	00:00	65	0.00%	0.23%	27956	28850	At 19:38 Hrs, during Line opening and isolation process of 220 kV CTPS-BTPS D/C line, bus fault occurred at 220 kV BTPS along with DC supply failure. Both ICTs tripped, causing load loss of approximately 65 MW . Power was extended through 132 kV BTPS-Barhi Line at 22:18 Hrs and subsequently other elements were normalized.	400KV/220KV 315 MVA ICT 1 & 2 AT Bokaro A TPS 220KV Bokaro -Jamshedpur-D/C 220KV/132 KV ATR 1 & 2 at Bokaro 220KV Bokaro -Ramgarh-D/C

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD I	Pasighat, Roing, Tezu, Namsai areas of Arunachal Pradesh Power System and Chapakhowa area of Assam Power System	03-07-2024 11:21	03-07-2024 11:48	00:27	0	10	0.00%	0.41%	3017	2421	i)220kV Baghpat(UP) has main and transfer bus scheme at 220kV level. ii)During antecedent condition, incoming power at Baghpat(UP) was approx. 80 MW through 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C, 220/132kV 160MVA ICT-1 and 220/132kV 100MVA ICT-2 were connected to 220kV main bus. 220 KV Baghpat(PG)-Baghpat(UP) (UP) D/C is on the same towers. iii)As per SCADA SOE, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped at 21:37:59.415hrs and 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 tripped at 21:37:59.535 hrs. iv)As reported, at 21:37 hrs, 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 tripped on Y-B-N double phase to earth fault and line tripped on zone-1 distance protection operation from both ends. As per Baghpat(UP) reporting, initially a B-ph fault occurred on 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 with fault current - 10.39kA in zone-1. At the same time, jumper at tower no 45 of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-2 got broken which created line-line (Y-B) fault due to which line got tripped from both ends. v)Further 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 also tripped on B-N phase to earth fault with fault current of ~9.7kA and fault distance of 11.8km from Baghpat(PG) end. As per DR of Baghpat(PG) end, B-N phase to earth fault with no A/R operation is observed and line tripped on zone-1 distance protection operation. Since there was no source remaining at 220kV Baghpat(UP) hence 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 didn't trip from Baghpat(UP) end. vi)As per PMU at Meerut(PG), Y-B phase to phase fault with fault clearance time of 80ms is observed. vii)Due to tripping of 220 KV Baghpat(PG)-Baghpat(UP) (UP) Ckt-1 & 2, Baghpat(UP) lost its connectivity from the grid and 220kV Baghpat(UP) S/s became dead.	132 kV Along - Pasighat Line 132 kV Roing - Pasighat Line
2	GD I	Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System and Tamu load of Myanmar Power System	05-07-2024 16:20	05-07-2024 16:35	00:15	0	28	0.00%	1.27%	3121	2202	Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System and Tamu load of Myanmar Power System were connected with rest of NER Grid via 132 kV New Thoubal - Kakching Line, 132 kV Old Thoubal - Kakching Line, 132 kV Kongba - Yaingangpokpi 1 Line and 132 kV Kongba - Yaingangpokpi 2 Line. 400kV-Imphal(PG)-Thoubal-1 line was under outage condition since 18.10.2021, 400kV-Imphal(PG)-Thoubal-2 line was under outage condition since 24.04.2024 and 132kV-Ningthoukhong-Churachandpur D/C lines were under outage condition since 15.06.2024. At 16:20 Hrs of 05.07.2024, 132 kV New Thoubal - Kakching Line, 132 kV Old Thoubal - Kakching Line, 132 kV Kongba - Yaingangpokpi 1 Line and 132 kV Kongba - Yaingangpokpi 2 Line tripped. Due to tripping of these lines, Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System and Tamu load of Myanmar Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Churachandpur Thanlon, Elangkangpokpi, Kakching, Chandel, Thoubal Old, New Thoubal and Kongba area of Manipur Power System and Tamu load of Myanmar Power System by first charging 132 kV Kongba - Yaingangpokpi 1 Line at 16:35 Hrs of 05.07.2024 and then subsequently charging 132 kV Kongba - Yaingangpokpi 2 Line, 132 kV New Thoubal - Kakching Line, 132kV Thoubal New – Kongba 1, 132kV Thoubal New – Kongba 2 and 132kV Thoubal Old – Thoubal New. 132 kV Old Thoubal - Kakching Line is not yet charged.	132 kV New Thoubal - Kakching Line, 132 kV Old Thoubal - Kakching Line, 132 kV Kongba - Yaingangpokpi 1 Line and 132 kV Kongba - Yaingangpokpi 2 Line
3	GD I	Karong area of Manipur Power System	07-07-2024 13:24	07-07-2024 13:52	00:28	0	8	0.00%	0.32%	2951	2478	Karong area of Manipur Power System was connected with rest of NER Grid via 132kV Karong - Kohima line. 132kV Imphal - Karong line was under outage condition since 28.06.2024. At 13:24 Hrs of 07.07.2024, 132kV Karong - Kohima line tripped. Due to tripping of this line, Karong area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Karong - Kohima line at 13:52 Hrs of 07.07.2024.	132 kV Karong - Kohima
4	GD I	Jirania area of Tripura Power System	07-07-2024 16:51	07-07-2024 17:42	00:51	0	4	0.00%	0.16%	2995	2525	Jirania area of Tripura Power System was connected with rest of NER Grid via 132kV Budhjunnagar - Jirania and 132kV Ambasa – Gamaitila– Baramura- Jirania lines. At 16:51 Hrs of 07.07.2024, 132kV Budhjunnagar - Jirania line and 132kV Gamaitila– Baramura - Jirania lines were tripped. Due to tripping of these lines, Jirania and Baramura area of Tripura Power System was isolated from NER Grid and collapsed due to no source available in these areas. However except Jirania area, no load loss or Generation loss observed in Baramura area of Tripura Power System during the event as reported. Power supply was extended to Jirania area of Tripura Power System by charging 132 kV Budhjunnagar - Jirania line at 17:42 Hrs of 07.07.2024.	132kV Budhjunnagar - Jirania line and 132kV Gamaitila– Baramura - Jirania
5	GD I	Rengpang area of Manipur Power System	08-07-2024 10:31	08-07-2024 12:12	01:41	0	3	0.00%	0.12%	3217	2571	Rengpang area of Manipur Power System was connected with rest of NER Grid through 132 kV Loktak-Rengpang line. 132 kV-Jiribam-Rengpang line was under long outage since 18:18 Hrs of 17.11.2023. At 10:31 Hrs of 08-07-2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Rengpang area of Manipur Power System by charging 132 kV Loktak-Rengpang line at 12:12 Hrs of 08.07.2024.	132 kV Loktak-Rengpang line
6	GD I	Longnak area of Nagaland Power System	08-07-2024 06:28	08-07-2024 12:04	05:36	0	2	0.00%	0.08%	3404	2391	Longnak area of Nagaland Power System was connected with rest of NER Grid through 132 kV Mokokchung longnak line. At 06:28 Hrs of 08.07.2024, 132 kV Mokokchung-Longnak line tripped. Due to tripping of this line,Longnak area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Longnak area of Nagaland Power System by charging 132 kV Mokokchung-Longnak line at 12:04 Hrs of 08.07.2024.	132 kV Mokokchung-Longnak line

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GD I	Longnak area of Nagaland Power System	08-07-2024 01:37	08-07-2024 05:27	03:50	0	8	0.00%	0.30%	3416	2650	Longnak area of Nagaland Power System was connected with rest of NER Grid through 132 kV Mokokchung longnak line. At 01:37 Hrs of 08.07.2024, 132 kV Mokokchung longnak line tripped. Due to tripping of this line,Longnak area of Nagaland Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Longnak area of Nagaland Power System by charging 132 kV Mokokchung Longnak Line at 05:27 Hrs of 08.07.2024.	132 kV Mokokchung-Longnak line
8	GD I	Pasighat, and Along areas of Arunachal Pradesh Power System	09-07-2024 12:38	09-07-2024 14:45	02:07	0	10	0.00%	0.38%	2963	2655	Pasighat and Along areas of Arunachal Pradesh Power System were connected with rest of NER Power system via 132 kV Roing - Pasighat Line, 132 kV Along-Pasighat line and 132 kV Basar-Along line. At 12:38 Hrs of of 09.07.2024, 132 kV Roing - Pasighat Line, 132 kV Along-Pasighat line and 132 kV Basar-Along line tripped. Due to tripping of these elements, Pasighat and along areas of Arunachal Pradesh Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power is restored to the Pasighat area by charging 132kV Roing-Pasighat line at 14:45 Hrs of 09.07.2024 and is further extended to the along area by charging 132kV Along-Pasighat line at 16:22 Hrs of 09.07.2024.	132 kV Roing - Pasighat Line, 132 kV Along-Pasighat line and 132 kV Basar-Along line
9	GD I	Blackout in Khupi, Tenga areas of Arunachal Pradesh Power System	09-07-2024 13:42	09-07-2024 15:20	01:38	0	23	0.00%	0.89%	3074	2582	Tenga area of Arunachal Pradesh Power System were connected with rest of NER Power system via 132 kV Balipara- Tenga line and Khupi area is connected via 132kV Khupi- Kameng line. Tenga and Khupi area are connected via 132 kV Tenga-Khupi line. At 13:42 Hrs of of 09.07.2024, 132 kV Balipara-Tenga line and 400/132kV 40 MVA ICT at Kameng tripped leading to blackout of Khupi and Tenga area of Arunachal Pradesh Power system. Dikshi generation also tripped due to difference in load and generation. Power was restored to the affected area by charging 132kV Balipara-Tenga line at 15:20 hrs of 09.07.2024.	132 kV Balipara-Tenga line
10	GD I	Rongkhon, Ampati, Phulbari and Ganol HEP of Meghalaya power system	11-07-2024 07:44	11-07-2024 07:55	00:11	0	23	0.00%	1.00%	3159	2305	Rongkhon, Ampati, Phulbari areas and Ganol HEP of Meghalaya Power System were connected to NER Power system via 132 kV Rongkhon - Nangalbibra line. At 07:44 Hrs of 11-07-2024, 132 kV Rongkhon – Nangalbibra Line tripped. Due to tripping of this line, Rongkhon, Ampati, Phulbari areas and Ganol HEP of Meghalaya Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Rongkhon, Ampati, Phulbari & Ganol areas of Meghalaya Power System by charging 132 kV Nangalbibra –Rongkhon line at 07:55 Hrs of 11-07-2024.	132 kV Rongkhon – Nangalbibra Line
11	GD I	Kokrajhar, Bilasipara, Gauripur areas of Assam Power System	11-07-2024 11:53	11-07-2024 13:01	01:08	0	55	0.00%	2.30%	2887	2393	Kokrajhar, Bilasipara, Gauripur areas of Assam Power System were connected with rest of NER Grid via132 kV BTPS Kokrajhar 2 Line. 132 kV BTPS Kokrajhar 1 Line was under outage. At 11:53 Hrs of 11.07.2024, 132 kV BTPS Kokrajhar 2 Line tripped. Due to tripping of this line, Kokrajhar, Bilasipara, Gauripur areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Kokrajhar, Bilasipara, Gauripur areas of Assam Power System by charging 132 kV Gauripur - Gossaigaon 1 line at 12:09 hrs of 11.07.2024.	132 kV BTPS Kokrajhar 2 Line
12	GD I	Rengpang area of Manipur Power System	11-07-2024 09:23	11-07-2024 10:22	00:59	0	1	0.00%	0.04%	2887	2393	Rengpang area of Manipur Power System was connected with rest of NER Grid via 132kV Loktak Rengpang line. 132kV Jiribam-Rengpang line is under long outage. At 09:23 Hrs of 11.07.2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this line, Rengpang area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Rengpang area of Manipur Power System by charging 132kV Rengpang-Loktak line at 10:22 Hrs of 11.07.2024.	132 kV Loktak-Rengpang line
13	GD I	Rokhia Generating Station of Tripura Power System	13-07-2024 00:27	13-07-2024 01:14	00:47	17	13	0.53%	0.45%	3197	2901	Rokhia Generating Station of Tripura Power System was connected with rest of NER Grid through 132 kV Rokhia-Agartala 1 line, 132 kV Rokhia-Agartala 2 line and 132kv Rokhia –Monarchak line. At 00:27 Hrs, 132 kV Rokhia-Agartala 1 line, 132 kV Rokhia-Agartala 2 line and 132kV Rokhia – Monarchak line tripped, leading to tripping of Rokhia Unit-7. This lead to blackout of 132kV Rokhia substation causing load loss of 13 MW and generation loss of 17 MW.Due to tripping of 132kV Rokhia – Monarchak line, SPS operated at Monarhak and it tripped Monarchak STG. Power was extended to Rokhia Generating Station of Tripura Power System by charging 132 kV Rokhia-Agartala 1 line at 01:14 Hrs of 14-07-2024.	132 kV Rokhia-Agartala 1 line, 132 kV Rokhia-Agartala 2 line and 132kV Rokhia – Monarchak line

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
14	GD II	Upper Assam areas of Assam Power System, AGBPP Generating stations, and Pasighat, Roing, Tezu and Namsai areas of Arunachal Power System	15-07-2024 19:45	15-07-2024 20:37	00:52	0	435	0.00%	13.85%	3434	3140	At 19:45 Hrs of 15.07.2024, 220 kV Mariani(AS)-Samaguri line, 220 kV Mariani(AS)- Mariani(PG) line, 220 kV AGBPP- Mariani(PG) line,132 kV Golaghat- Mariani(AS) line, and 132 kV Along-Pasighat line tripped. Due to tripping of these elements, Upper Assam power system consisting of Mariani(AS), Jorhat(Garmur), Jorhat(west), Bokakhat, Teok Nazira, Lakwa(LTPS), Moran, Sonari, NTPS, Bordubi, Namrup PS-1, Tinsukia, Amguri JACKSON SP, Dibrugarh, Margherita, Rupai, Margherita Chapakowa areas, and Deomali, Pasighat,Roing, Tezu and Namsai areas of Arunachal Power System of NER and AGBPP Generating Stations were isolated from NER Grid and collapsed due to load generation mismatch in these areas. Power was extended to Upper Assam power system consisting of Mariani(AS), Jorhat(Garmur), Jorhat(west), Bokakhat, Teok Nazira, Lakwa(LTPS), Moran, Sonari, NTPS, Bordubi, Namrup PS-1, Tinsukia, Amguri JACKSON SP, Dibrugarh, Margherita, Rupai, Margherita Chapakowa areas by charging 220 kV Mariani(AS)-Samaguri line at 20:37 Hrs and to Pasighat, Roing, Tezu and Namsai areas of Arunachal Power System by charging 132 kV Along Pasighat line at 20:56 Hrs and to Deomali area of Assam Power System by charging 220 kV Mariani(AS)-Samaguri line.	220 kV Mariani(AS)-Samaguri line, 220 kV Mariani(AS)-Mariani(PG) line, 220 kV AGBPP- Mariani(PG) line,132 kV Golaghat- Mariani(AS) line, and 132 kV Along-Pasighat line
15	GD I	Kokrajhar, Bilasipara and Gauripur areas of Assam Power System	15-07-2024 18:10	15-07-2024 21:00	02:50	0	45	0.00%	1.56%	3224	2878	Kokrajhar, Bilasipara and Gauripur areas of Assam Power System were connected with rest of NER Grid via 132 kV BTPS Kokrajhar 1& 2 Line. At 18:10 Hrs of 15.07.2024, 132 kV BTPS Kokrajhar 1 &2 Line tripped. Due to tripping of this line,Kokrajhar, Bilasipara, Gauripur areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Kokrajhar, Bilasipara, Gauripur areas of Assam Power System by charging 132 kV BTPS Kokrajhar 2 line at 21:00 Hrs of 15-07-2024.	132 kV BTPS Kokrajhar 1 &2 Line
16	GD I	Doyang Generating station of NEEPCO Power System	16-07-2024 10:08	16-07-2024 11:21	01:13	73	0	3.03%	0.00%	2408	2546	Doyang generating station of Nagaland Power System was connected with rest of NER Grid with 132 kV Dimapur-Doyang 2 and 132 kV Doyang-Mokokchung and 132 kV Doyang-Sanis lines. Prior to the event, 132 kV Dimapur-Doyang I was uner planned shutdown. At 10:08 Hrs of 16-07-2024, 132 kV Dimapur-Doyang 2,132 kV Doyang-Mokokchung and 132 kV Doyang-Sanis lines tripped and subsequently all three units of Doyang tripped leading to generation loss of 73 MW. Power was extended to Doyang Generating Station by charging 132 kV Doyang-Mokokchung line at 11:21 Hrs of 16.07.2024.	132 kV Dimapur-Doyang 2,132 kV Doyang-Mokokchung and 132 kV Doyang-Sanis lines
17	GD I	Kokrajhar, Bilasipara and Gauripur areas of Assam Power System	16-07-2024 12:23	16-07-2024 12:28	00:05	0	64	0.00%	2.40%	2300	2664	Kokrajhar, Bilasipara and Gauripur areas of Assam Power System were connected with rest of NER Grid via 132 kV BTPS Kokrajhar 2 Line. 132 kV BTPS Kokrajhar 1 Line was under outage since 18:10 hrs of 15.07.2024. At 12:23 Hrs of 16.07.2024, while charging 132 kV BTPS Kokrajhar 1 Line, 132 kV BTPS Kokrajhar 1 & 2 Lines tripped. Due to tripping of these lines, Kokrajhar, Bilasipara, Gauripur areas of Assam Power System were isolated from NER Grid and collapsed due to no source available in these areas. Power was extended to Kokrajhar, Bilasipara, Gauripur areas of Assam Power System by charging 132 kV BTPS Kokrajhar 2 line at 12:28 Hrs of 16.07.2024.	132 kV BTPS Kokrajhar 1 & 2 Lines
18	GD I	Rongkhon,Ampati and Ganol area of Meghalaya Power System	17-07-2024 12:29	17-07-2024 12:37	00:08	22.5	24	0.89%	1.00%	2521	2401	Tenga area of Arunachal Pradesh Power System were connected with rest of NER Power system via 132 kV Balipara- Tenga line and Khupi area is connected via 132kV Khupi- Kameng line. Tenga and Khupi area are connected via 132 kV Tenga-Khupi line. At 13:42 Hrs of of 09.07.2024, 132 kV Balipara-Tenga line and 400/132kV 40 MVA ICT at Kameng tripped leading to blackout of Khupi and Tenga area of Arunachal Pradesh Power system. Dikshi generation also tripped due to difference in load and generation.	132 kV Balipara-Tenga line
19	GD I	Ganol HEP of Meghalaya power system	17-07-2024 10:11	17-07-2024 10:16	00:05	15	9	0.56%	0.32%	2702	2805	Ganol HEP of Meghalaya Power System is connected to NER Power system via 132kV Ganol –Rongkhon line. At 10:11 hrs, 132 kV Ganol – Rongkhon line tripped leading to tripping of Ganol HEP. Power was extended to Ganol Generating Station of Meghalaya Power System by charging 132kV Ganol – Rongkhon line at 10:16 Hrs of 17-07-2024.	132 kV Ganol – Rongkhon line

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
20	GD I	NEIGRIHMS & IIM Area of Meghalaya Power System	17-07-2024 14:50	17-07-2024 15:15	00:25	0	2	0.00%	0.07%	2643	2965	132 kV IIM S/S of Meghalaya Power System was connected with rest of NER Grid via 132 kV IIM - NEIGRIHMS line At 14:50 hrs of 17.07.2024, 132 kV IIM - NEIGRIHMS line tripped. Due to tripping of this lines, IIM area of Meghalaya Power System was isolated from NER Grid and collapsed due to no source in the area. Subsequently, 132 kV NEIGRIHMS S/S of Meghalaya Power System was connected with rest of NER Grid via 132 kV NEHU -NEIGRIHMS line & 132 kV Khelihriat – NEIGRIHMS. At 14:56 hrs of 17.07.2024, 132 kV NEHU - NEIGRIHMS line & 132 kV Khelihriat – NEIGRIHMS. Due to tripping of these lines, NEIGRIHMS area of Meghalaya Power System was isolated from NER Grid and collapsed due to no source in the area. Power supply was extended to NEIGRIHMS area of Meghalaya Power System by charging 132 kV NEHU - NEIGRIHMS at 15:15 hrs of 17-07-2024. Power supply was extended to IIM area of Meghalaya Power System by charging 132 kV IIM - NEIGRIHMS at 15:34 hrs of 17-07-2024.	132 kV NEHU - NEIGRIHMS, 132 kV Khelihriat – NEIGRIHMS & 132 kV IIM - NEIGRIHMS lines
21	GD I	Umiam Stage 2 of Meghalaya Power System	17-07-2024 15:03	17-07-2024 15:48	00:45	14	0	0.51%	0.00%	2720	2715	Umiam Stage 2 of Meghalaya Power System was connected with rest of NER Grid via 132 kV Umiam Stg 2-Umiam Stg 1 line. At 15:03 Hrs of 17-07-2024, 132 kV Umiam Stg 2-Umiam Stg 1 line tripped. Due to tripping of this line,Umiam Stage 2 of Meghalaya Power System was isolated from NER Grid. Power supply was extended to Umiam Stage 2 of Meghalaya Power System by charging 132kV Umiam Stg 2-Umiam Stg 1 line at 15:48 hrs of 17-07-2024.	132 kV Umiam Stg 2-Umiam Stg 1 line
22	GD I	Khupi, Tenga areas of Arunachal Pradesh Power System	19-07-2024 10:43	19-07-2024 12:51	02:08	20	1	0.82%	0.04%	2432	2801	Khupi area of Arunachal Pradesh Power System were connected with rest of NER Power system via 132 kV Khupi-Tenga line and 132 kV Khupi- Kameng line. Dikshi is connected to NER grid grid via 132 kV Dikshi-Tenga line. At 10:43 Hrs of of 19.07.2024, 132 kV Bus coupler tripped at Kameng leading to under voltage at khupi end. 132kV Tenga- Khupi line tripped and 132kV Kameng- Khupi tripped leading to blackout of Khupi area. 132kV Dikshi-Tenga line tripped leading to Generation loss of Dikshi. Power is restored by charging 132 kV Tenga Khupi line at 12:51 Hrs of 19.07.2024.	132kV Tenga- Khupi & 132kV Kameng- Khupi lines
23	GD I	Deomali Area of Arunachal Pradesh Power system	19-07-2024 11:17	19-07-2024 22:19	11:02	0	1	0.00%	0.03%	2483	2882	Deomali area of Arunachal Pradesh Power System is connected to NER Power system via 220 kV AGBPP-Deomali line. At 11:17 hrs of 19.07.2024, 220 kV AGBPP-Deomali line tripped leading to blackout of Deomali area of Arunachal Pradesh power system. Power was extended to Deomali area by charging 220 kV AGBPP-Deomali line at 22:19 Hrs of 19.07.2024.	220 kV AGBPP-Deomali line
24	GD I	Dhemaji & Silapathar Areas of Assam Power system	19-07-2024 11:19	19-07-2024 11:33	00:14	0	36	0.00%	1.25%	2500	2886	Dhemaji & Silapathar areas of Assam Power System are radially connected to NER Power system via 132 kV North Lakhimpur-Dhemaji line & 132 kV Dhemaji- Silapathar line. At 11:19 hrs of 19.07.2024, 132 kV North Lakhimpur-Dhemaji line tripped leading to blackout of Dhemaji & Silapathar areas of Assam power system. Power was extended to Dhemaji area of Assam power system by charging 132 kV North Lakhimpur-Dhemaji line at 11:33 hrs of 19.07.2024.	132 kV North Lakhimpur-Dhemaji line
25	GD I	Rengpang area of Manipur Power System	20-07-2024 18:05	22-07-2024 19:18	49:13	0	3	0.00%	0.10%	3408	3004	Rengpang area of Manipur Power System was connected with rest of NER Grid through 132 kV Loktak-Rengpang line. 132kV-Jiribam-Rengpang line was under long outage since 18:18 Hrs of 17.11.2023. At 18:05 Hrs of 20-07-2024, 132 kV Loktak-Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in this area. Power was extended to Rengpang area of Manipur power system by charging 132 kV Loktak-Rengpang line at 19:18 hrs of 22.07.2024.	132 kV Loktak-Rengpang line
26	GD I	Karong area of Manipur Power System of NER	21-07-2024 10:15	21-07-2024 11:36	01:21	0	9	0.00%	0.35%	2496	2564	Karong area of Manipur Power System was connected with rest of NER Grid through 132 kV Karong Kohima Line. 132 kV Yurembam-Karong line was idle charged condition since 17:05 hrs of 20.07.2024. At 10:15 Hrs of 21-07-2024, while closing the breaker at Karong end for 132 kV Yurembam-Karong line, 132 kV Karong-Kohima Line tripped due to tree branches which were is in touch with the Bus isolator as informed. Due to tripping of this element, Karong area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in this area. Power is extended to Karong area of Manipur Power System by charging 132 kV Karong-Kohima Line at 11:36 Hrs of 21.07.2024.	132 kV Karong-Kohima Line

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event (GI 1or GI 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
27	GD I	Kohima area of Nagaland Power System and Karong area of Manipur Power System of NER	21-07-2024 12:19	21-07-2024 12:59	00:40	0	14	0.00%	0.53%	2562	2618	Kohima area of Nagaland Power System and Karong area of Manipur Power System were connected with rest of NER Grid through 132 kV Dimapur(PG)-Kohima Line, 132 kV Kohima – Zhadima Line. 132 kV Yurembam-Karong line was idle charged condition since 11:42 hrs of 21.07.2024. 132kV Kohima – Meluri was under outage since 10:05 Hrs of 27-09-2023 as S/D taken by Kohima trans. Div. for dismantling of Tower no. AP 130. At 12:19 Hrs of 21-07-2024, 132 kV Dimapur(PG)-Kohima Line and 132 kV Kohima – Zhadima Line tripped. Due to tripping of these elements, Kohima area of Nagaland Power System and Karong area of Manipur Power System were isolated from NER Grid and collapse due to no source available in these areas. Power is extended to Kohima area of Nagaland Power System and Karong area of Manipur Power System by charging 132 kV Dimapur(PG)-Kohima Line at 12:59 Hrs of 21.07.2024.	132 kV Dimapur(PG)-Kohima Line and 132 kV Kohima – Zhadima Line
28	GD I	Karong area of Manipur Power System	21-07-2024 10:15	21-07-2024 11:36	01:21	0	9	0.00%	0.35%	2496	2564	Karong area of Manipur Power System is connected with rest of NER Grid through 132 kV Imphal (MSPCL)-Karong and 132 kV Karong-Kohima lines. Prior to the event, 132 kV Imphal(MSPCL)-Karong line was under idle charged condition since 17:05 Hrs of 20.07.2024. At 10:15 Hrs of 21.07.2024, while closing the breaker at Karong end for 132 kV Imphal(MSPCL)-Karong line, 132 kV Karong-Kohima Line tripped due to tree branches which came in touch with the Bus isolator as informed by MSPCL. Due to tripping of this line, Karong area of Manipur Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Karong-Kohima Line at 11:36 Hrs of 21.07.2024.	132 kV Karong-Kohima Line
29	GD I	Kohima area of Nagaland & Karong area of Manipur power system	21-07-2024 12:19	21-07-2024 12:59	00:40	0	24	0.00%	0.92%	2562	2618	Kohima area of Nagaland and Karong area of Manipur Power System are connected with rest of NER Grid through 132 kV Imphal (MSPCL)-Karong, 132 kV Dimapur-Kohima & 132 kV Kohima-Zhadima lines. Prior to the event, 132 kV Yurembam-Karong line was under idle charged condition since 11:42 Hrs of 21.07.2024; 132 kV Kohima – Meluri line was under outage since 10:05 Hrs of 27-09-2023 as S/D was taken by Kohima trans. Div. for dismantling of Tower no. AP 130. At 12:19 Hrs of 21.07.2024, 132 kV Dimapur-Kohima & 132 kV Kohima-Zadima lines tripped. Due to tripping of these lines, Kohima and Karong areas got isolated from NER Grid and collapsed due to no source available in these areas. Power supply was extended to Kohima area of Nagaland & Karong area of Manipur Power System by charging 132 kV Dimapur-Kohima Line at 12:59 Hrs of 21.07.2024.	132 kV Dimapur-Kohima & 132 kV Kohima-Zadima lines
30	GD I	Pasighat area of Arunachal Power System	26-07-2024 10:50	26-07-2024 11:17	00:27	0	2	0.00%	0.07%	2444	2894	Pasighat area of Arunachal Pradesh Power System was connected with rest of NER Power system via 132 kV Along – Pasighat & 132 kV Roing-Pasighat lines. Prior to the event, 132 kV Roing-Pasighat line tripped at 10:42 Hrs of 26.07.2024. At 10:50 Hrs of 26.07.2024, 132 kV Along - Pasighat Line tripped. Due to tripping of these elements, Pasighat area of Arunachal Pradesh Power System got isolated from NER Grid and collapsed due to no source available in this area. Power is restored at Pasighat area by charging 132 kV Along-Pasighat line at 11:17 Hrs of 26-07-2024.	132 kV Along - Pasighat Line
31	GD I	Udaipur area of Tripura Power System	26-07-2024 11:25	26-07-2024 12:10	00:45	0	25	0.00%	0.87%	2471	2873	Udaipur area of Tripura Power System is connected with rest of NER Grid through 132 kV Palatana-Udaipur & 132 kV Monarchak-Udaipur lines. At 11:25 Hrs of 26.07.2024, 132 kV Palatana-Udaipur & 132 kV Monarchak-Udaipur lines tripped. Due to tripping of these lines, Udaipur area of Tripura Power System got isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Udaipur area of Tripura Power System by charging 132 kV Palatana-Udaipur line at 12:10 Hrs of 26.07.2024.	132 kV Palatana-Udaipur & 132 kV Monarchak-Udaipur lines
32	GD I	Rengpang area of Manipur Power System	27-07-2024 08:23	28-07-2024 15:07	30:44	0	1	0.00%	0.04%	2592	2758	Rengpang area of Manipur Power System was connected with rest of NER Grid through 132 kV Loktak – Rengpang line. 132 kV Jiribam-Rengpang line was under outage since 18:18 Hrs of 17.11.2023. At 08:23 Hrs of 27-07-2024, 132 kV Loktak – Rengpang line tripped. Due to tripping of this element, Rengpang area of Manipur Power System was isolated from NER Grid and collapsed due to no source available this area. Power supply was extended to Rengpang area of Manipur Power System by charging 132 kV Loktak – Rengpang line at 15:07 Hrs on 28-07-2024.	132 kV Loktak – Rengpang line
33	GD I	Tuirial generating station of NEEPCO power system	27-07-2024 19:45	27-07-2024 19:51	00:06	55	0	1.57%	0.00%	3514	3757	Tuirial generating station of NEEPCO Power System was connected with rest of NER Grid through 132 kV Tuirial-Kolasib Line. At 19:45 Hrs of 27-07-2024, 132 kV Tuirial-Kolasib line tripped. Due to tripping of this element, Tuirial generating station of NEEPCO got isolated from NER Grid and collapsed due to loss of evacuation path. Power supply was extended to Tuirial S/S by charging 132 kV Tuirial-Kolasib line at 19:51 Hrs on 27-07-2024.	132 kV Tuirial-Kolasib line

20.Details of Grid Events during the Month of July 2024 in North Eastern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
34	GD I	Nongstoin area of Meghalaya power system	30-07-2024 15:45	30-07-2024 16:06	00:21	0	8	0.00%	0.30%	2500	2657	Nongstoin area of Meghalaya power system is connected with rest of NER grid through 132 kV Nongstoin-Mawngap & 132 kV Nangalbibra-Nongstoin lines. At 15:45 Hrs of 30.07.2024, 132 kV Nongstoin-Mawngap & 132 kV Nangalbibra-Nongstoin lines tripped. Due to tripping of these elements, Nongstoin area of Meghalaya power system got seperated from NER grid and collapsed due to no source available in this area. Power was extended to Nongstoin area of Meghalaya power system by charging 132 kV Nongstoin-Mawngap line at 16:06 Hrs of 30.07.2024.	132 kV Nongstoin-Mawngap & 132 kV Nangalbibra-Nongstoin lines
35	GD I	Karong area of Manipur Power System	31-07-2024 12:04	31-07-2024 12:37	00:33	0	6	0.00%	0.21%	2535	2816	Karong area of Manipur Power System was connected with rest of NER Grid through 132 kV Karong Kohima Line. 132 kV Yurembam-Karong line was under outage condition since 03:50 hrs of 30.07.2024. At 12:04 Hrs of 31-07-2024, 132 kV Karong-Kohima Line tripped. Due to tripping of this element, Karong area of Manipur Power System was isolated from NER Grid and collapsed due to no source available in this area. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Karong-Kohima Line at 12:37 Hrs of 31.07.2024.	132 kV Karong-Kohima Line

21. NEW ELEMENTS COMMISSIONED* IN JULY 2024

COMMISSIONING OF NEW GENERATING UNITS

REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE@
NR	1	AYANA RENEWABLE POWER THREE PRIVATE LIMITED (ARP3PL)	Ayana Renewable Power Three Private Limited(Solar)/Rajasthan	58.3	-	272	25-07-2024
WR	2	TP SAURYA LIMITED (TPSL)	TP Saurya Limited (Solar)/MP	140.0	II	140	18-07-2024
	3	ADANI RENEWABLE ENERGY FORTY ONE LIMITED (ARE41L)	Adani Renewable Energy Forty One Limited /Gujarat	249.6	I	249.6	24-07-2024
SR	4	JSW_NEO	JSW_NEO(Wind)/Tamilnadu	72.9	LOT-II	124.2	06-07-2024
	5	JSW RENEW ENERGY TWO LIMITED	JSW Renew Energy Two Limited(Wind)/Tamilnadu	13.5	LOT-XIV	291.6	24-07-2024
	6	PVG_IRCON_S	PVG_IRCON_S(Wind)/Karnataka	50	LOT-I	50	23-07-2024

@RE (Solar & Wind) generation FTC for complete capacity is in phases. Date mentioned is the date by which complete (Added) capacity synch./COD

COMMISSIONING OF NEW ICTs

REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	PBTSL	Bikaner_II	400/220/33	IV	500	02-07-2024
	2	RENEW HANS URJA PVT LTD	M/s Renew Hans Urja Pvt Ltd	400/33	III	200	26-07-2024
	3	RENEW HANS URJA PVT LTD	M/s Renew Hans Urja Pvt Ltd	400/33	II	200	26-07-2024
	4	RENEW HANS URJA PVT LTD	M/s Renew Hans Urja Pvt Ltd	400/33	I	200	26-07-2024
	5	POWERGRID	Fatehgarh_II	765/400/33	V	1500	29-07-2024
SR	6	TANTRANSCO	Kayathar	400/110/33	IV	200	05-07-2024
	7	POWERGRID	Nagarjuna Sagar	400/220/33	II	500	08-07-2024
	8	POWERGRID	Munirabaad	400/220/33	II	315	21-07-2024
ER	9	NTPC	North Karanpura	400/220	I	315	09-07-2024

COMMISSIONING OF NEW TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
NR	1	UPPTCL	400kV Aligarh-Shamli	I	400	03-07-2024
	2	UPPTCL	400kV Aligarh-Shamli	II	400	03-07-2024
	3	PBTL	765kV Fatehgarh_II-Bhadla_II	III	765	12-07-2024
	4	RENEW HANS URJA	400kV Jaisalmer-Renew Hans Urja Pvt Ltd	I	400	24-07-2024

LILO OF TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE
NR	1	PKTCL,PSTCL	400kV Ropar-Ludhiana Ckt-I (400kV Koldam - 400kV Ludhiana)	Ropar	400	20-07-2024
	2	PKTCL,PSTCL	400kV Koldam-Ropar Ckt -I (400kV Koldam - 400kV Ludhiana)	Ropar	400	20-07-2024
SR	3	TANTRANSCO	400kV SEPC-Ottapidaram Ckt-I (400KV SEPC-Thennampatty)	Ottapidaram	400	25-07-2024
	4	TANTRANSCO	400kV SEPC-Ottapidaram Ckt-II (400KV SEPC-Thennampatty)	Ottapidaram	400	25-07-2024
	5	TANTRANSCO	400kV Thennampatty -Ottapidaram Ckt-I (400KV SEPC-Thennampatty)	Ottapidaram	400	25-07-2024
	6	TANTRANSCO	400kV Thennampatty -Ottapidaram Ckt-II (400KV SEPC-Thennampatty)	Ottapidaram	400	25-07-2024

COMMISSIONING OF BUS/LINE REACTORS

REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE
NR	1	POWERGRID BHADLA TRANSMISSION LIMITED (PBTL)	Fatehgarh_2 -Bhadla_2 Ckt-III @ Fatehgarh_2	765	240	01-07-2024
	2	POWERGRID BHADLA TRANSMISSION LIMITED (PBTL)	Fatehgarh_2 -Bhadla_2 Ckt-IV @ Fatehgarh_2	765	240	01-07-2024
	3	UPPTCL	Aligarh- Shamli Ckt-I @ Shamli	400	50	03-07-2024
	4	UPPTCL	Aligarh- Shamli Ckt-II @ Shamli	400	50	03-07-2024
	5	POWERGRID BHADLA TRANSMISSION LIMITED (PBTL)	Fatehgarh_2 -Bhadla_2 Ckt-III @ Bhadla_2	765	240	07-07-2024

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM

REGION	S.NO.	ELEMENT NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE

NIL

*Includes first time charging (FTC) as well

22.Voltage Profile

विद्युत विभव रूपरेखा माह - जुलाई 2024 VOLTAGE PROFILE - July 2024

Date of reporting: 05-Aug-24

क्र.स. Sl. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1		अंगुल ANGUL	0%	100%	0%	0%	0	798	753	774
2		दालिपली DARLIPALI	0%	100%	0%	0%	0	793	764	777
3		गया GAYA	0%	100%	0%	0%	0	793	746	768
4	पूर्वी क्षेत्र ER	जीरत JEERAT	0%	100%	0%	0%	0	783	736	760
5		झारसुगुडा JHARSUGUDA	0%	100%	0%	0%	1	803	770	785
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	796	746	769
7		राँची RANCHI	0%	100%	0%	0%	0	794	755	776
8		सासाराम SASARAM	0%	100%	0%	0%	0	784	744	764
1		आगरा AGRA	0%	100%	0%	0%	0	801	750	777
2		आगरा (फ़तेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	0	801	740	774
3		अजमेर AJMER	0%	99%	1%	1%	10	804	752	785
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	801	742	772
5		अनपरा सी ANPARA-C	0%	100%	0%	0%	0	782	757	768
6		अनपरा डी ANPARA-D	0%	100%	0%	0%	0	778	753	765
7		अंता ANTA	0%	100%	0%	0%	0	797	764	781
8		बलिया BALLIA	0%	100%	0%	0%	0	793	735	764
9		बारा BARA	0%	100%	0%	0%	0	780	751	766
10		भाडला BHADLA	0%	100%	0%	0%	1	802	741	778
11		भाडला-2 BHADLA-2	0%	100%	0%	0%	0	800	748	775
12		भिवानी BHIWANI	0%	99%	1%	1%	8	805	748	779
13		बीकानेर BIKANER	0%	100%	0%	0%	3	805	738	780
14		बरेली BAREILLY	0%	100%	0%	0%	1	801	741	774
15		चित्तौड़गढ़ CHITTORGARH	0%	99%	1%	1%	11	804	769	788
16		फतेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	795	739	775
17		फ़तेहपुर FATEHPUR	0%	100%	0%	0%	0	791	743	770
18	उत्तरी क्षेत्र NR	घाटमपुर GHATAMPUR	0%	100%	0%	0%	2	804	751	781
19		ग्रेटर नोएडा GREATER NOIDA	0%	100%	0%	0%	1	806	744	776
20		हापुड़ HAPUR	0%	100%	0%	0%	0	800	733	770
21		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	0	799	741	769
22		झाटिंकरा JHATIKARA	0%	100%	0%	0%	0	799	742	771
23		कानपुर जीआईएस KANPUR GIS	0%	100%	0%	0%	0	798	740	769
24		खेतरी KHETRI	0%	100%	0%	0%	0	803	741	777
25		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	799	747	776
26		ललितपुर LALITPUR	0%	100%	0%	0%	0	773	732	760
27		लखनऊ LUCKNOW	0%	100%	0%	0%	0	795	739	769
28		मैनपुरी MAINPURI	0%	100%	0%	0%	0	792	738	764
29		मेरठ MEERUT	0%	100%	0%	0%	2	806	739	779
30		मेरठ(UP) MEERUT (UP)	0%	100%	0%	0%	2	803	740	778
31		मोगा MOGA	0%	99%	1%	1%	5	804	734	779
32		ओबरा सी OBRA C	0%	100%	0%	0%	0	777	748	761
33		उरई ORAI	0%	100%	0%	0%	2	804	751	781
34		फागी PHAGI	0%	99%	1%	1%	5	803	756	781
35		रामपुर RAMPUR	0%	99%	0%	1%	4	808	717	767
36		उन्नाव UNNAO	0%	100%	0%	0%	0	784	731	760
37		वाराणसी VARANASI	0%	100%	0%	0%	0	794	747	772
1		अकोला AKOLA	0%	100%	0%	0%	0	794	751	776
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	796	752	777
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	789	754	771
4		भुज BHUJ	0%	100%	0%	0%	0	795	747	780
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	787	745	769
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	787	758	770
7		बीना BINA	0%	100%	0%	0%	0	802	752	778
8		बनसकंठा BANASKANTHA	0%	100%	0%	0%	3	802	758	787
9		चांपा CHAMPA	0%	100%	0%	0%	0	803	764	781
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	797	756	777
11		धर्मजयगढ़ DHARAMJAIGARH	0%	100%	0%	0%	0	788	758	771
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	791	758	778
13		एकतुनी EKTUNI	0%	100%	0%	0%	0	796	745	775
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	2	803	766	787
15		ग्वालियर GWALIOR	0%	100%	0%	0%	2	800	752	779
16		इंदौर INDORE	0%	100%	0%	0%	0	789	752	771
17	पश्चिमी क्षेत्र WR	जबलपुर JABALPUR	0%	100%	0%	0%	3	806	760	785
18		खंडवा KHANDWA	0%	100%	0%	0%	0	791	756	773
19		कोरडी KORADI	0%	100%	0%	0%	0	784	752	768
20		लकाडिया LAKADIYA	0%	100%	0%	0%	0	795	754	777
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	793	767	780
22		पड़घे PADGHE	0%	100%	0%	0%	0	801	748	780
23		परली PARLI	0%	97%	3%	3%	24	806	761	789
24		पुणे PUNE	0%	100%	0%	0%	3	804	742	779
25		राजनन्दगांव RAJNANDGAON	0%	100%	0%	0%	2	802	767	787
26		सासन SASAN	0%	100%	0%	0%	0	794	760	778
27		सतना SATNA	0%	100%	0%	0%	2	804	761	783
28		सिवनी SEONI	0%	100%	0%	0%	0	796	762	779
29		सीपत SIPAT	0%	100%	0%	0%	0	787	760	771
30		सोलापुर SOLAPUR	0%	99%	1%	1%	5	803	761	786
31		तिरोड़ा TIRORA	0%	100%	0%	0%	0	779	752	766
32		तामनार TAMNAR	0%	100%	0%	0%	0	792	758	780
33		वड़ोदरा VADODARA	0%	98%	2%	2%	13	809	755	778
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	798	762	780
35		वर्धा WARDHA	0%	100%	0%	0%	0	799	755	781
36		वरोरा WARORA	0%	99%	1%	1%	4	802	759	787
1		अरियलूर ARIYALUR	0%	95%	5%	5%	36	811	744	783
2		कड़पा CUDDAPAH	0%	96%	4%	4%	27	812	755	786
3		चिल्कलूरीपेट CHILAKALURIPETA	0%	99%	1%	1%	9	809	754	782
4		कर्नूल KURNOOL	0%	100%	0%	0%	1	803	748	782
5		महेश्वरम MAHESWARAM	0%	96%	4%	4%	28	807	756	787
6		निज़ामाबाद NIZAMABAD	0%	92%	8%	8%	63	807	752	789
7	दक्षिणी क्षेत्र SR	नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	794	750	776
8		नार्थ चेन्नै पूलिंग NORTH CHENNAI PS	0%	85%	15%	15%	109	817	741	788
9		रायचूर RAICHUR	0%	100%	0%	0%	0	800	748	780
10		श्रीकाकुलम SRIKAKULAM	0%	99%	1%	1%	4	802	759	786
11		तिरुवलम THIRUVALEM	0%	100%	0%	0%	0	798	745	777
12		वेमागिरी VEMAGIRI	0%	94%	6%	6%	43	807	758	786
13		वारंगल WARANGAL	0%	85%	15%	15%	109	817	741	788
1		बालिपारा BALIPARA (400 kV)	0%	100%	0%	0%	0	417	396	405
2		बिस्वनाथ चरिआली BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	414	399	406
3		बोंगाईगांव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	416	402	409
4		बोंगाइगांव टीपीएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	414	395	404
5		इम्फ़ाल IMPHAL (400 kV)	0%	100%	0%	0%	0	412	396	402
6		बर्निहाट BYRNIHAT (400 kV)	0%	100%	0%	0%	0	412	396	402
7		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	416	391	403
8		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	410	401	406
9	पूर्वोत्तर क्षेत्र NER	मीसा MISA (400 kV)	0%	100%	0%	0%	0	415	394	403
10		न्यू मरियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	416	392	406
11		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	417	395	407
12		पालाटाना PALATANA (400 kV)	0%	100%	0%	0%	0	417	398	407
13		पूर्वकंचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	416	391	402
14		रंगानदी RANGANADI (400 kV)	0%	100%	0%	0%	0	416	391	402
15		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	418	398	407
16		सुबानसिरी SUBANSIRI (400 kV)	0%	100%	0%	0%	0	417	396	405
17		सूर्यामणिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	413	391	402
18		थौबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	415	384	401

All listed stations are 765 kV stations unless otherwise mentioned

*Percentage is calculated w.r.t. Time of one month.

23.ALL TIME HIGHEST

31-07-2024

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	91215 19-06-2024	82312 23-07-2024	1990 18-06-2024	443 01-08-2023	86 07-08-2023	186 01-05-2024
WR	74974 01-06-2024	67482 30-05-2024	1648 06-05-2024	167 18-12-2014	310 28-05-2024	102.85 30-04-2024
SR	68735 13-03-2024	54798 29-04-2024	1416 05-04-2024	208 31-08-2018	323 26-07-2024	148.7 31-03-2024
ER	32531 10-06-2024	29695 29-05-2024	692 10-06-2024	157 14-09-2022	-	7 14-07-2024
NER	3764 27-07-2024	3710 26-07-2024	76 27-07-2024	43 27-06-2024	-	2 22-06-2022
All India	250070 30-05-2024	227354 29-05-2024	5466 30-05-2024	877 30-08-2022	619 28-05-2024	426 30-04-2024

Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)	
			As on date		As on date
NR	Punjab	15980	29-06-2024	366.8	21-07-2024
	Haryana	14524	31-07-2024	293.4	19-06-2024
	Rajasthan	17949	20-01-2024	379.1	30-05-2024
	Delhi	8568	18-06-2024	177.7	18-06-2024
	UP	30032	13-06-2024	658.8	17-06-2024
	Uttarakhand	2863	14-06-2024	62.1	14-06-2024
	HP	2235	20-01-2024	41.2	26-06-2024
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	66.8	26-01-2024
	Chandigarh	443	13-06-2024	9.1	18-06-2024
	Railways_NR_ISTS	-	-	-	-
WR	Chhattisgarh	6347	05-04-2024	144.4	06-04-2024
	Gujarat	25502	07-06-2024	514.2	07-06-2024
	MP	17819	29-12-2023	335.9	10-11-2023
	Maharashtra	28969	07-02-2024	648.3	30-04-2024
	Goa	803	19-04-2024	17.6	18-05-2024
	DD & DNH	1353	25-06-2024	32.1	05-06-2024
	AMNSIL	1083	10-01-2024	21.0	31-05-2022
	Balco	-	-	-	-
SR	Andhra Pradesh*	13712	04-05-2024	263.8	16-06-2023
	Telangana*	15623	08-03-2024	311.0	13-03-2024
	Karnataka	17220	12-03-2024	336.9	05-04-2024
	Kerala	5797	02-05-2024	116.1	03-05-2024
	Tamil Nadu	20830	02-05-2024	443.6	30-04-2024
	Pondy	530	31-05-2024	11.8	31-05-2024
ER	Bihar	7879	28-07-2024	172.8	30-07-2024
	DVC	3751	18-04-2023	81.2	22-04-2022
	Jharkhand	2351	09-07-2024	50.4	10-06-2024
	Odisha	7104	13-07-2023	148.5	19-04-2024
	West Bengal	12756	30-04-2024	264.0	30-04-2024
	Sikkim	137	03-02-2022	2.5	28-01-2020
	Railways_ER_ISTS	-	-	-	-
NER	Arunachal Pradesh	198	26-05-2024	4.8	30-07-2024
	Assam	2524	26-07-2024	52.8	27-07-2024
	Manipur	258	31-01-2022	3.7	10-02-2022
	Meghalaya	415	09-02-2022	7.8	31-01-2022
	Mizoram	162	27-01-2024	2.6	02-03-2024
	Nagaland	188	22-07-2024	3.42	29-07-2024
	Tripura	386	04-05-2024	7.3	02-06-2024
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014

24.System Reliability Indices Report for the month of July 2024

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	16	4.00	0.54
2	ER-NR	1	0.25	0.03
3	Import of NR	149	37.25	5.01
4	NEW-SR	1	0.25	0.03
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	1	0.25	0.03
2	ER-NR	1	0.25	0.03
3	Import of NR	6	1.50	0.20
4	NEW-SR	1	0.25	0.03
4	NER Import	0	0.00	0.00

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पॉवर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
Katwaria Sarai, New Delhi-110016.